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Learning Robot-Aware Object Affordances for Manipulation

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Abstract

A key challenge in autonomous mobile robotics lies in enabling robots to interact effectively with previously unseen objects across diverse tasks. Addressing this problem requires a profound understanding of object functionality to identify actions that are both feasible and lead to the intended outcomes. The concept of affordances provides a powerful framework for this understanding. Affordances describe the potential actions that an agent can perform with an object, determined jointly by the object’s physical and functional properties and by the agent’s own morphology and control capabilities.

This thesis examines different approaches to affordance learning aimed at enabling effective object interaction in robotic systems. In contrast to conventional object-centric approaches, we introduce robot-centric learning strategies that incorporate the robot’s morphology and control capabilities. This perspective enables affordance prediction and action planning that are tailored to the capabilities and constraints of a specific embodied agent, resulting in feasible, robust, and adaptive interaction strategies.

In the first part of this work, we present a simulation-based framework for affordance learning, with a focus on articulated object manipulation tasks such as door opening and closing. We propose a novel closed-loop manipulation pipeline that integrates point-level affordance inference from visual input with a whole-body, sampling-based control strategy. The proposed agent-aware network incorporates the full robot model and controller to generate interaction data in simulation. During execution, affordance inference is employed in a closed-loop manner, enabling the agent to recover from failures and unexpected states. We show that this capability contributes to bridging the sim-to-real gap and improves task success rates in articulated object manipulation.

In the second part of this work, we shift the focus from low-level, point-wise affordances of isolated objects to the prediction of object-level affordances within complex scenes. Understanding the potential interactions a robot can perform with objects in its environment is essential for effective high-level planning. To address this, we propose an approach for learning robot-centric affordances through interactive exploration. The method integrates an object-level mapping

module into a reinforcement learning-based exploration framework, enabling the agent to track and re-identify object instances across different viewpoints. This capability enhances exploration efficiency and results in denser affordance annotations, thereby improving the overall performance of affordance prediction.

In the third part of this work, we investigate an alternative to simulation-based affordance learning, which is often computationally expensive, by exploiting the capabilities of pretrained Vision-Language Models (VLMs). While these models exhibit strong contextual reasoning about objects and possible actions, they lack awareness of the local, task-specific physical setup encountered by a real robot. To bridge this gap, we propose a calibration method based on conformal risk control to estimate model uncertainty using data collected from the target environment. Real-world experiments involving the manipulation of fragile objects demonstrate that the proposed approach enables the robot to adapt its manipulation strategy according to the inferred object properties. This adaptability leads to a favorable trade-off between speed and safety, outperforming baseline methods.

Finally, we investigate affordance learning in the context of tasks that require precise interactions between multiple objects, such as hanging or stacking, rather than solely between the robot and a single object. We review state-of-the-art approaches and identify key limitations, including task-specific design, overfitting to specific object categories, and extensive training requirements. This analysis underscores the need for more generalizable methods capable of jointly learning relevant geometric and physical properties to enable robust understanding of multi-object interactions.

In summary, this thesis presents a comprehensive exploration of how affordances can be learned, represented, and exploited in robotic systems, with an emphasis on explicitly incorporating the robot’s embodiment into the learning process. To this end, multiple approaches are explored, including finetuning large pretrained models with real-world data and simulating the complete robot during training. The findings demonstrate that explicitly accounting for the robot’s morphology improves performance across a range of tasks, ranging from articulated object manipulation to scene-level affordance prediction and property-aware interaction. Furthermore, we analyze the requirements and review current state-of-the-art methods for tasks involving precise geometric interactions between multiple objects. The thesis concludes with a discussion of current progress in affordance learning and an outlook on promising directions for future research.

Zusammenfassung

Eine der zentralen Herausforderungen für autonome mobile Roboter besteht darin, mit zuvor unbekannten Objekten zu interagieren, um eine Vielzahl von Aufgaben zu erfüllen. Dies erfordert ein tiefes Verständnis der Funktionalität von Objekten, um bestimmen zu können, welche Handlungen machbar sind und zum gewünschten Ergebnis führen. Das Konzept der Affordanzen beschreibt die möglichen Handlungen, die ein Agent mit einem Objekt ausführen kann, und hängt sowohl von den Eigenschaften des Objekts als auch von den Fähigkeiten des Roboters ab.

Diese Arbeit untersucht Methoden zum Erlernen von Affordanzen, um Robotern eine effektive Interaktion mit Objekten zu ermöglichen. Im Gegensatz zu traditionellen Ansätzen, die Affordanzen als rein objektzentrierte Eigenschaften betrachten, schlagen wir robotikzentrierte Lernstrategien vor, die die spezifische Morphologie und die Fähigkeiten des Roboters berücksichtigen. Diese Perspektive ermöglicht die Vorhersage von Interaktionsstrategien, die für einen spezifischen Roboter durchführbar und robust sind.

Im ersten Teil dieser Arbeit stellen wir eine Methode zum Lernen von Affordanzen vor, die eine Simulationsumgebung nutzt, um Interaktionen zu simulieren. Spezifisch fokussieren wir uns auf die Manipulation artikulierter Objekte, wie das Öffnen und Schliessen von Türen. Wir führen eine neuartige Manipulationsstrategie ein, die punktbasierte Affordanzerkennung aus visuellen Daten mit einem sample-based Controller kombiniert. Unsere Pipeline nutzt das vollständige Robotermodell und den Controller, um Trainingsdaten von Interaktionen in der Simulation zu sammeln. Während der Ausführung der Interaktion verwenden wir eine Affordanzerkennung im Closed-Loop, wodurch der Agent auf unerwartete Zustände reagieren kann. Wir zeigen, dass diese Fähigkeit hilft, die Diskrepanz zwischen Simulation und Realität zu überbrücken, und die Erfolgsrate bei der Interaktion mit artikulierten Objekten verbessert.

Über punktbasierte Affordanzen einzelner Objekte hinaus konzentrieren wir uns im nächsten Schritt auf die Vorhersage von objektbasierten Affordanzen innerhalb grösserer Szenen. Das Verständnis der Interaktionsmöglichkeiten, die ein Roboter mit den Objekten in seiner Umgebung hat, ist entscheidend um die

nötigen Schritte zur Ausführung einer Aufgabe zu planen. Daher schlagen wir im zweiten Teil einen Ansatz zum Lernen robotikzentrierter Affordanzen durch interaktive Erkundung vor. Unser Ansatz integriert ein objektbasiertes Mapping-Modul mit einer Reinforcement-Learning-Pipeline, um eine effiziente Exploration zu ermöglichen. Die erstellte Map ermöglicht es dem Agenten, Objektinstanzen über verschiedene Blickwinkel hinweg zu verfolgen und wiederzuerkennen. Dies steigert die Effizienz der Exploration und führt zu dichteren Affordanzen-Labels, was wiederum die Vorhersagegenauigkeit der Affordanzen verbessert.

Anstatt Affordanzen durch Interaktion in einer Simulation zu lernen, was rechenintensiv ist, kann alternativ auf vortrainierte Vision-Language-Modelle (VLMs) zurückgegriffen werden. Während diese Modelle breites, kontextuelles Wissen über Objekte und Aktionen haben, fehlt ihnen das Bewusstsein für die lokale, reale Aufgabenumgebung. Im dritten Teil dieser Arbeit adressieren wir die Herausforderung, VLM-Vorhersagen an die lokale Umgebung anzupassen. Zu diesem Zweck führen wir eine Kalibrationsmethode basierend auf Conformal Risk Control ein, um die Modellunsicherheit auf Grundlage von im Zielumfeld gesammelten Daten abzuschätzen. In realen Experimenten mit delikaten Objekten passt unser System seine Manipulationsstrategie basierend auf den vorhergesagten Objekteigenschaften erfolgreich an und erreicht so eine bessere Balance zwischen Geschwindigkeit und Sicherheit im Vergleich zu Referenzansätzen.

Abschliessend untersuchen wir das Lernen von Affordanzen für Aufgaben, die nicht nur die Interaktion zwischen Roboter und einem Objekt erfordern, sondern präzise Interaktionen zwischen mehreren Objekten, wie zum Beispiel das Aufhängen von einem Objekt oder das Stapeln von Objekten. Wir analysieren den aktuellen Stand der Forschung und heben bestehende Einschränkungen hervor, darunter die Aufgabenspezifität, die eingeschränkte Generalisierung zwischen Objektkategorien sowie den hohen Trainingsaufwand. Wir betonen die Notwendigkeit allgemeinerer Methoden, die relevante geometrische und physikalische Eigenschaften gemeinsam lernen können, um eine robuste Interaktionsvorhersage zu ermöglichen.

Insgesamt untersucht diese Arbeit, wie Affordanzen erlernt und in der Robotik angewendet werden können. Der Fokus liegt auf der Integration der Roboterfähigkeiten in das Affordanzen-Lernen. Zu diesem Zweck werden verschiedene Ansätze untersucht, wie zum Beispiel das Finetuning eines grossen vortrainierten Modells mit realen Daten oder die Simulation des gesamten Roboters während des Trainings. Wir zeigen, dass die Berücksichtigung der Roboteranatomie bei einer Vielzahl von Aufgaben von Vorteil ist, von der Interaktion mit artikulierten Objekten über das Lernen von Szenen-Affordanzen bis hin zur Manipulation von fragilen Objekten. Ausserdem analysieren wir die Anforderungen und den Stand

der Forschung für Aufgaben, die präzise geometrische Interaktionen zwischen zwei Objekten erfordern. Die Arbeit schliesst mit einer Diskussion des aktuellen Stand des Affordanzen-Lernens und einem Ausblick auf zukünftige Forschungsrichtungen.

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Preface

This doctoral dissertation documents the research that I carried out as part of my doctorate under the supervision of Prof. Dr. Roland Siegwart at the Autonomous Systems Lab at ETH Zurich. The thesis is written as a monograph and is based on two published articles and one manuscript currently in submission.

Chapter 1 introduces and motivates the topics of our research and states the main objectives and contributions of this work. Chapters 2 to 5 provide an in-depth description of each of the research contributions presented in this thesis. Finally, Chapter 6 summarizes and discusses the overall contribution of this dissertation and discusses potential future research directions.

Statement on the Use of Generative AI. During the writing of this dissertation, generative AI ¹ was occasionally employed for language reformulation, grammar checks and image generation. These tools were not used for content generation, analysis, or any form of substantive academic research. Since their role was limited to the aforementioned uses, no explicit citations are provided. The intellectual and academic content remains entirely my own work.

¹ChatGPT, Writefull

Chapter 1

Introduction

1.1 Motivation

A central goal of autonomous mobile robotics is to relieve humans of monotonous manual tasks. Achieving this goal in unstructured environments, such as hospitals or homes, requires robots capable of perceiving and interacting with previously unseen objects while adapting to dynamically changing surroundings. This introduces challenges that differ fundamentally from those faced by traditional industrial robots, which operate in highly structured and predictable settings. In industrial domains, robots could rely on predefined object models and fixed task routines. By contrast, robots designed for unstructured environments must have high-level reasoning abilities that allow them to adapt to variations in object appearance, environmental context, and task requirements. To successfully perform diverse tasks, such as cleaning up a kitchen or assembling parts, a robot must have a comprehensive understanding of the objects within its environment. This understanding extends beyond object geometry to physical properties and knowledge of object functionality. To plan the actions needed to complete a high-level task, the robot must first infer which interactions are feasible for each encountered object (see Figure 1.1).

The question of how an object can be used by an agent and which functions it provides can be described by the term *affordances*. Introduced by psychologist Gibson [41], affordances define the set of possible actions that an agent can perform with an object, determined jointly by the object's properties and the agent's own capabilities. Examples for common affordances are: A chair is '*pushable*' and '*sittable*' while a cup is '*pickutable*' and '*pourable*'. Affordance prediction is a highly complex task as it depends not only on the geometry and physical

attributes of the object, but also on the agent’s capabilities. Moreover, unlike traditional classification problems, a single object can possess multiple affordances simultaneously, which may change dynamically as a task progresses. Consequently, affordance-based reasoning remains an open and challenging research problem, particularly with respect to generalization to unseen instances [25, 50, 89].

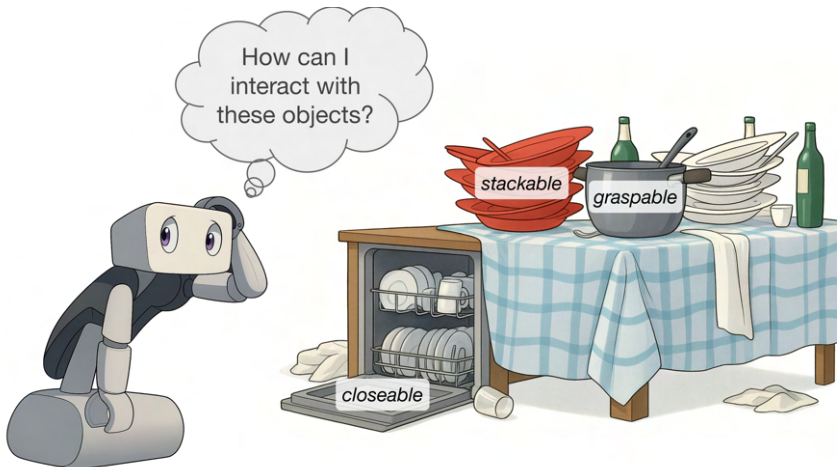


Figure 1.1: Affordances describe how an agent can interact with an object and which functionality it provides.

In robotics literature, the term *affordance* has been interpreted at varying levels of abstraction, from low-level, point-wise grasp affordances to object- or part-level functional properties, which may be either agent-independent or agent-dependent. The geometric shape of an object is often closely related to its function and can therefore be used to detect object affordances, for instance a handle-shaped object part typically indicates graspability. Most approaches in literature rely on object-centric affordance datasets, in which either the entire object or specific parts are labeled with an affordance [26, 97]. However, these object-centric methods predict affordances independently of the interacting agent, despite the fact that many affordances are only meaningful when defined relative to an agent; for example, the affordance ‘*grasp*’ depends strongly on the gripper design.

Since affordances inherently describe the potential interactions between an agent and an object, a natural approach to exploring them is through physical interaction. Some approaches use a physical simulator to generate training data

with the robot and then train a network that learns to predict affordances from this dataset [93, 136]. Others use reinforcement learning (RL) to actively explore interactions and collect informative data more efficiently from meaningful interactions [63, 98]. While previous work demonstrates that learning affordances through interaction is highly effective, these approaches are typically tailored to a single task and limited to specific object categories. Scaling these interaction-based models to generalize across diverse tasks and objects remains a significant challenge and requires large-scale models trained on vast amounts of costly training data.

To address these challenges, Vision-Language Models (VLMs) have increasingly been applied in robotics to provide high-level, contextual understanding of objects and actions [125]. Trained on web-scale image-text datasets, they enable robots to interpret natural-language instructions such as “place the red mug next to the green plate”. VLMs can be used to decompose high-level instructions to select appropriate predefined skills, as demonstrated by ‘SayCan’ [53]. Alternatively, end-to-end models trained on multi-robot interaction data, such as RT-2 [22], predict actions directly from the VLM latent space. While these large pretrained models are effective for simple, generalist tasks, they often struggle with complex 3D tasks due to mostly being trained on 2D images. They also require large-scale models and extensive, costly training data. It remains an open challenge how to best adapt their generalist knowledge to a specific robot’s morphology and capabilities.

Overall, we argue that affordances should be understood as relational properties defined jointly by the specific agent and the object. Consequently, in this thesis we investigate several approaches to learn affordances, including exploring affordances through interaction in simulation and fine-tuning large pretrained models with real-world data. Our results show that explicitly incorporating the robot’s morphology improves performance across a diverse set of tasks, from articulated object manipulation to scene-level affordance prediction and property-aware interaction.

1.2 Objectives

The overarching goal of this thesis is to develop methods for learning affordances that enable autonomous robots to reason about and interact effectively with objects. Rather than treating affordances as purely object-centric attributes, we emphasize their agent-dependent nature and explore how they can be learned in relation to the robot’s specific embodiment. The work is organized around three core objectives:

- **O1: Learning affordances through interaction in simulation:** Develop efficient affordance learning methods that move beyond passive observation by using simulated physical interactions to infer feasible actions.
- **O2: Adapting large pretrained models to local robot setup:** Adapt the broad, contextual understanding of large pretrained models to specific robotic embodiments and environments through targeted fine-tuning using local interaction data.
- **O3: Investigating manipulation strategies for object-object interactions:** Explore how affordances can be learned and represented to enable generalization across tasks that require precise object-object interactions, integrating both geometric and physical understanding.

1.3 Methodology

In this thesis, we explore how affordances can be used across the different stages of a robotic perception-action pipeline. Figure 1.2 provides an overview of the developed approaches and illustrates at which stage each of them could contribute to the overall system.

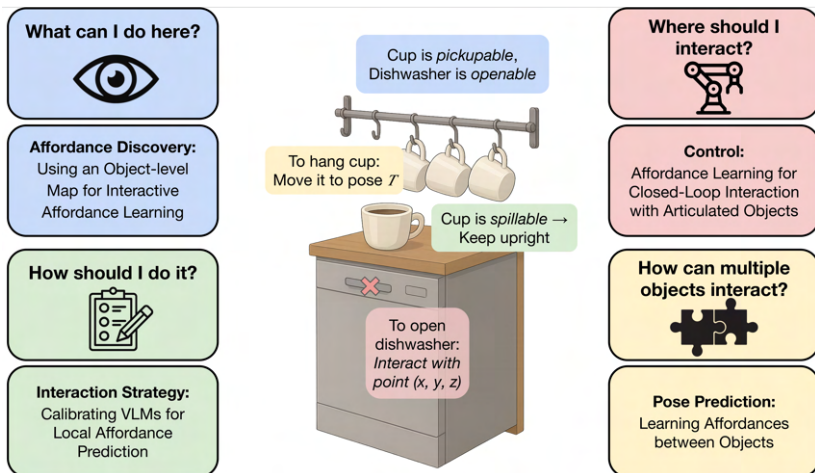


Figure 1.2: Overview of the proposed approaches and their roles within the robotics pipeline.

First, after perceiving a scene, a robot must identify which actions are feasible for the objects it observes. This object-level affordance prediction can provide valuable information to a downstream planning module by constraining its choices to meaningful and possible interactions. Our approach for efficiently learning such an affordance prediction model is presented in the section *Using an Object-level Map for Interactive Affordance Learning*. Next, the robot needs to understand how specific affordances influence the way objects should be manipulated. For instance, a spillable cup should be handled upright. In the section *Calibrating VLMs for Local Affordance Prediction*, we describe how a large pretrained Vision-Language Model can be adapted to predict local, agent-dependent handling cues. Affordance predictions can also guide low-level control. In the section *Affordance Learning for Closed-Loop Interaction with Articulated Objects*, we introduce an affordance model capable of predicting the precise point of interaction on an object, for example, where to apply force to open a door. Finally, affordances can inform interactions between multiple objects. In the section *Learning Object-Object Affordances*, we explore several state-of-the-art approaches that use affordances to predict the required relative object poses for tasks such as hanging a cup.

This chapter provides an overview of the methodologies developed for each of these approaches.

Affordance Learning for Closed-Loop Interaction with Articulated Objects

We target our first objective *O1*, which aims to learn affordances through interactions in a simulation environment. The resulting policies are conditioned on the robot’s capabilities, ensuring that the predicted actions are feasible. In our first work, we address the task of interacting with articulated objects, such as opening and closing of doors. This is a challenging task for a robotic agent because the object’s degrees of freedom and the kinematic constraints of the robot have to be considered.

We introduce the concept of agent-aware affordances, which explicitly account for the full capabilities and embodiment of the agent. In contrast, the state-of-the-art counterparts at the time of publishing were only conditioned on the end-effector geometry, restricting the transferability to the full robot used for testing [93]. To address this limitation, we introduce a novel closed-loop control framework that combines manipulation priors from affordance estimation with sampling-based whole-body control. The affordance network is trained to predict point-wise scores on the object surface, identifying the most suitable regions for interaction to accomplish the given task. Our closed-loop affordance inference approach is found to allow the agent to divide a task into multiple non-continuous motions and recover from failure and unexpected states.

Contribution: We propose a novel closed-loop manipulation framework that integrates visual affordance predictions with a sampling-based control strategy.

Result: This method allows the robot to perform dynamic mobile manipulation tasks such as opening and closing doors with high success rates.

Using an Object-level Map for Interactive Affordance Learning

Previously, we predicted point-wise affordance labels of isolated objects to use for the low-level manipulation controller. Here, we focus instead on predicting object-level affordances in a larger scene. Knowing which interaction possibilities a robot has with the objects in the scene is crucial to making a high-level plan. The goal of this work is thus to efficiently learn a model that can predict which affordances apply to an unseen object in a scene. Instead of simply tying affordance directly to the semantic class of the object (e.g. “mugs are pickupable”), we use a simulator to let the robot interact with objects to experience their affordances, thereby addressing *O1*. This ensures that the affordances are learned given the capabilities of this specific robot. Moreover, we simulate a realistic household environment with the objects in them, instead of considering the isolated objects as in most existing affordance datasets [26, 97].

We propose an interactive exploration pipeline which allows the robot to collect interaction experience while exploring an unknown environment. We integrate an object-level map to a reinforcement learning exploration pipeline. This allows the robot to track and re-identify object instances across diverse viewpoints. This results in denser and more accurate affordance annotations compared to state-of-the-art methods, which do not incorporate a map [98]. We show that our affordance exploration approach makes exploration more efficient and results in more accurate affordance prediction models compared to baseline methods.

Contribution: We propose to integrate an object-level map in an interactive affordance learning pipeline.

Result: Using an object-level map enables the robot to explore a scene more efficiently, improving affordance predictions that can subsequently inform downstream planning.

Calibrating Vision-Language-Models for Local Affordance Prediction

The first two works leveraged a simulation environment to incorporate robot-specific knowledge into the learning pipeline, but this approach is computationally intensive and can suffer from the sim-to-real gap. In this work, we investigate how to leverage large pretrained models, like VLMs, and adapt them for local robotic tasks, thus addressing *O2*. VLMs offer the advantage of providing high-level, contextual reasoning that can guide decision-making across a wide range of scenarios. However, these models can only provide generic predictions since they are lacking the information about the local physical setup and the robot’s morphology. As a result, the predictions of the VLM may be inaccurate, as they are not tailored to the specific task. In particular, the model may propose actions that are infeasible given the robot arm’s strength or reach.

In this work, we address the challenge of leveraging the high-level reasoning skills of VLMs while adapting the output to the physical reality of the robotics setup. Our key idea is to use a small interaction dataset collected in the target environment to calibrate the model’s uncertainties. It is relevant for safety and reliability to know how uncertain the prediction of an object property is, since it allows the robot to err on the side of caution in uncertain cases. To obtain uncertainty estimates grounded in the real robot scenarios, we introduce a novel calibration method based on conformal risk control (CRC) [4], which is a method typically used in machine learning to ensure that predictions are statistically reliable. By incorporating active perception, where the robot can acquire additional images from different viewpoints if uncertain, we extend conformal prediction to a sequential decision-making setting. In real-world experiments, our system successfully adapts its manipulation strategy based on the object properties, thus achieving a better balance between speed and safety compared to baseline approaches that use fixed strategies.

Contribution: We present a method to adapt the predictions of a large pretrained Vision-Language Model to a local robot setup using a novel uncertainty calibration approach.

Result: This approach enables the prediction of safe and effective interaction strategies, allowing the robot to avoid potentially unsafe actions when handling delicate objects.

Learning Object-Object Affordances

While previous chapters focused on the interaction between an agent and a single object, the final chapter extends this perspective to affordance learning for tasks

that involve interactions between two objects, thereby addressing *O3*. Object-object affordances can be seen as an extension of agent-object interactions: they require not only understanding each object’s geometry and functionality but also learning how they relate to each other. Examples include stacking, insertion, and hanging, all of which demand precise alignment of local geometric details as well as overall shape compatibility to successfully complete the task. For hanging, for example, the hanger has to support the hanging object such that the center of mass of the object is below the contact point, while for stacking the center of mass has to lie above the contact surface. These intricate tasks that require an understanding of geometry as well as of the underlying physics are challenging to solve.

Conventional approaches typically develop task-specific methods, each tailored to a single application [81, 165]. More recently, methods have been proposed that can be trained across multiple tasks [128, 129, 170]. However, many of these approaches face limitations in generalization, often performing well only within a single object category (e.g., mugs). Hence, it remains an open research topic how to efficiently design a system that can extract geometric and physical information and utilize it to successfully solve complex interaction tasks. In this thesis, we investigate different methods to design, supervise and train such object-object affordances learning methods. We discuss the limitations and advantages of different approaches and suggest ideas for future work.

Contribution: We investigated various approaches for tasks involving complex interactions between two objects, such as hanging, stacking, and inserting.

Result: We show that existing methods still require further research on effectively extracting task-relevant local geometries and predicting suitable placement poses to achieve greater generalizability and reliability.

1.4 Academic and Scientific Contributions

This section presents an overview of published papers and other contributions to the scientific community, including supervised student projects and participation in outreach events.

1.4.1 List of Publications

During the doctoral studies, the following publications were co-authored by the candidate and are part of this thesis:

1. G. Schiavi*, **P. Wulkop***, G. Rizzi, L. Ott, R. Siegwart, and J. J. Chung.

Learning agent-aware affordances for closed-loop interaction with articulated objects. In *International Conference on Robotics and Automation (ICRA)*, 2023.

2. **P. Wulkop***, H. U. Özdemir*, A. Hübner, J. J. Chung, R. Siegwart, and L. Ott. Learning affordances from interactive exploration using an object-level map. In *International Symposium of Robotics Research (ISRR)*, 2024.
3. J. Gerigk*, **P. Wulkop***, P. Bajaria, R. Siegwart, F. Shkurti, M. Itkina, and I. Gilitschenski. Handle with Care: Calibrating Vision-Language Models for safe and efficient Object Manipulation. In submission, 2025.

(*) Indicates Equal Contribution.

My personal contribution to the papers written as shared first author are the following:

1. For this work, Giuseppe Rizzi and I conceived the main idea together, where my expertise was in affordance learning and his in sampling-based control. Giulio Schiavi worked on the project throughout his master’s thesis, during which he implemented the core components of the project under our close guidance and with our continuous input. To strengthen the work for publication, I designed and carried out additional real-world experiments in collaboration with Giulio Schiavi. Finally, I took the primary responsibility for writing the paper.
2. The core idea of this project was conceived by me, based on my previous research and an extensive review of the literature on affordance learning. I guided the overall approach and together with Antonia Hübner ensured the project’s progress through close supervision of Halil Umut Özdemir, who implemented the main code as part of his master’s thesis. To move the work toward publication, I took the initiative to design, implement, and evaluate additional experiments in close collaboration with Halil Umut Özdemir.
3. For this project, Jasper Gerigk and I jointly developed the idea and defined the overall goal. I was responsible for developing and implementing the robot skills on the real-world system and conducting tests across different object types, while Jasper Gerigk implemented the VLM calibration method. We collaborated closely on writing the manuscript.

Additionally, during my PhD I co-authored another publication that was not included in the thesis: J. J. Chung, J. Förster, P. Wulkop, L. Ott, N. Lawrance, and R. Siegwart. It’s Just Semantics: How to Get Robots to Understand the World the Way We Do. *Robotics Research*, 27:3, 2023 [21].

1.4.2 Conferences and Outreach

During my doctoral studies I presented the main publications included in this thesis at the following conferences:

- 2023 IEEE International Conference on Robotics and Automation (ICRA), London, United Kingdom
- 2024 International Symposium of Robotics Research (ISRR), Long Beach, United States

Additionally, I had the opportunity to attend the following conferences:

- 2022 International Symposium of Robotics Research (ISRR), Geneva, Switzerland. I participated as a co-author of our paper [21].
- 2023 Conference on Neural Information Processing Systems, New Orleans, United States. I was a co-organizer of the 6th Robot Learning Workshop on *Pretraining, Fine-Tuning, and Generalization with Large Scale Models*. I contributed to the writing of the proposal, reached out to potential speakers and reviewers, planned the workshop schedule and acted as area chair to review the submissions.

As part of my activity within the European Union Horizon 2020 Project “Harmony: Enhancing healthcare with assistive robotic mobile manipulation”, I developed, integrated and tested a dynamic obstacle tracking pipeline on a mobile manipulator. I participated in several dissemination and collaborative events, including integration sessions with other partners and outreach workshops.

Additionally, I conducted demonstrations at several public outreach events hosted by our lab, including *Schülerinnentag*, an initiative designed to inspire female high school students to pursue studies in engineering.

Finally, I had the opportunity to undertake a research collaboration at the Toronto Intelligent Systems Lab, University of Toronto, under the supervision of Prof. Dr. Igor Gilitschenski, from August to December 2024.

1.4.3 Teaching and Supervision

During my doctoral studies, I had the opportunity to share my knowledge through teaching and supervising student projects. I served as a lecturer for the Mobile Robotics Bachelor course at Fachhochschule Graubünden (FHGR) for two semesters, where I delivered lectures and led exercise sessions. Additionally, I assisted in the Autonomous Mobile Robotics Master course at ETH, supporting course administration and leading several exercise sessions. Finally, throughout my PhD, I supervised a total of 16 student projects at ETH, which are listed below.

Master Thesis The Master's Thesis is a six-month, full-time project.

- Giulio Schiavi (Spring 2022): *Learning robot-aware affordances for interaction with articulated objects*
- Halil Umut Özdemir (Spring 2023): *Object-Level Mapping for Interactive Exploration*
- Robin Ofner (Spring 2024): *Path Planning for Apple Picking in real life*
- Xinyi Chen (Spring 2024): *Learning to place objects space-efficiently via the synergy of pushing and placing*
- Samuel Bamert (Spring 2025): *Learning Geometric and Relational Features from Point Clouds for Object Manipulation Tasks*

Semester Thesis The Semester Thesis is a semester-long, part-time project for Master students.

- Paul Fitz (Spring 2021): *Teleoperation for Teaching Robots*
- David Helm (Autumn 2021): *Inter-Object Affordance Prediction*
- Gerasimos Maltezos (Autumn 2023): *Pose estimation and modeling of articulated objects for manipulation*
- Luca Schwenker (Spring 2023): *Learning Stability-Aware Packing of Irregular Objects for Robotic Shelf Stocking*
- Roman Lemp (Spring 2024): *Real world shelf stocking*

Bachelor Thesis The Bachelor's Thesis is a semester-long, part-time project.

- Axel Wagner (Spring 2022): *Implementation of a people and moving object detector on a mobile robot*
- Henriette Stadler (Autumn 2022): *Pole detection using depth cameras for the Geranos-Project*

Perception and Learning for Robotics (PLR) Projects The Perception and Learning for Robotics course consists of a semester-long, part-time project for Master students.

- Boyan Duan and Xinyi Chen (Spring 2023): *Learning to Place Objects Space-Efficiently*
- Cafer Mertcan Akçay and İrem Kaftan (Spring 2022): *Interactive Exploration for Mapping*
- Halil Umut Özdemir (Spring 2022): *Shape-aware Predicate Learning*

Focus Project The Focus Project is a year-long, part-time project for a group of bachelor students to design and build a full robotic system.

- Samuel Bamert, Rafael Cathomen, Nicolas Gorlo, Gabriel Käppeli, Mario Müller, Tim Reinhart, Hua Shen, Henriette Stadler (2021-2022): *Geranos: Autonomous Aerial Transportation of Poles*

1.5 Organization

This thesis is organized into six chapters. Chapter 2 introduces a pipeline for learning affordances that predict suitable interaction points on articulated objects. Chapter 3 extends affordance learning to the scene-level, addressing how affordances can be efficiently learned by using map-based exploration. Next, Chapter 4 introduces a method to adapt a Vision-Language Model (VLM) for affordance prediction in a local task setup. Subsequently, Chapter 5 explores how affordances can facilitate tasks requiring interactions between multiple objects. Finally, Chapter 6 summarizes the key findings and discusses promising directions for future research.

Affordance Learning for Closed-Loop Interaction with Articulated Objects

In this chapter, we introduce the first project of this thesis. The content of this chapter is based on our paper “Learning Agent-Aware Affordances for Closed-Loop Interaction with Articulated Objects” [122] as published at the IEEE International Conference on Robotics and Automation (ICRA) 2023.

2.1 Introduction

This chapter investigates how to learn agent-aware affordances that enable a robot to interact with articulated objects. To achieve the long-term vision of service robots autonomously performing tasks in human environments, it is essential for them to learn how to open and close the doors of objects such as dishwashers and ovens. However, this remains a challenging task for robotic agents due to the complex system dynamics caused by both the robot and the object’s degrees of freedom and kinematic constraints. A common approach for this task is to estimate the kinematic and semantic properties of articulated objects from visual data [1, 55, 82, 142] and leverage this information for planning and control [6, 91]. This two-staged approach often requires heuristics, for example defining the grasping point on the handle, limiting the flexibility to deal with unseen articulation types and object geometries.

A more generic approach using affordances, i.e. where and how an agent can interact with an object, was recently explored Mo et al. [93], Wu et al. [150]. Given a point cloud of an articulated object, the authors use neural networks to predict point-wise interaction scores (*actionability*), which are used as priors for a downstream robotic controller. While these approaches show promising initial

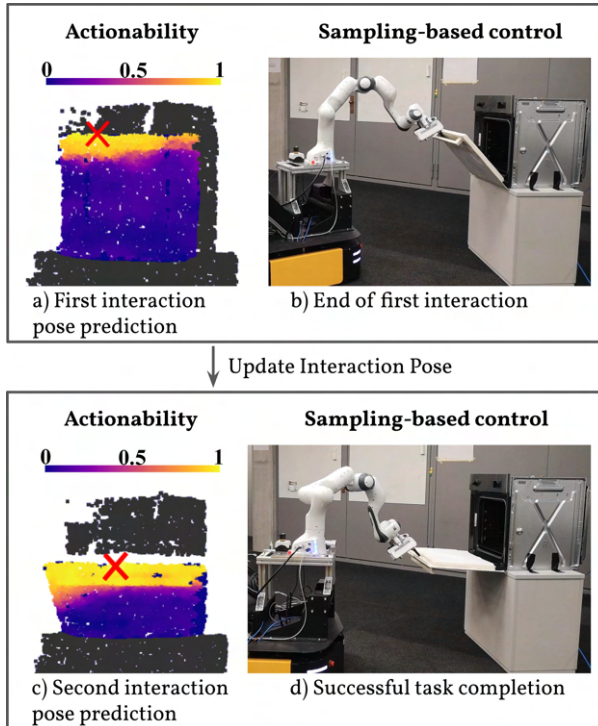


Figure 2.1: Real-world experiment of opening an oven in two motions. a) & c): Estimated actionability map where the red cross represents the selected interaction point. b) The first interaction pose becomes unfavorable, therefore an update is triggered. d): Successful task completion after the second interaction.

results, they neglect that affordances are always dependent on the agent’s capabilities, which are defined by the hardware and the controller. Instead, these models are trained on a disembodied gripper, disregarding the robot kinematics and joint limits. This can lead to predicting motions that are in reality infeasible for the real robot. Furthermore, these pipelines query the affordance module only once and then keep the interaction pose and planned trajectory fixed. Due to this open-loop perception and planning setup, they cannot perform long-term tasks requiring a change of interaction pose when the robot reaches kinematic or joint limits.

To overcome these limitations, we propose a novel closed-loop pipeline combin-

ing agent-aware affordance perception with sampling-based whole-body control. Concretely, this work deals with the non-prehensile manipulation of articulated objects with one degree of freedom using a mobile manipulator equipped with a single, fixed cylindrical finger. Taking inspiration from the *Where2Act* framework [93], we train an artificial neural network to estimate point-level affordances from visual data, indicating the success likelihood for interactions at each point. The pose proposal with the highest affordance score is passed on to a sampling-based controller (based on [118, 148]), which then iteratively determines the best interaction trajectory. This setup is well suited for non-prehensile manipulation since the interaction location and trajectory are continuously adapted based on real-time feedback, enabling adaptive and robust task execution. In contrast to previous end-effector-aware approaches, we train our affordance inference network with data collected using the full model of our target robot platform, making our predictions fully agent-aware. While this implies the need to repeat data collection and training for a different agent, we find that it significantly improves the number of successful interactions. Additionally, unlike previous approaches with a fixed interaction pose, our pipeline is able to re-evaluate the affordance model at any point during task execution, which allows the robot to execute long-term tasks where a change of interaction pose is required.

In this work, we show in ablation experiments that agent-aware training significantly increases the quality of pose proposals, and that allowing the agent to change the interaction pose during the task increases the robustness of the integrated affordance-control pipeline. Additionally, we benchmark our method against VAT-Mart [150] as a state-of-the-art work and we perform experiments in the real world (Figure 2.1), reaching a success rate of more than 70% for both fully opening and closing an oven door using a mobile manipulator. In summary, our contributions are:

- We formulate the concept of agent-aware object affordances which are conditioned on the full shape and kinematics of the robot.
- We propose a novel closed-loop manipulation framework that combines the concept of visual affordances with a sampling-based controller.

2.2 Related Work

Control strategies for manipulating articulated objects A common approach for the manipulation of articulated objects is to extract object properties like part segmentation and joint kinematics from visual data [1, 55, 82, 119, 140, 142] and use this to plan an interaction trajectory [54, 88, 131]. Mittal et al. [91] recently successfully used this approach by implementing a Model Predictive Controller (MPC) to track the feed-forward trajectory plan. However, MPC struggles with

discontinuities caused by contact dynamics, such that a simplification of the problem is required, e.g. splitting the task into first achieving a stable grasp at a fixed interaction position and then executing a predefined motion based on known articulation kinematics.

To alleviate these limitations, sampling-based control has recently emerged, allowing a more task-specific and complex interaction representation [74, 147, 148]. In previous work by Rizzi et al. [118], a sampling-based controller was successfully applied as a whole-body closed-loop controller for the non-prehensile manipulation of articulated objects. However, to decrease the size of the sampling space, a fixed end-effector interaction pose still had to be provided a priori for each object.

Affordances for manipulating articulated objects An affordance is defined as the ability of an agent to perform an action with a target object in a given environment [5, 42]. In robotics, the concept of affordances is commonly applied either on an object-level, e.g. a dishwasher is *openable* [79, 98], or on a point-level to encode information about the object geometry and grasp possibilities [94, 110, 164, 167]. In the current literature, affordances are usually learned from labeled visual data [26, 97, 121] or from self-supervised interactions [79, 94, 98, 145].

Recent works have applied this concept to the manipulation of articulated objects. VAT-Mart [150] and Where2Act [93] propose frameworks which learn affordances for robotic manipulation from interactions generated in a photo-realistic simulator [153]. A network is then trained to infer interaction points and trajectory proposals for downstream manipulation tasks from visual data. However, these approaches generate training data by simulating the interaction between a disembodied end-effector and a unit-sphere scaled object. This results in affordances that are not only unaware of the kinematics and control of the actual robot platform, but also of the realistic object size. Consequently, the trajectories derived from these approaches might be infeasible in real life, e.g. due to joint limits or collisions of the arm with the object. Additionally, they only query the visual model at the beginning of the interaction in an open-loop fashion, keeping the planned grasp location and trajectory constant during the interaction.

UMPNet [161] on the other hand proposed a closed-loop affordance-based manipulation framework that can update the interaction direction during task execution. However, it still uses a fixed interaction point, determined at the start of the interaction using only end-effector-aware (agent-agnostic) affordances.

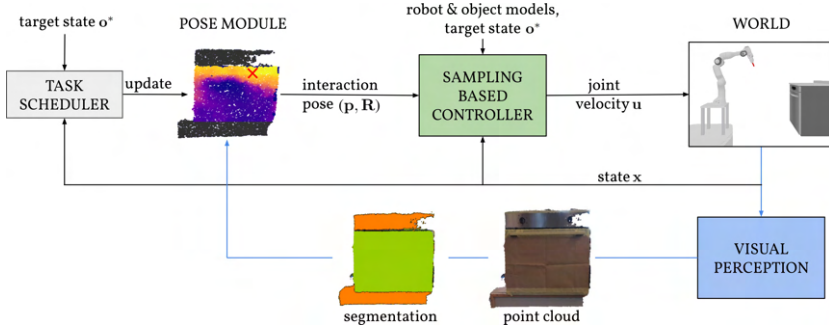


Figure 2.2: Block diagram of the control pipeline. The pose module uses affordance estimation to choose the optimal end-effector reference pose. The sampling-based controller generates velocity commands to interact with the object. The task scheduler can trigger an update of the interaction pose if required.

2.3 Problem Formulation

We consider a physical system consisting of a mobile manipulator and an articulated object which is composed of two rigid bodies connected by a rotational or translational joint. We define the state $\mathbf{x} = [\mathbf{q}, \mathbf{o}, \dot{\mathbf{q}}, \dot{\mathbf{o}}]$ as the configurations of the robot and object and their time derivatives, where $\mathbf{q}$ consists of the position of the mobile base relative to the object as well as the joint angles of the manipulator. The robot kinematics and the object geometry and articulation are assumed to be provided a priori. Given a desired object configuration $\mathbf{o}^*$, the point cloud from the robot’s point of view $\mathcal{C}$ and the current state observation $\mathbf{x}$, the overall objective is to find a robot control policy that generates joint velocity commands $\mathbf{u}$ such that $\mathbf{o} = \mathbf{o}^*$.

2.4 Agent-aware Affordance Learning

We propose a novel control pipeline combining an agent-aware affordance network with a sampling-based whole-body controller (Figure 2.2). Given an object point cloud $\mathcal{C}$, from which the movable object part is extracted by applying a segmentation mask, and target configuration $\mathbf{o}^*$, we use an affordance neural network to select a favorable end-effector interaction pose (point $\mathbf{p} \in \mathbb{R}^3$ and orientation $\mathbf{R} \in SO(3)$). The interaction pose is then used by the sampling-based controller κ to generate joint velocity commands $\mathbf{u} = \kappa(\mathbf{x} \mid \mathbf{o}^*, \mathbf{p}, \mathbf{R})$. When the interaction fails or stops leading to any improvement of $\mathbf{o}$, a task scheduler triggers a pose update as described in Subsection 2.4.2.

2.4.1 Affordance-based Pose Module

Pose inference Let $J(\mathbf{x}_{0:N}, \kappa(\mathbf{x}_{0:N-1}) \mid \mathbf{o}^*)$ be a reward function that, given the target object configuration $\mathbf{o}^*$, maps a trajectory of system states $\{\mathbf{x}_{0:N}\}$ and controller inputs $\{\mathbf{u}_{0:N-1}\}$ to a binary reward value. Given an object and its target configuration $\mathbf{o}^*$, we define the affordance A of an interaction pose $(\mathbf{p}, \mathbf{R})$ as the expected value of J over all the state-input trajectories induced by the controller $\kappa(\mathbf{x} \mid \mathbf{o}^*, \mathbf{p}, \mathbf{R})$ from initial state $\mathbf{x}_0$. Inferring the optimal interaction pose by maximizing the affordance function A is impractical, as the set of all possible interaction positions and orientations is extremely large. We therefore follow [93, 150] and define two auxiliary functions. The orientation proposal function Q generates end-effector orientation proposals given an interaction point. The actionability function α models the point-wise expected affordance A over a distribution of interaction orientations. At test time, we first select the interaction point that maximizes α . We then generate multiple orientation proposals and score their affordance. Finally, the highest-scoring pose proposal $(\mathbf{p}, \mathbf{R})$ is selected and passed on to the sampling-based controller.

Network architecture We use a neural network based on [93] to learn the functions required by the optimization problem, namely affordance A , orientation proposal Q and actionability α . The network architecture, shown in Figure 2.3, combines a PointNet++ [112] feature encoder with three Multilayer Perceptron (MLP) decoding heads, one for each function. All three networks take as input the point position $\mathbf{p}$, its feature vector $\mathbf{f}_{\mathbf{p}}$ and the target object configuration $\mathbf{o}^*$. The orientation proposal module additionally takes a sampled Gaussian noise vector $\mathbf{z} \in \mathbb{R}^{10} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ as input to generate proposals $\mathbf{R}$ which are then passed on as an input to the affordance prediction module.

Data collection The training data is collected in a simulation setup where we simulate the complete control pipeline as well as the kinematic chain, collision bodies and dynamics of the target robot platform in order to create agent-aware affordances. A baseline agent samples a random interaction pose $(\mathbf{p}, \mathbf{R})$ on the movable part of the object, where the orientations are sampled only within a 45° cone of the surface normal at point $\mathbf{p}$ (based on [150]). Given the sampled pose and the desired configuration $\mathbf{o}^*$, the sampling-based controller attempts to interact with the object for 10s. A binary reward J rates the interaction as successful if the object configuration was changed by more than 5° towards the target $\mathbf{o}^*$. Finally, we balance the dataset to contain the same number of successful and unsuccessful samples. One advantage of using the sampling-based controller to generate training data is the relatively high rate of successful interactions (13.6%), which makes it much more data efficient than first having to train a control policy (as done in [150]).

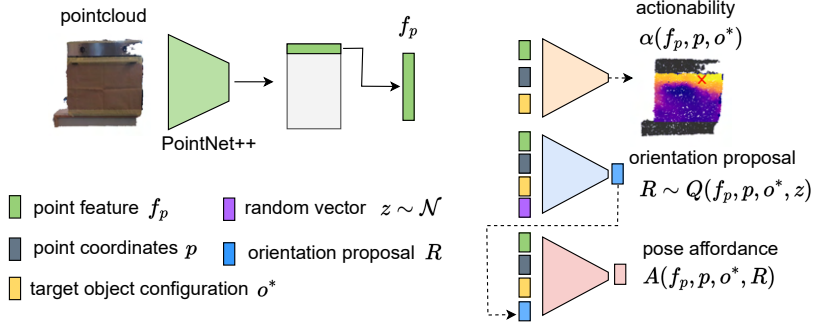


Figure 2.3: The pose module, based on [93], combines a PointNet++ [112] feature encoder with three MLP decoding heads, outputting actionability α , orientation proposal Q and pose affordance A .

Training and losses Given the point cloud and the interaction sample $(\mathbf{p}, \mathbf{R}, \mathbf{o}^*, J)$, we train the components of the pose module jointly where the affordance module learns to predict the reward realization J using a binary cross-entropy loss, while the orientation proposal module uses the mean cosine similarity loss on quaternion predictions. The actionability module is trained on an L1 loss, where the ground truth actionability for a given pose proposal can be calculated using predictions from the affordance and orientation modules. We use an Adam optimizer and employ learning rate scheduling and early stopping to ensure convergence. Training a network takes around 3 hours on a low-grade GPU (NVIDIA GTX 1050 Ti with 4 GB RAM).

2.4.2 Control Pipeline

Sampling-based controller The sampling-based controller uses a physics engine to forward simulate the system dynamics of the robot and object to iteratively refine the motion trajectory [118]. At every time step, it samples multiple robot joint velocity references $\mathbf{u}$ around the current best guess and simulates the resulting system dynamics for a time horizon of 1s. The best trajectory is then selected based on a cost function, which is made up of multiple components. The object cost encodes the offset from the target $\mathbf{o}^*$, the collision cost punishes robot-object collision, the joint limit cost and the arm reach cost prevent mechanically unreachable configurations. Finally, the pose reach cost is defined as the distance between the end-effector and the reference pose $(\mathbf{p}, \mathbf{R})$, which is required to make the optimization problem feasible. A low-level proportional-integral controller then converts the velocity reference $\mathbf{u}$ to the executed joint

torques commands.

Task scheduler To increase the robustness of the system, we implement a closed-loop affordance inference setup, allowing the robot to change its interaction pose when the current one is no longer favorable. In practice, the task scheduler triggers a pose update whenever the object state cost stagnates. To update the pose, the robot moves to a configuration from where it can observe the full object, records a point cloud, segments it and passes it through the pose module to obtain the new interaction pose.

2.5 Experiments

To evaluate the performance of our pipeline, we conducted three different experiments in simulation and additionally validated our approach in real life. For training, we collect for each task 12000 simulations with eleven different object models from the PartNet-Mobility dataset categories *oven* and *dishwasher*, which both have a revolute joint [92]. First, we compared affordances conditioned only on the end-effector (as in [93, 150]) against affordances conditioned on the full robot. Second, we evaluated the advantage of allowing the agent to update the interaction pose when the current one is no longer advantageous, i.e. split one task into multiple non-continuous motions. These two experiments were performed with seven unseen object models from the training categories since they are especially challenging in terms of reach and collision and are therefore well suited to study the advantages of our method. Finally, we show that our approach is also capable of generalizing to unseen object categories and articulation types. To this end, we tested our pipeline on 140 object models from the categories *safe*, *table* and *washing machine* containing both revolute and prismatic joints and compare it to VAT-Mart [150] as a state-of-the-art benchmark.

2.5.1 Agent and Environment

The robot platform consists of a seven degree-of-freedom manipulator mounted on a mobile base. The manipulator hand is equipped with a single, fixed cylindrical finger. One of the advantages of this simple finger design is that it allows the physics engine RAISIM to simulate the contact dynamics, which is a computationally demanding task [10], smoothly in real-time with a simulation timestep of 0.0015 s. To generate the point clouds, the robot and the object are rendered in the SAPIEN environment [153], where the object’s initial position and scale are sampled uniformly at random within predefined, category-specific bounds.

2.5.2 Baselines

As a minimum baseline, we employ the same random agent used during training for data collection. To evaluate the importance of agent-aware affordances, we compare against a purely end-effector-aware baseline where we collected the training data with a disembodied, freely moving gripper initialized directly at the interaction pose and thus ignoring the issue of reach. This is similar to the affordance-learning paradigm in Where2Act [93].

As an additional baseline, we enhance the end-effector-aware network with a reachability filter using a Jacobian-based inverse kinematics (IK) check as well as with a robot-object collision filter. These filters are implemented such that each interaction pose proposed by the end-effector-aware network has to pass the feasibility check before being executed. Next, to evaluate the effect of enabling a change of interaction pose, we created for both the end-effector-aware and the agent-aware network one version where the pose stays fixed during the interaction and a closed-loop setup that allows pose updates.

2.5.3 Results

Agent-aware vs end-effector-aware We first evaluate the quality of the proposed agent-aware interaction poses with a simplified version of the pipeline which keeps the interaction pose fixed. As in [93], we report the sample success rate as the percentage of interaction poses creating movement towards the target $\mathbf{o}^*$, i.e. $|\Delta\mathbf{o}| \geq 5^\circ$, as well as the sample reach rate defined as the percentage of pose references that are reachable by the agent. The considered tasks are to *open* or *close* articulated objects from the testing dataset using the full robot model. Over the 500 trials of each task, the initial object configuration $\mathbf{o}_0$ is such that half of the trials start with the object fully closed or fully open (for the *open* and *close* tasks, respectively), the other half start with the object open to an angle sampled randomly over the full joint range.

In the *open* task, both networks learned to interact with a similar set of points (Figure 2.4a, leading to similar average performance (Table 2.1). In the *close* task on the other hand, while the end-effector-aware network considers the entire outer surface of the door to be actionable, the agent-aware network strongly prefers interaction points close to the top edge (Figure 2.4b). As the end-effector-aware network was trained on a disembodied gripper, it often predicts interaction poses that are unreachable for the full agent due to kinematic or collision constraints. This results in a significantly lower reach and success rate compared to the agent-aware version.

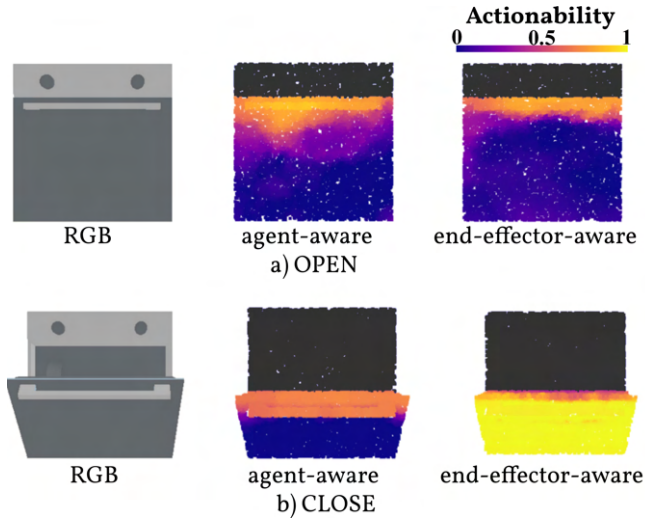


Figure 2.4: a) *Open* task: both networks learn to interact with a similar set of points. b) *Close* task: the agent-aware network strongly prefers reachable points while the end-effector-aware network, which was trained on a disembodied gripper, also predicts points on the door surface which are unreachable for the full agent.

Table 2.1 shows that even though the feasibility checks progressively improve the reachability and success rate of poses, the performance is still lower than for our proposed agent-aware network. From this experiment we conclude that there is no simple heuristic filter that can restore the agent-awareness a posteriori. The failure cases of the agent-aware model on the other hand are mostly due to suboptimal actionability predictions of the network, i.e. it proposes poses that can be reached but do not result in successful interactions. Additionally, we compared multiple network instances trained with different random seeds and found that while stochasticity has a visible effect on the affordance map, the resulting performance is consistent across networks. In general, we observe that the sampling-based controller provides a certain robustness against small errors of the predicted interaction pose, as it locally adapts the pose while minimizing the controller cost.

Closed-loop pose update In this experiment we evaluate the performance on the full opening ($\mathbf{o}_0 = 0^\circ, \mathbf{o}^* = 90^\circ$) and closing ($\mathbf{o}_0 = 90^\circ, \mathbf{o}^* = 0^\circ$) tasks and re-

OPEN	sample success rate	sample reach rate
Random	15.5%	68.5%
EE-aware	91.9%	99.8%
Agent-aware	92.1%	99.9%
CLOSE	sample success rate	sample reach rate
Random	21.0%	61.5%
EE-aware	38.5%	54.2%
EE-aware + IK check	51.7%	65.0%
EE-aware + IK + collision check	59.5%	76.6%
Agent-aware	76.1%	92.0%

Table 2.1: Sample success rate and sample reach rate using the full robot model for interaction trials of the *open* and *close* task. We compare our agent-aware network to a random baseline, the end-effector-aware network (EE-aware) and enhanced versions of the EE-aware network combined with an inverse kinematic (IK) and a collision check.

port the task success rate. Over 1000 trials we compare our closed-loop pipeline, which allows the robot to change its interaction pose during the task, with an ablated fixed pose version. The agent-aware closed-loop pipeline achieves an extremely good task success rate of 88.6% on both *open* and *close* tasks (Table 2.2). We observe two advantages of the closed-loop setup: Firstly, when an initially successful interaction pose becomes infeasible at an intermediate state, the task scheduler triggers an interaction pose update. Secondly, when the sampling-based controller is unable to perform the task at all by interacting with the current reference pose, the agent is able to recover by trying again at a different reference pose. In this simulated setup, we predominantly observed the second effect, because the sampling-based controller is able to locally adapt the interaction pose such that the task can be fulfilled in one motion, e.g. by sliding or rotating. Finally, the closed-loop setup also improves the results of the end-effector-aware network, but it is still outperformed by the agent-aware network.

Generalization capabilities To benchmark against VAT-Mart [150], we evaluated our pipeline on their test objects (*safe*, *table* and *washing machine*) and use their task definition which is to sample the target $\mathbf{o}^*$ and the initial object configuration $\mathbf{o}_0$ randomly over the full joint space. Qualitative results in Figure 2.5 show that our pipeline learned interaction strategies that generalize well to unseen object categories despite the limited object variability seen during training. Table 2.3 shows that we outperform the VAT-Mart benchmark in all tasks and categories,

	OPEN		CLOSE	
	Agent-aware	EE-aware	Agent-aware	EE-aware
Closed-loop	88.6%	76.6%	88.6%	64.3%
Fixed pose	83.6%	64.7%	69.2%	44.1%

Table 2.2: Task success rate of the agent-aware and the end-effector-aware pipeline for the long-horizon open and close tasks. While the closed-loop setup improves the performance for both pipelines, the agent-aware version clearly outperforms its ablation.

even when only using a fixed interaction pose. Our pipeline is explicitly designed to profit from available information about the object and the robot which VAT-Mart does not use, namely the collision shape and articulation model as well as the real-time feedback of robot joints and articulation angle which are required by the controller. For our targeted service robotics applications, object CAD models are often readily available [52], while generating novel models would also only require a one-time effort.

	close door	open door	close drawer	open drawer
VAT-Mart [150]	36.5%	14.3%	38.3%	31.1%
Ours (Fixed Pose)	44.0%	49.0%	68.0%	46.5%
Ours (Closed-loop)	65.8%	66.7%	80.5%	59.1%

Table 2.3: Task success rate, defined according to [150] as within a tolerance of 15% of the task, of our framework on unseen object categories compared to an ablated fixed pose version of our method and the VAT-Mart benchmark.

2.5.4 Real-world experiments

We deployed our pipeline on a mobile manipulator interacting with an oven (Figure 2.1). Collision bodies and articulation type of the oven are provided to the controller a priori, while the articulation joint angle is obtained by integrating the angular velocity measurement of an IMU mounted on the back of the oven door. The pose of the robotic platform is tracked by an OptiTrack motion capture system and the point cloud data is collected from an onboard Azure Kinect sensor. The reflective glass surface of the oven door was covered to make it visible to the RGB-D camera. For part segmentation, the Iterative Closest Point (ICP) algorithm is used to match the real-world point cloud to the point cloud rendered in simulation, allowing a projection of the segmentation mask.

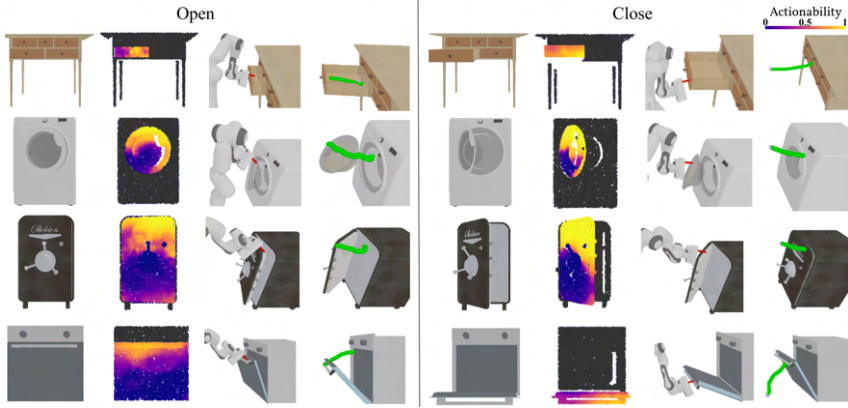


Figure 2.5: Qualitative analysis of the interactions with different articulated objects. Each interaction block shows from left to right: The RGB image, the predicted actionability map, the robot interacting with the object and the full trajectory of the end-effector (shown in green).

The closed-loop pipeline was tested over 25 trials of both the *open* ($\mathbf{o}_0 = 0^\circ$, success if $\mathbf{o}_N \geq 75^\circ$) and *close* ($\mathbf{o}_0 \geq 75^\circ$, success if $\mathbf{o}_N \leq 2^\circ$) tasks, where the joint limit of the real oven was at 85° . In Table 2.4, real-world results are compared to simulation trials with the same setup and testing object. We report the task success rate (*task s. r.*), the average number of interaction attempts (*n. int.*) and the average number of successful motions ($|\Delta \mathbf{o}| \geq 5^\circ$) needed to complete a task (*n. mot.*).

Results The network is able to produce reasonable actionability maps and interaction poses from real-world point cloud data (Figure 2.1). For the *close* task, the behavior transfers very well from simulation to the real world: The robot is able to complete the task 72% of the time and often in one motion (Table 2.4). In the *open* task, unlike in simulation, the real robot typically needs multiple attempts to slide the finger into the handle due to errors introduced by the state estimation and model mismatches. Once the end-effector is placed such that it can exert enough force, it opens the door to an intermediate state. This end-effector pose usually becomes unfavorable at a certain articulation angle (as seen in Figure 2.1b), such that a pose update is triggered and the robot uses a second and sometimes third motion to fully open the door. The recovery capabilities of the closed-loop pipeline are therefore essential to overcome the described sim-to-real gap, such that we achieve an overall task success rate of 71% for opening.

OPEN	task s. r.	time (s)	n. int.	n. mot.
real	71%	112±41	4.3±1.6	3.0±0.9
sim	100%	50±52	2.0±1.6	1.0±0.2
CLOSE	task s. r.	time (s)	n. int.	n. mot.
real	72%	70±53	2.1±1.6	1.4±0.7
sim	93%	39±29	1.9±1.2	1.2±0.5

Table 2.4: Results from real-world and simulation testing for fully *opening* and *closing* an oven. The average number of interaction attempts (*n. int.*) shows how often the task scheduler triggers a pose update and the number of motions (*n. mot.*) is the subset of interactions that leads to an improvement of the articulation state.

2.6 Conclusion

We introduced a novel closed-loop manipulation pipeline combining point-level affordance inference from visual data with whole-body control. The pipeline is based on sampling-based control, using affordances to estimate the end-effector interaction pose. Current state-of-the-art methods only condition affordance predictions on end-effector geometry. In contrast, we show that our agent-aware network is able to exploit the full robot model and low-level controller to improve the quality of the inferred interaction priors. Our pipeline also enables the agent to re-evaluate the pose model and update the interaction pose at any time during task execution. This allows the agent to split a task into two or more non-continuous motions and recover from failure and unexpected states. We find this to be crucial especially in the real-world experiments, partially compensating for the effects of the sim-to-real gap and allowing our pipeline to perform long-horizon mobile manipulation tasks with high success rates.

Limitations and future work The sampling-based controller used in our pipeline requires precise measurements of the object joint state, currently provided by an external sensor (IMU). While this is not an issue in simulation and in controlled real-world environments, for general deployment the object pose estimation should be performed using onboard sensors, e.g. from RGB-D data [27, 82, 95, 107, 146]. The controller also requires a precise object model, which could be estimated from visual data and optionally refined using interaction data [55, 57, 168]. Additionally, in future work the pipeline could be extended to allow for grasping, since not all tasks can be solved through non-prehensile manipulation.

Using an Object-level Map for Interactive Affordance Learning

In this chapter, we introduce the second project of this thesis. The content of this chapter is based on our paper “Learning Affordances from Interactive Exploration using an Object-level Map” [152] as published at the International Symposium of Robotics Research (ISRR) 2024.

3.1 Introduction

In the previous chapter, we addressed the task of learning affordances to guide a low-level controller in determining how to best interact with articulated objects. In this chapter, we take a higher-level perspective and consider affordances as a means of helping the robot understand the potential interactions it can perform with objects in a scene. For instance, in a task such as finding and retrieving a laptop, the robot must first pick up the laptop before placing it on a table. Object-level affordances can encode this information, indicating that the laptop is an object that can, in fact, be picked up by this robot.

Most recent works on visual affordance estimation are supervised learning approaches that rely on pre-existing single-object affordance datasets [26, 28, 60, 97, 101]. In contrast, we take the view that affordances for interactions are not only an object property but also depend on the robot’s intrinsic capabilities [20]. For example, whether an object is pickupable depends on the strength and the gripper type of the robot arm. Therefore, a model learned using the experiences of one robot is not directly applicable to another robot with a different morphology. Additionally, to obtain good performance transfer between training and testing, the training data should ideally consist of objects contextualized in the

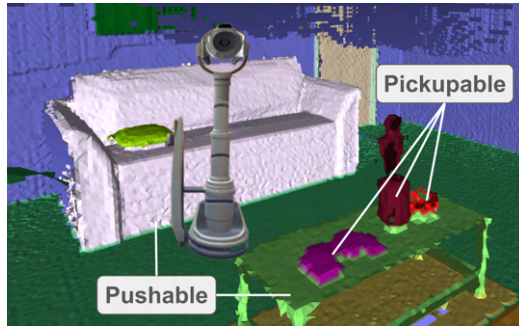


Figure 3.1: An object-level map of a living room scene generated by TSDF++ [45] with the robotic agent from the iTHOR simulation framework [68].

environment and not of isolated objects, as is the case in most existing affordance datasets. Consequently, this work explores how to leverage self-supervised learning to create a robot-specific, scene-level affordance dataset without relying on manual labels.

With this in mind, a promising approach is to learn affordances from data generated by a robot interacting with its environment in a photo-realistic simulator. Recently, Nagarajan and Grauman [98] proposed such an approach for learning affordances from interactions. They use deep reinforcement learning (RL) to explore a new 3D environment while, in parallel, training an affordance model based on the results of the executed interactions. The authors showed that affordance-based exploration leads to faster discovery of interactable regions. Additionally, their learned affordance model proved helpful in complex downstream tasks like washing the dishes. However, since the exploration framework of Nagarajan and Grauman [98] operates on each frame individually, it is unable to exploit structural scene information to expedite the agent’s affordance exploration and learning, i.e. the learned exploration agent has no sense of object permanence.

In contrast, modern robotic mapping pipelines provide representations at the object level [45, 123] and explicitly offer dense object segmentations that can be associated across multiple viewpoints. This information can be directly applied to extrapolate observed interaction data across the full exploration sequence and substantially improve learning efficiency by providing higher interaction success rates.

In this work, we create and use an explicit map of the environment in which

each discovered object and interaction is stored (see Figure 3.1). We integrate TSDF++ [45], a dynamic object-level mapping framework, into the pipeline to enable the agent to identify different object instances and track their movements across different viewpoints. Our approach consists of an RL agent exploring the simulated training environments by interacting with various objects and storing this data in the object-level map. From this, we periodically extract a dataset to concurrently train an affordance classifier that learns to accurately predict agent-specific object affordances. While the RL policy ensures that the robot efficiently explores the scene during training, the final output of the system is the learned affordance predictor.

Our contribution is the introduction of an object-level map into an interactive affordance learning pipeline. The benefits of an explicit map representation are twofold: Firstly, storing the interaction experiences in the object-level map permits the propagation of interaction outcomes across different viewpoints, creating more accurate and complete annotations. Secondly, explicitly including object information as state information improves the interaction success rate and efficiency, leading to higher-quality training data. Overall, we show that the usage of an object-level map improves the quality of the affordance predictions, therefore allowing a robot to explore a new scene more efficiently.

3.2 Related work

3.2.1 Visual Affordance Learning

Affordances are defined as the actions that an agent can apply to an object [41, 56]. They are usually learned from visual inputs by either segmenting the objects and extracting the parts that are relevant for an interaction [67, 77], or by performing dense, pixel-wise affordance predictions [26, 28, 97, 101]. Most state-of-the-art approaches focus on affordance learning via supervised learning, for which annotated datasets of object affordances are required [26, 28, 97, 101]. However, most datasets only contain single-object affordances [60, 69, 97, 102] and do not consider the context in a scene. Additionally, due to the physical constraints of each robot, affordance definitions can vary from robot to robot. Therefore labels extracted either from human annotation or from demonstrations [3, 24, 30, 99, 100, 174] do not always generalize across different robots. Rather than learning affordances from annotated datasets, we therefore propose to learn affordances from data generated by an agent interacting with different objects in a scene-level simulation environment.

3.2.2 Interactive Exploration

Most works on the exploration of unknown environments focus on navigation actions, using either learning-based methods like reinforcement learning [14, 103, 113, 114, 134] or classical SLAM-based methods [11, 133]. Recently, Liang et al. [84] proposed an embodied learning framework that uses the data collected during active exploration of a 3D environment to learn visual representations. However, beyond autonomous navigation, effective interactions with objects, e.g. opening the fridge or picking up a pillow, are also important for using robots in our daily lives. Recently developed simulators with high-level interaction capabilities [37, 68, 111] enable pipelines where agents can follow instructions or answer questions that require interaction [23, 37, 43, 127]. While these works focus on task-driven goals, we aim to use the interactions in order to build up an explicit knowledge of affordances.

3.2.3 Map Representations for Exploration

A structured representation of visited locations is beneficial for effectively exploring a new environment. Gervet et al. [40] show that the usage of explicit maps as a state compared to an implicit memory improves the generalization performance of the agent. [14, 114] use a top-down occupancy map to keep track of explored areas and the obstacles in them. In [169] 3D information is crucial since the goal is to find a specific object in a scene, so they use fused 3D point clouds as observations. Similarly to our approach, Chaplot et al. [12] build a 3D semantic map for self-supervised label propagation. Specifically, the labels are stored in a 3D semantic map and then projected onto the agent’s frame-wise observations from different viewpoints, thereby generating pixel-wise instance labels for new frames. In this work, we will use a similar approach to generate additional affordance labels.

3.2.4 Learning Affordances from Interactions

In previous works, interactive exploration has been used to learn affordances for simple objects in table-top environments [9, 75] and for block pushing tasks [64]. Wang et al. [143] focus on learning affordances for moving obstacles out of the way when navigating through a cluttered scene. Most relevant to our work, Nagarajan and Grauman [98] recently proposed a method to interactively learn generic object affordances on a scene-level. Their approach consists of a reinforcement learning algorithm that learns the interactive exploration strategy and an affordance network that is trained with the interactions executed during exploration. For each executed interaction, they mark the point of interaction and back-project it to the previous frames in order to annotate objects from different viewpoints. However, since they do not create an explicit map, they

cannot re-identify an object instance, leading to very sparse annotations and a low interaction success rate.

3.3 Approach

Our approach aims to train an affordance model using an agent’s previous experiences, which are stored in an object-level map. Figure 3.2 shows an overview of our approach. The first component on the top shows the simulation environment where the RL agent can collect experiences by moving around and attempting to either *pick up* or *push* the objects in the scene. The simulator then returns the result of whether the interaction was successful or not. Once an RL episode finishes, the collected experience, in conjunction with the object-level map and the RGB-D data from the onboard sensor, is used to generate new labels to improve the affordance network. In the following, we provide detailed descriptions of these components.

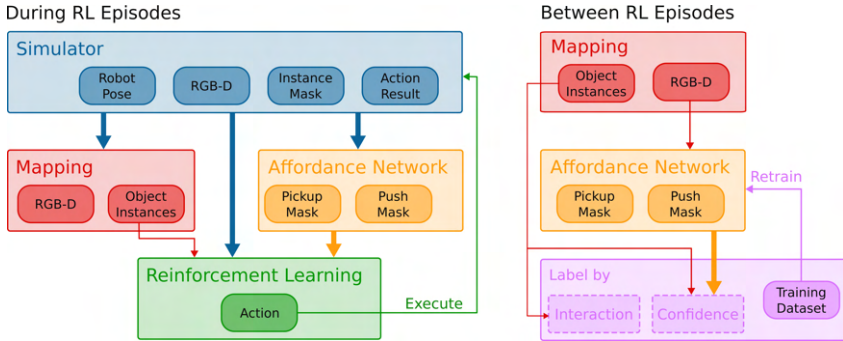


Figure 3.2: Overview of our method during training. At each time step, an action is executed and the simulator (blue) outputs if the action was successful, as well as the RGB-D image, ground truth instance segmentation mask, and robot pose. The mapping module (red) updates the map with this data, while the affordance module (yellow) predicts the affordances. The RL exploration policy module (green) estimates the next optimal action using the current state of the object-level map, RGB-D image, and affordance estimation as input. At the end of the episode, the label module (purple) annotates each object instance based on the interaction data and estimations from the current affordance network. Finally, the affordance network is retrained after every episode with the new data.

3.3.1 Mapping Framework

An object-level map is used to reconstruct the environment during exploration. At every step, the RGB-D frame and the instance segmentation mask are used to update the map. Additionally, the affordance label is stored in the map for those objects which have already been annotated. We use TSDF++ [45] which is a voxel-based mapping framework that models each object instance as a separate truncated signed distance function (TSDF) layer. An important feature of TSDF++ is that it is a dynamic object-level framework, allowing it to track object movements when the agent interacts with an object. The advantage of tracking the object is that the affordance label can be applied to the same object observed from different viewpoints. One limitation of TSDF++ is that it can only be used with rigid objects, i.e. affordances such as *slice* or *open* cannot be represented in the map. With a more flexible mapping framework however, the approach presented here would also extend to such actions.

3.3.2 Interactive Exploration Policy

We model this exploration task as a Markov decision process and use reinforcement learning to learn the best policy. The action space, state space, and reward are described in the following.

State space The state space of the agent is a combination of the raw perception input, affordance estimation, spatial representations, and action history. Figure 3.3 shows the visual representation of the state space. From the onboard sensor the RGB image as well as the depth image are used, from which we create a so-called distance image by projecting the depth image into the robot arm’s base frame to indicate if an object is within reach. The output of the current affordance estimation network indicates whether an object is likely to be interactable. Additionally, the state space contains the action history of the last five actions with their obtained success and reward as well as the inventory indicating if the robot currently holds a picked up object. In addition to these states derived from the current RGB-D input, we add states based on the object-level map to improve exploration efficiency. An egocentric 2D grid map of the previously visited positions as well as the interacted objects mask with the labels *interaction successful*, *interaction unsuccessful* and *no interaction attempted* are used as states. Furthermore, we create an occupancy map and an interacted and a non-interacted objects map by projecting the 3D voxels into 2D. While the compression into 2D space, which is necessary to ensure computational feasibility of the RL pipeline, inevitably leads to a loss of information about the vertical arrangement of objects, it captures the planar layout which is crucial for navigation. All maps are used once as global maps with a size of 10 m × 10 m and once

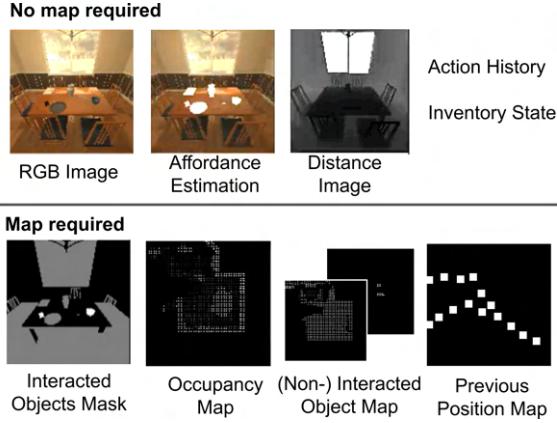


Figure 3.3: Visual representation of the state space. The elements in the top row do not require a map, while the states in the bottom are obtained through the map.

as local egocentric maps of size $2\text{ m} \times 2\text{ m}$.

Action space The action space of the robot in the iTHOR [68] simulation environment consists of navigation actions and interactions with objects. The discrete navigation actions are: *forward movement by 0.25 m*, *rotate right by 30°* , *rotate left by 30°* , *look up by 15°* , *look down by 15°* . The possible interactions with objects are *pick up*, *drop* and *push*. Once the robot picks up an object, it can attempt to move the object by $\pm 0.1\text{ m}$ along the three axes.

Interaction selection For every robot pose there are usually multiple objects that the robot could try to interact with. Nagarajan and Grauman [98] solve this ambiguity by always selecting the object that is most centered in the current RGB frame. We however want to enable the robot to actively select the object to interact with as we believe that this will lead to a more targeted exploration. Since the number of objects varies from frame to frame but the RL action space has a fixed size, we implemented the following approximation for the object selection: The center region of the image frame is split up in a grid of nine equal-sized cells and the objects are assigned to the grid cell that they are closest to. In this way, the robot selects the object by selecting the corresponding grid cell, where empty grid cells cannot be selected due to a mask and for cells with multiple objects the largest object will be selected. Additionally, we use the output of the affordance network to filter out objects with a low confidence score.

Reward To train the agent, we use a reward function with three components, as shown in (3.1). R_{nav} encourages spatial exploration by rewarding each newly visited position as well as each new orientation at a previously visited position. R_{int} rewards each interaction with a new object instance, preventing exploitation of known objects. R_{fail} punishes each failed action, for example if the agent tries to pick up a couch which is non-pickupable, discouraging repeat negative interactions.

$$R = \alpha_1 R_{nav} + \alpha_2 R_{int} + \alpha_3 R_{fail} \quad (3.1)$$

$$R_{nav} = \begin{cases} 1 & \text{if novel position} \\ 0.3 & \text{if novel orientation at prev. pose} \\ 0 & \text{otherwise} \end{cases} \quad (3.2)$$

$$R_{int} = \begin{cases} 1 & \text{if successful new interaction} \\ 0 & \text{otherwise} \end{cases} \quad (3.3)$$

$$R_{fail} = \begin{cases} -1 & \text{if action failed} \\ 0 & \text{otherwise} \end{cases} \quad (3.4)$$

Exploration policy learning The policy network is trained using Proximal Policy Optimization (Proximal Policy Optimization (PPO)) [124] with rollouts of 600 time steps. All the elements of the state space are encoded using 2D convolutional encoders for the image-like inputs with a dimension of (128, 128) and MLPs for 1D states. These encodings are concatenated and then passed through two three-layer MLPs to separately predict the action function and the value function, with MLP layer dimensions of (64, 32, 21) and (64, 32, 1), respectively. Additionally, to obtain faster convergence, we apply an action mask which disables actions that are infeasible given the current state, e.g. the action *drop* is disabled if the agent does not hold any object.

3.3.3 Affordance Learning

The affordance estimation network is trained on the interaction data collected during exploration, as shown in Figure 3.2. In the following, we will explain how the interaction data collected by the robot during an episode is labeled and used to retrain the affordance model. The improved affordance model is then used in the exploration pipeline, thereby ensuring that the two components are helping each other to converge.

Data generation and labeling To generate data, the exploration policy executes at each RL time step an action and the simulator returns whether it was successful or not. Rather than annotating the object directly on a single RGB-D frame after each interaction, we annotate the object instance in the map. This method

is similar to SEAL [12] which also uses a map to propagate annotations. The labeled map allows us to annotate each frame of the episode in which this object instance has been visible, thereby applying the information of a single interaction to multiple frames. For each frame we use the segmentation mask to create dense labels for all pixels of an annotated object. In our implementation, the segmentation mask comes directly from the simulator, but it could also easily be obtained by a separate state-of-the-art segmentation network, e.g. [66]. The baseline [98] instead does not use a segmentation mask and instead labels a sphere of radius 20 cm centered on the interaction position (see Figure 3.4). Since this sphere is of a fixed size, this strategy can lead to false positive labels for small objects and false negative labels for large objects.

Overall, the instance mapping allows us to consistently annotate objects across different viewpoints and robot movements, thereby creating more densely labeled datasets. The first and most straightforward annotation approach is to perform *annotation by interaction*, meaning that if the agent performed a successful or unsuccessful interaction with an object instance, it is annotated accordingly. However, since these interactions are sparse, we propose to additionally apply a self-supervised annotation strategy which we call *annotation by confidence*. This approach takes advantage of the classifier likelihoods outputted by the affordance network to label additional data. After the affordance network starts to converge, the affordance prediction is queried for each frame F_i and is used to further annotate objects with which the robot has not interacted. For each frame F_i in which the object appears, the percentiles of the affordance predictions are computed. If the maximum of the 95th percentile over all these frames is higher than a threshold $\xi_T = 0.9$, or if the minimum of the 5th percentile is lower than $\xi_F = 0.1$, then the object instance is annotated accordingly as a successful or unsuccessful interaction, i.e.:

$$\text{label} = \begin{cases} 1 & \text{if } \max \{ \{F_i\}_{P95} \forall i \} > \xi_T \\ 0 & \text{if } \min \{ \{F_i\}_{P5} \forall i \} < \xi_F \\ \text{none} & \text{otherwise.} \end{cases} \quad (3.5)$$

Note that since neural networks are known to be over-confident in their predictions, it could be beneficial to apply a calibration method, as suggested in [48]. At the end of each episode a partially annotated RGB-D affordance dataset is created by combining the object interaction annotations, RGB-D images and segmentation masks for each frame. In order to ensure a balanced dataset, we only use frames that contain pixels with both positive and negative labels.

Training procedure The affordance network is based on a U-Net architecture [120] with a loss function that is a combination of binary cross entropy and dice loss.



Figure 3.4: An example image of the annotation approach used by [98] and the No Map + No Seg ablation (middle) compared to our approach which uses object segmentation and is therefore able to annotate the full object (right). Annotation masks are shown in green.

The U-Net implementation contains an encoder and a decoder with 4 downscale layers each, with output sizes of (24, 48, 92, 164). To train the affordance network, we use the Adam optimizer with weight decay. After each episode, the newly generated dataset is split up into training, validation, and testing sets and added to a global affordance dataset. The training of the affordance network is then continued with this global dataset, whereby the dataset of an episode is removed from the global dataset after its 35th usage in the training loop. If the newly trained affordance network outperforms the previous one, it replaces it.

3.4 Experiments

In this section we evaluate the effect of using an object-level map for the affordance learning. First, we evaluate if it increases the efficiency of the policy learning. Second, we evaluate the accuracy of the resulting affordance model.

3.4.1 Experimental Setup

Simulation environment We evaluate our pipeline on the iTHOR [68] simulation environment, which supports high-level interactions with various types of objects in realistic 3D indoor scenes. We use living room scenes such as the one shown in Figure 3.1 since they contain a large variety of objects that the agent can interact with and also have large areas to explore. The scenes contain up to 50 different object categories which support 0 to 2 interactions (*push*, *pick up*). We split the 30 living room scenes into 25 training scenes (20% of this data is used for validation) and five testing scenes.

Baseline We compare our method to the best-performing method in the state-of-the-art paper by Nagarajan and Grauman [98] which they call “IntExp(PT)”. Note that they also evaluated a different version of their method that uses object segmentation instead of spherical labels (“IntExp(Obj)”), but this method was outperformed by “IntExp(PT)”. For this reason, we compare against the “IntExp(PT)” approach, which we implemented in our framework and call it here IntExp [98]. The following adaptations were necessary to make their approach directly comparable to our results: We retrained their code on the two affordances *pick up* and *push*, since they had used additional affordances which are not covered by our approach. Additionally, we used the AI2-THOR living room environments instead of the kitchen and we increased the image size slightly from 80x80 pixels to 128x128 pixels, since these settings were also used in our approach. Finally, our computational hardware—which we also used to train our approach—limited us to run 8 parallel processes during the RL training instead of 16 as in the original implementation, while still keeping the total number of training steps the same.

Ablations To evaluate how the integration of an object-level map impacts the affordance prediction and the exploration rate, we compare our approach to two ablations of our full method:

- Ours: The full method using the object-level map as described in Section 3.3. It uses the ground truth segmentation mask for labeling and applies annotation by interaction and confidence.
- No Map + Seg: This ablation evaluates the contribution of the map, therefore the state space contains only elements that do not require a map (RGB image, distance image, affordance estimates). As in IntExp [98], instead of a map we use a Gated Recurrent Unit (GRU) module in the RL part to aggregate observations over time.
- No Map + No Seg: This ablation aims to evaluate the importance of object segmentation. Instead of the ground truth segmentation mask used in our full approach, we apply the annotation strategy from IntExp [98]. Annotations from multiple viewpoints are generated by projecting 3D interaction points obtained during the episode (successful and unsuccessful) into all captured images where the object was visible shortly before and after the interaction. As objects have a volume, a sphere of radius 20 cm centered on these interactions is used to generate interaction labels.

3.4.2 Affordance-based Interactive Exploration

In this section, we address the question of whether the interactive exploration strategy is made more efficient by using an explicit map. To evaluate the perfor-

mance during training of our pipeline compared to the ablation baselines without a map, we use the following metrics:

- Affordance IoU: Intersection over Union of affordance predictions compared to ground truth labels per frame.
- Object-level Accuracy: The ratio of objects with correctly predicted affordance labels over all visible objects in a frame. Object-level labels are obtained by performing a majority vote of all pixels within the object boundaries.
- Interaction Success Rate: The ratio of successful interactions over all attempted interactions during an episode.
- Interacted Object Rate: The ratio of the objects that the robot interacted with during an episode over all interactable objects in the scene.
- Interactable Annotation Rate: The ratio of pixels that are correctly annotated as *interactable* over all pixels.
- Non-interactable Annotation Rate: The ratio of pixels that are correctly annotated as *non-interactable* over all pixels.

In each episode, the object positions, object states, initial position of the agent, and initial camera viewpoint are sampled randomly. At every time step, the agent executes an action from its action space (Subsection 3.3.2) and obtains feedback from the simulator on whether or not the action was successful. Figure 3.5 shows the behavior of the system over the course of the RL training episodes. For our approach the agent not only interacts with more distinct objects per episode, but it is also more successful when trying out interactions with objects (Figure 3.5a and b). This shows that our approach leads to a more targeted exploration which results in more informative labels. Interestingly, annotation by confidence, which is activated after an empirically determined number of 400 000 steps, seems to boost the performance of the Interaction Success Rate since more informed interactions can be executed by leveraging the knowledge of the pretrained affordance model.

Figure 3.5c and d show that our approach leads to a higher ratio of annotated pixels compared to the approaches that do not accumulate information in a map. Our approach can annotate more pixels as *pickutable* for multiple reasons: Not only does our agent interact with more objects successfully, it also processes these annotations more efficiently by propagating them to multiple frames using the map. Similarly, the high non-interactable annotation rate of our approach is mostly caused by the annotation of large objects like the floor and walls which can be annotated either through interaction or through annotation by confidence and then re-identified in every frame thanks to the map. Overall this suggests

Table 3.1: Affordance prediction (Median and [10th, 90th] percentile), averaged over 611 frames of the test scenes.

	IntExp [98]	No Map + No Seg	No Map + Seg	Ours
Pick up				
Precision	0.03 [0.00, 0.22]	0.00 [0.00, 0.06]	0.10 [0.00, 0.75]	0.20 [0.00, 0.99]
Recall	0.07 [0.00, 0.51]	0.0 [0.00, 0.15]	0.22 [0.00, 0.96]	0.37 [0.00, 0.99]
F1	0.04 [0.00, 0.26]	0.00 [0.00, 0.08]	0.13 [0.0, 0.65]	0.22 [0.00, 0.85]
OAcc	0.86 [0.74, 1.00]	0.82 [0.69, 1.00]	0.87 [0.77, 1.00]	0.89 [0.78, 1.00]
Push				
Precision	0.36 [0.06, 0.68]	0.30 [0.09, 0.52]	0.52 [0.15, 0.85]	0.99 [0.23, 1.00]
Recall	0.52 [0.24, 0.69]	0.93 [0.52, 1.0]	0.83 [0.12, 0.99]	0.48 [0.01, 0.96]
F1	0.41 [0.09, 0.63]	0.44 [0.14, 0.67]	0.58 [0.12, 0.83]	0.60 [0.01, 0.96]
OAcc	0.65 [0.47, 0.85]	0.70 [0.50, 0.86]	0.80 [0.60, 0.92]	0.76 [0.57, 0.92]

that retaining knowledge about object instances, which is only possible when using a map, provides an advantage over just annotating images by backprojecting interaction points as done by [98].

Finally, we evaluate the quality of the resulting affordance predictions over time by looking at the pixel-level score Affordance IoU and the Object-level Accuracy (Figure 3.5e and f). Our approach shows a faster and larger increase of the Affordance IoU than for the baselines, which means that the model quickly learns to predict the affordances accurately. The Object-level Accuracy converges quickly to a high level in all approaches, which is partially due to the fact that the predictions on an object level are more forgiving to pixel misclassifications.

To visualize the annotation quality, Figure 3.6 shows a qualitative comparison between our approach and the ablation baselines. As highlighted by the cyan colored circle, the labeling of the No Map + No Seg baseline especially leads to wrong annotations for very small objects since the labeling sphere has a fixed size. The magenta colored circle highlights an object that is annotated in our approach but not in the baselines. This is due to the annotation by confidence process which leverages the pretrained affordance model to label new objects.

3.4.3 Affordance Prediction

In this section, we evaluate the performance of the learned affordance model in comparison to our ablations as well as to the state-of-the-art work IntExp [98]. To evaluate the frame-wise affordance predictions, we created a test dataset by manually steering the agent through five unseen iTHOR living room scenes and recording each frame of the trajectories, resulting in 611 frames. In addition to the previously introduced Object-level Accuracy (OAcc), we use Precision, Re-

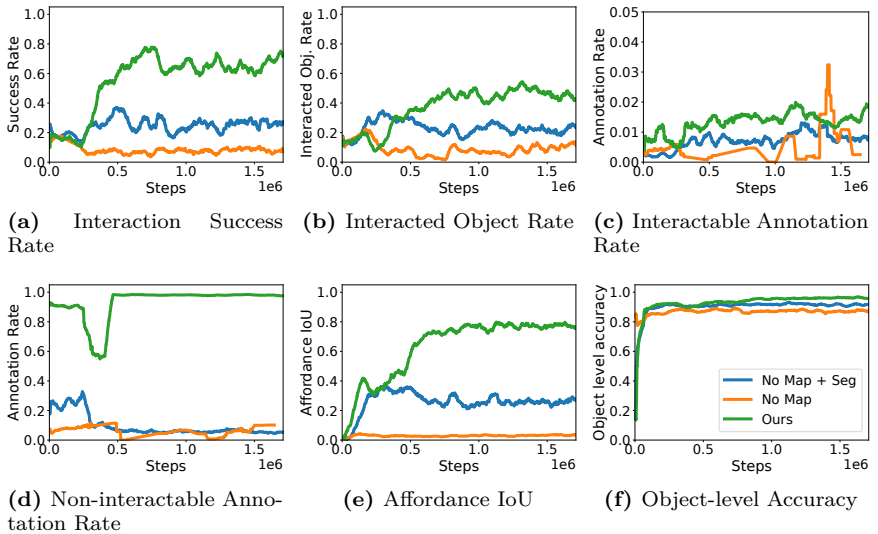


Figure 3.5: The training curves for the *pick up* affordance show that our approach leads to a higher interaction success rate and a better affordance estimation performance.

call, and F1 Score. Table 3.1 shows that for the *pick up* affordance our approach outperforms the baseline and the ablations for all metrics. As expected, the IntExp [98] method and the No Map + No Seg ablation behave similarly, since the ablation was designed to mimic the assumptions of [98]. They obtain low scores in precision and recall, which can be explained by the fact that the pick-upable objects are usually sparse and small and therefore difficult to learn with the sphere-based annotation strategy. The comparison of the ablations shows clearly that the usage of both the segmentation masks and the map have a positive impact on all metrics. Figure 3.7 shows the qualitative prediction results for one example test frame. The pickupable objects on the table are mostly correctly identified by our approach, whereas the IntExp [98] and the No Map + No Seg approaches misclassify all of these small pickupable objects. While our approach predicts some false positives on the table surface, those will tend to be averaged out when evaluating the predictions on an object-level.

For the *push* affordance, Table 3.1 shows a trade-off between precision and recall. Our approach reaches a precision of 0.99, which means that it has a very low number of false positives. The baseline and the ablations on the other hand



Figure 3.6: *Pickupable* annotations for an example frame. The cyan circle shows a small pickupable object for which the No Map + No Seg baseline annotates a fixed big area since it does not use the segmentation mask. The magenta circle shows an object that the baseline methods fail to label but our approach is able to annotate by confidence.

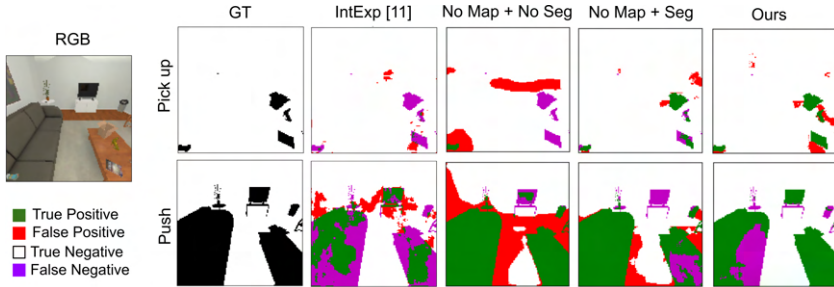


Figure 3.7: Affordance predictions for an example test frame. For the *pick up* affordance our approach results in more true positives and fewer false negatives. Similarly, for the *push* affordance our approach results in fewer false positives.

have a lower precision but a higher recall instead, meaning they are overly likely to predict *pushable*. This is clearly visible in Figure 3.7, where the baseline approaches wrongly predict parts of the floor as pushable, while our approach only labels pixels that belong to pushable objects. Interestingly, in terms of object-level accuracy the No Map + Seg ablations outperforms our approach, which could be caused by the fact that for larger objects the misclassification of a few pixels is averaged out on an object-level.

Finally, we also evaluate the Interaction Success Rate to show whether the trained affordance model can correctly predict the feasibility of an interaction. We collected a dataset by placing the robot in front of every object in the test scenes and attempting the *pick up* and *push* actions, storing the outcome as

Table 3.2: Interaction Success Rate, calculated over the 236 objects of the test scenes.

	IntExp [98]		No Map + No Seg		No Map + Seg		Ours	
	Pick up	Push	Pick up	Push	Pick up	Push	Pick up	Push
Accuracy	0.72	0.58	0.69	0.76	0.75	0.72	0.81	0.61
Precision	0.75	0.76	0.50	0.77	0.76	0.94	0.85	0.90
Recall	0.12	0.51	0.03	0.91	0.30	0.60	0.47	0.43
F1	0.21	0.61	0.05	0.83	0.43	0.73	0.60	0.58

ground truth labels. The affordance prediction modules are queried on the image frame and an object level prediction is calculated as the mean of the pixel-wise predictions of the object. In Table 3.2 we report the Accuracy, Precision, Recall and F1 score of the predicted object-wise interaction labels compared to the ground truth labels. The results show that for the *pick up* affordance our approach outperforms the baselines. For the *push* affordance, the No Map + No Seg baseline achieves a higher Recall and Accuracy than our approach, while the Precision is lower. Similar to the results from Table 3.1, this indicates that they are overly likely to predict *pushable*, which could be caused by the spherical annotations spilling over onto other objects.

3.5 Conclusion and Future work

In this work, we proposed a reinforcement learning pipeline to learn robot-centric affordances from interactions. Our approach allows an interactive exploration agent to learn how to efficiently explore an unseen environment while collecting a robot-specific dataset to train an object affordance model. To enable the agent to identify object instances and track object movements, we integrated an object-level map into our pipeline. This allows the agent to re-identify different object instances, thereby obtaining denser annotations from different viewpoints. Additionally, leveraging the information from the object-level map during exploration increases the interaction success rate and efficiency, leading to higher-quality training data. As a result, our method successfully learns to predict robot-centric object affordances.

The next step would be to transfer the learned affordance model to a real-world scenario. Given the domain gap of both the sensor data and the robot morphology, we envision a fine-tuning step in the real-world. We propose to use the pretrained affordance model and RL policy from simulation to collect additional interaction data in the real-world, using a modular robot-specific skill library to execute the high-level interaction commands. Another future step could be to

introduce additional actions such as *break* or *open*. While our architecture in general is modular and easily allows adding additional affordances, our chosen TSDF++ representation would have to be developed further to also handle articulated, separable or deformable objects.

Calibrating Vision-Language Models for Local Affordance Prediction

In this chapter, we introduce the third project of this thesis. The content of this chapter is based on our paper “Handle With Care: Calibrating Vision-Language Models for Safe and Efficient Object Manipulation” [39], which is currently in submission.

4.1 Introduction

In the previous chapter, we explored a method for learning affordances through interaction with objects in a simulation environment. Since running a simulation is computationally expensive, in this chapter we investigate how we can use large pretrained models as an alternative. The recent advent of Vision-Language Models (VLMs) provides us with a powerful tool to reason about properties of novel objects in open-world settings [78]. However, while a VLM can infer object properties from an image, it does not incorporate any knowledge about the capabilities of the specific robot in use. A robot with a rigid gripper for example might not be able to pick up a banana without damaging it. Additionally, based on the text prompt the VLM might have a different understanding of the handling constraint than the user.

The goal of this chapter is to address the challenge of aligning the VLM’s output to the real world context in order to safely manipulate objects based on their object properties. Accurately identifying relevant object properties is crucial, since this work focuses on fragile objects that would be damaged if handled using a generic pick-and-place manipulation pipeline [171]. For example, a ripe banana requires gentle handling to prevent bruising, while a cup of coffee must

maintain its upright orientation to prevent spillage. This diversity necessitates an adaptive approach that can identify and follow object-specific handling constraints.

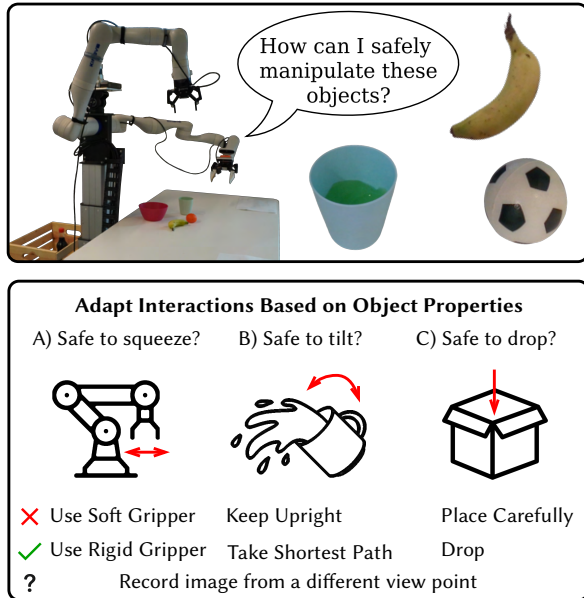


Figure 4.1: Our manipulation pipeline uses a Vision-Language Model to carefully handle objects based on their physical properties. For each object in the scene, the VLM predicts the constraints relevant for manipulation. Our conformal risk control method provides us with calibrated uncertainty estimates which are used to limit the failure rate below a predefined threshold. If uncertain, the robot records additional images of the object from different viewpoints. Once the properties have been determined, the robot uses the appropriate manipulation strategy to grasp and place the object in a box.

To this end, we developed a method that can safely manipulate objects based on their properties. Our system predicts these properties using a Vision-Language Model (VLM), whose broad general knowledge facilitates manipulation in open-world settings with novel items [78]. However, VLMs have a tendency of being overconfident [46] which can lead to neglecting safety-relevant object properties. To address this problem, we propose a novel method to extract and integrate ac-

curate estimates of how confident the VLM is about its prediction. Our method is based on conformal risk control (CRC) [4], a technique to ensure statistical reliability of black box predictions. It allows us to bound the false negative rate of object property prediction by performing an uncertainty calibration with labeled ground truth data. This enables safe, trustworthy manipulation.

While prior work has applied conformal prediction methods to robotics, we are the first to use conformal risk control in a closed-loop robotics task with sequential decision-making or consider multi-label classification. We demonstrate our approach on the task of safely transferring diverse household objects from a table to a box (ref. Figure 4.1). Our setup considers the following three constraints, of which either none, one, or multiple could apply to an object: delicateness (requiring soft gripper use), non-droppability (requiring careful placing), and possibility of spillage (requiring maintenance of upright orientation). These constraints require different handling strategies, and incorrect detection would have negative consequences either in terms of safety or efficiency. We leverage the closed-loop robotics task setup to use active perception. In cases where the uncertainty is too high, the robot can record an additional image from different viewpoints. If this does not decrease the uncertainty enough, a slow but safe interaction method is selected. For calibrating the uncertainties, we use a dataset of diverse household objects that we recorded and labeled with their robot-specific handling constraints.

While most related works consider handling constraints to be only dependent on the object itself (e.g. [34, 116, 117]), it is important for many policy decisions to also take the robot’s capabilities and its operating environment into account. Determining, for example, if an object should be treated as “delicate” depends not only on the object but also on the force of the gripper. This causes effective calibration to be more difficult, as VLMs can exhibit systematic biases and misunderstandings of certain properties [33]. To address this issue, we investigate how we can adapt the generic predictions of a VLM to a specific task setup using calibration data previously collected on this task.

With a focus on decreasing the manual effort for practitioners building systems for similar tasks, we develop and investigate three such methods to align the VLM output with a specific task: Firstly, we show that an iterative prompting approach improves the VLM’s accuracy and decreases hallucinations by ensuring that it agrees with its own prior answer. Secondly, we show how automatic prompt optimization can significantly improve upon hand-crafted prompts [70]. Lastly, we introduce the idea of using a small model to “tune” the VLM’s predictions to alleviate the systematic biases a VLM exhibits, dramatically improving weak prompts and enabling more complex approaches. We validate the effective-

ness of each of the three modular approaches on a varied set of prompts written by humans.

We evaluate our system on a diverse set of 22 household objects, achieving 67% accuracy in property detection while maintaining a false negative rate below 10%. The dataset of household objects that was collected and labeled with their affordances will be released to encourage further research on the topic of learning physical object properties. In real-world experiments across 10 test scenarios, our method successfully adapts its manipulation strategy based on predicted properties, achieving a better balance between speed and safety compared to fixed-strategy baselines.

In summary, this work presents a property-aware manipulation framework that adapts robot behavior based on predicted object constraints. We make the following contributions: (1) an uncertainty calibration method based on conformal prediction to prevent unsafe actions in multi-label decision-making settings; (2) a comprehensive evaluation of strategies for eliciting reliable property predictions from VLMs; (3) a novel dataset of common household items annotated with manipulation-relevant properties.

4.2 Related work

(Vision) Language Models for Manipulation

The recent advancements of Large Language Models (LLMs) and Vision-Language Models (VLMs) make it possible to incorporate semantics into robotic manipulation. Language models are used for different tasks ranging from perception and visual reasoning to task planning and even low-level control [78]. For instance, Mirjalili et al. [90], Vuong et al. [138] utilize VLMs to identify the optimal grasp location by considering the functionality of the object’s components. LLMs’ semantic understanding can be used to generate code to execute the required interaction policy [83]. Similarly, Zhou et al. [172] focuses on failure detection by requiring a VLM to identify possible failure causes and then generating code to monitor them.

However, VLM’s understanding of physical properties is not necessarily aligned with human intuition [33]. To improve this understanding, Gao et al. [33] collected the PhysObjects dataset, which contains physical concept annotations of common household objects. They then use the VLM in task planning to identify the object requested via its properties (“bring me the *ceramic* cup”). In our work, we go one step further and use properties to not only identify objects but to adapt the interaction strategy based on the properties.

Adapting Interaction based on Object Properties

Some recent works focus on obtaining the properties of the objects and incorporating them in their interaction strategy. For navigation, Peng et al. [108] uses a VLM to explicitly estimate the friction coefficient of upcoming terrain. The estimated value is provided to the RL policy, enabling the robot to adapt its behavior to the surface type before reaching it. For manipulation, Baek et al. [7] proposes an online learning-based framework to estimate the inertial parameters of manipulated objects to enhance the model-based controller. Most similar to our work, DeliGrasp [156] uses a VLM to estimate the mass, friction coefficient and spring constant of an object before manipulating it. While this allows them to adapt the grasp policy for delicate objects, they do not consider more complex object properties like spillability and do not get any safety guarantees on their predictions.

Uncertainty Calibration of VLMs

Hallucinations and overconfidence are a common issue in modern VLMs [163], which limits their ability to support or make trustworthy decisions. Ye et al. [163] benchmark various models in the 7-14B range and find that models' accuracy and certainty is not generally correlated. Various works [13, 135, 155] have aimed to improve the calibration of large model outputs. Tu et al. [135] show that temperature scaling, i.e. optimizing the temperature value of the final softmax for calibration, is effective for VLMs. For LLMs, Xie et al. [155] extend temperature scaling to be adaptive on the prompt by learning to predict the certainty of the next emitted token. However, none of these techniques provides statistical guarantees on the quality of their calibration.

Combining Conformal Prediction, LLMs and Robotics

LLMs and Conformal Prediction have been combined in robotics to detect whether an LLM is uncertain about the task and ask the user for clarification if necessary [116]. This was achieved by requesting the LLM to select the correct plan in a multiple-choice setting and using the probabilities of each option as their score. As it is possible that multiple rounds of clarifications are required for a single task, the Split-Conformal Prediction framework [106, 137] was extended from calibrating a single decision to calibrating a sequence of decisions for the same task. Various works have built upon this approach, including Ren et al. [117], which designed a system in which an embodied agent would explore the environment until it could confidently answer the question asked. Mullen Jr. and Manocha [96], on the other hand, decreased the number of clarification questions by introducing a feasibility measure. A more comprehensive review can be found in Shorinwa et al. [126]. In line with these works, we demonstrate

how conformal prediction can be used in a multi-step multi-label classification problem. We go beyond these prior approaches by addressing the challenge of aligning VLM predictions with the robot’s physical capabilities (motion types, grip strength) and real-world constraints (drop height), ensuring that property predictions are calibrated to the specific manipulation context rather than just task or physical-world understanding.

4.3 Background

Learned prediction models are typically trained to maximize accuracy. This allows for and often leads to miscalibration, i.e., the predicted confidence level does not match the true likelihood of being correct [49]. Conformal Prediction addresses this by providing a method to calibrate arbitrary models’ outputs to ensure statistical validity of their predictions. The basic idea is to construct a predictor that, for a given input, produces a set of possible outputs that satisfy some statistical correctness guarantee. This is achieved by using a calibration dataset and treating the original predictor p as a heuristic confidence measure, assuming that a higher output value for some option is correlated with it being correct.

The most common approach, split-conformal prediction [106, 137], aims to ensure that the true output is in the set of possible outputs with a probability of at least $1 - \alpha$ with α being a user-defined target error rate. Given a classifier p and a calibration dataset $\{X_i, Y_i\}_{i=0}^n$ where X_i represents inputs and Y_i their true labels, the method constructs a prediction function $\mathcal{C}$ that outputs a set of labels satisfying

$$P(Y_{n+1} \notin \mathcal{C}(X_{n+1})) < \alpha,$$

where (X_{n+1}, Y_{n+1}) represents a new test point. The prediction function $\mathcal{C}$ selects labels whose prediction scores exceed a threshold λ . Typically, the original uncalibrated model serves as p but any function whose output is correlated with confidence can be used.

Split-Conformal Risk control cannot be applied when more than one correct option must be selected. Conformal Risk Control [4] addresses this shortcoming and allows controlling any arbitrary loss function ℓ , which measures the error of a prediction between 0 (perfect) and its maximum L (completely wrong). This allows applications to control losses such as the false negative rate or the F1-score in scenarios with multiple true labels [4].

For effective calibration, CRC requires that ℓ be non-increasing with respect to λ .

4.4 Method

Our approach addresses the challenging task of a robot interacting with unknown objects which can have different context-dependent properties (Subsection 4.4.1). The system identifies which properties apply by querying a VLM based on one or more images of the object. These images are taken by the robot from different locations relative to the object. Since it takes time for the robot arm to move the camera to a new location, images should only be taken if needed. The primary goal is to obtain per object guarantees on the false negative rate given that a varying number of images can be taken (Subsection 4.4.2). Additionally, we want to improve the VLMs performance by better aligning the VLM’s understanding of the properties and how they relate to the robot’s environment (Subsection 4.4.3).

4.4.1 Manipulation Pipeline

The considered task consists of moving various household objects from a table to a collection box using a dual-armed robot. Each arm is equipped with a wrist-mounted RGB-D camera and a gripper, one arm having a standard rigid gripper and one using a custom soft gripper (ref Figure 4.3). We consider three key object properties which are relevant for enabling safe and efficient manipulation: delicateness (requiring soft gripper use), spillability (requiring maintenance of upright orientation), and non-droppability (requiring careful placing). These were selected to affect all three phases of a manipulation task: pick-up, movement, and drop-off.

We use the following pipeline to address this task: The robot first records an image of the entire scene, from which all objects are detected and segmented using SAM2 [115]. The VLM then determines which handling constraints from the set of three possible constraints apply for each object. If the uncertainty of the prediction is high, the robot can take two additional pictures: one directly from above and one at an angle. They are particularly useful for objects like containers where contents may only be visible from certain angles. Finally, the objects are manipulated based on the determined constraints.

4.4.2 VLM Constraint Selection with Uncertainty Calibration

We use a VLM for constraint selection because it can reason about constraints for novel objects by leveraging its general knowledge of object properties. Since pre-trained VLMs can be overconfident, we calibrate the uncertainties using our knowledge of the task. For this calibration, we collected a dataset of RGB images of scenes with different household objects using the same robot setup as in our experiments. Each object was manually labeled with its applicable constraints. CRC allows us to use this calibration dataset to determine a decision threshold

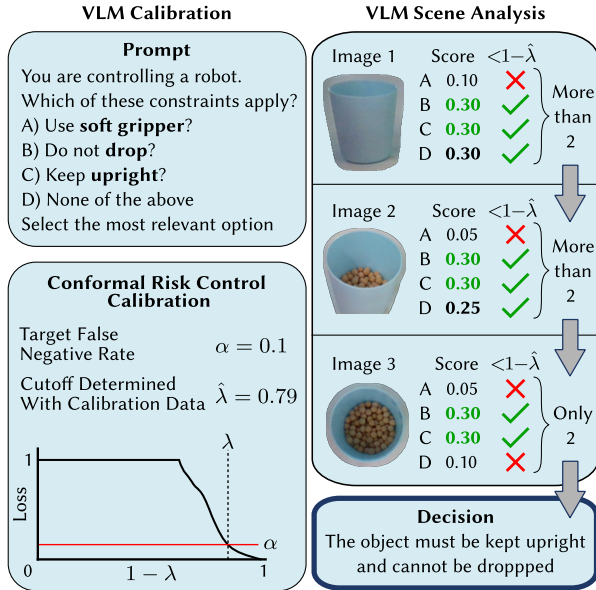


Figure 4.2: For a fixed prompt, the Conformal Risk Control Calibration Process uses calibration dataset to determine the $1 - \lambda$ score cutoff to ensure the desired error rate. During inference the cutoff is used to determine which of the options the VLM has selected. Additional viewpoints are collected until either only 2 options remain or 3 images have been taken. At that point all selected manipulation constraints determine the executed action.

that ensures a fixed false-negative rate for predicted constraints. For this work, a false-negative rate limit of 10% is used.

The three constraints are presented to the VLM as a letter-based multiple-choice question along with a fourth option, in case none of the constraints apply. We explicitly instruct the VLM to select the most likely option rather than all applicable ones in order to enforce a single token reply and avoid having to aggregate the probability distributions of a sequence of tokens. Then, to extract the probabilities, we use the relative probabilities assigned to the tokens for each of the answers in this one token reply. If the predicted probability of a given property is above a threshold, this constraint is selected and applied during the manipulation. Rather than optimizing solely for accuracy, we focus on controlling the false negative rate, as missing a critical constraint (e.g., not identifying a

delicate object) is significantly more consequential than a false positive.

This approach differs from traditional split-conformal prediction, which is designed for single-label classification. In our setting, where we need to identify all applicable constraints simultaneously, we can employ CRC to convert the VLM’s uncalibrated predictions into rigorous statistical guarantees, as visualized in Figure 4.2.

Given the input space $\mathcal{X}$ representing the images of the object available, the label space $\mathcal{Y}$ representing the active constraints, and a VLM that outputs scores $s_i \in [0, 1]$ for each constraint i , our goal is to find a threshold λ such that the predictor of selected constraints $\mathcal{C}(x) = \{i : s_i \geq 1 - \lambda\}$ has a false negative rate below $\alpha = 0.1$ on the test dataset.

Formally, CRC allows us to find λ such that for a new test point X_{n+1}, Y_{n+1} ,

$$\mathbb{E}[\ell(\mathcal{C}(X_{n+1}), Y_{n+1})] \leq \alpha$$

where ℓ is the false negative rate or more generally any loss function that does not decrease as λ increases. The optimal threshold λ^* is found by solving

$$\lambda^* = \inf\{\lambda : L(\lambda) \leq \hat{\alpha}\}$$

where L is average loss on the calibration dataset and $\hat{\alpha}$ is α adjusted for the number of data points used in calibration [4].

For the VLM’s scores s_i , we follow Ren et al. [116] and use the model’s output probabilities as a direct measure of prediction confidence. The standard CRC statistical guarantee holds for a single prediction. However, our system can gather a variable number of views per object, creating a sequence of predictions for each object. We stop collecting new images once one constraint option is eliminated (confidence drops below threshold) or two additional images have been collected. This multi-view approach creates a challenge, as we need to guarantee a false negative rate below 10% for the entire sequence rather than for individual predictions. Additionally different thresholds λ cause the stopping rule to terminate the sequence at different points, which changes the images the VLM uses for its predictions. We therefore modify the standard CRC approach to handle sequences that terminate when a stopping rule is met. We integrate the rule into the calibration, by modifying $\mathcal{C}$ to return constraint sets based on the stopping rule and all possible images rather than just properties selected by a single prediction. This means that $\mathcal{C}$ uses the smallest set of images that provides confident predictions: the first image when that is sufficient, the first two if possible, and only when necessary all three.

This approach is only viable when the resulting loss function ℓ stays non-decreasing with respect to λ . However, VLM predictions on later images are not

guaranteed to be more accurate than earlier ones. This can create cases where a larger λ yields a lower false negative rate than a smaller λ — for instance, when the first prediction is uncertain but the second image yields a confidently wrong prediction. To address the lack of monotonicity, we follow Ren et al. [116] and use the minimum score for each option across all predictions made up to the current point: $s_i^{\min}(t) = \min_{k=1}^t s_i^{(k)}$, where $s_i^{(k)}$ is the score for constraint i from the k -th image. This ensures that as more images are collected, our loss increases monotonically with respect to λ . Unlike Ren et al. [116], our calibration includes the stopping rule, allowing more aggressive early termination by utilizing intermediate states rather than always calibrating with the final values.

4.4.3 Improving VLM Alignment with Robot Capabilities

Simply relying on generic VLM prompts leads to suboptimal performance, as the VLM’s interpretation of the properties may not match the context in which it is applied. To address this alignment challenge, we take inspiration from recent LLM literature [70] and explore two distinct approaches to improve the prompts: Iterative Prompting for Self-Consistency and Prompt Optimization. After determining the best prompts, we perform another step to further improve the performance, which is Post-Tuning of the VLM Confidence Scores. Each of these three methods aims to refine the VLM’s output to better reflect the robot’s operational context and improve the reliability of property predictions for manipulation tasks.

Iterative Prompting for Self-Consistency

To remove the limitation of the VLM producing only a single output token, we investigate the use of an iterative prompt approach [70]. After first obtaining a prediction from the VLM, we prompt the VLM with its last prediction and the original prompt and ask it to check if it still agrees with its previous prediction. This setup no longer requires us to restrict the output to a single token, and we ask it to list all the options which apply. Laban et al. [73] have shown that asking the VLM “Are you sure?” directly decreases performance. Instead, following Krishna et al. [70], we ask it if it still agrees with its prior answer upon further reflection. This process is repeated for the same prompt and image until either the VLM agrees with itself or a maximum of 10 iterations is reached. For each constraint option, the score is the fraction of iterations in which that option was included in the VLM’s answer. We further take advantage of the additional information on the VLM’s reasoning process this approach provides, and utilize more information to make better calibrated predictions rather than simply using the last probabilities as our scores.

Prompt Optimization

To develop better aligned prompts than the human-created ones, we apply automated prompt optimization. We use an iterative sampling approach [70] to incrementally improve the prompt. A large language model, in our case LLama-3.3-70B [44], receives the original user prompt as input and generates ten similar prompts. The new prompts’ conformal prediction performance is then evaluated using our calibration dataset. For the next iteration, the current best five prompts along with the results from the last iteration are provided as additional input. To allow the LLM to track its plans and insights gained, a scratchpad is provided. We perform 10 iterations and at the end select the prompt with the highest overall score.

Post-Tuning VLM Confidence Scores

In practice VLMs’ often have a systematic misunderstanding of certain options. To avoid having to fine-tune the entire VLM, we note that the VLMs often have consistent biases in their rate of choosing certain options. These biases can be addressed by training a small model which takes in original VLM output scores and uses them to predict new adjusted scores for each of the constraints. Since the standard CRC approach includes multiple non-differentiable steps, it has to be adapted to be differentiable and thus directly optimizable. Stutz et al. [132] developed a differentiable version of Conformal Split-Predictions which allows training neural networks to directly minimize average set sizes. Since their method uses Conformal Split-Predictions while we use Conformal Risk Control, we cannot use their method directly, but we can transfer some of the general ideas. For training data, we use the calibration dataset and split each batch into two parts to ensure that the exchangeability assumption is not violated [132]. One half is used to estimate the value of the threshold λ , while the other half is used to calculate the loss. To make a soft version of calibration for CRC, we developed a differentiable version of the false negative rate and the root-finding step which determines the optimal lambda value. The smooth false negative rate is calculated using the ground truth correct labels y and soft-predictions $\hat{y}$, as $\frac{(1-y) \cdot \hat{y}}{|y|}$. We can then use a differentiable root-finder to make the calibration step differentiable. In the soft-assignment step from Stutz et al. [132], which uses a sigmoid to determine the assignments, a very small temperature value (0.05) is required for the model to converge.

The model we use to predict the adjusted scores is a two-layer MLP with a hidden dimension of size four and a ReLU non-linearity. It is trained using the Classification loss from [132] with an SGD optimizer and a learning rate of 0.01 with Cosine Annealing for 200 epochs. For effective learning, the MLP is

initialized to maintain the approximate scale of the inputs by setting the weights to an identity matrix. Since it is not feasible to integrate the stopping rule for the sequential image acquisition into the optimization approach, we treat each individual image of our training data as an independent data point.

Further, when applying Post-Tuning together with iterative prompting, we can incorporate additional features such as the number of iterations required to reach consistency and the average probabilities for each options across all iterations as inputs to the model. These additional signals help the calibration process by providing richer information about the VLM’s certainty.

4.5 Experiments

We first investigate the benefit of our conformal calibration approach for controlling the false negative rate using data collected from our real-world setup. We then measure the impact of our prompt improvement and post-tuning strategies on the alignment of the VLM output with the robot capabilities. Finally, we demonstrate our system’s effectiveness in real-world safe object manipulation.

4.5.1 Experimental Setup

Robot Setup Our system as shown in Figure 4.3 consists of two Kinova Gen 3 robotic arms, each equipped with a Robotiq 2F-140 gripper. One gripper is augmented with two custom 3D printed soft fingers in a fin-ray inspired design for delicate object manipulation. We implemented three robot skills: grasp object, drop object and place object. These use heuristics based on the point clouds from the wrist cameras to determine how to grasp an object and where to place it in the box. All skills can be executed either with the soft gripper or with the standard rigid gripper. While the soft gripper is required to lift up delicate objects without squishing them, it is not capable of lifting up heavy objects due to the limited pinch force. For path planning, we use the ROS MoveIt implementation of the RRTConnect planner [71]. To handle spillable objects, we add an orientation constraint to the path planner to keep the object upright while it is being moved from the table to the box. While this prevents spilling, it increases the path lengths, leading to longer execution times. For perception, each arm is equipped with an Intel RealSense D415 camera to provide high-quality RGB and depth images.

Dataset We collected a data set of 22 household objects covering a range of manipulation challenges:

- 6 delicate objects (e.g., banana, ping pong ball, chocolate Santa Claus)
- 3 containers (evaluated both empty and filled)

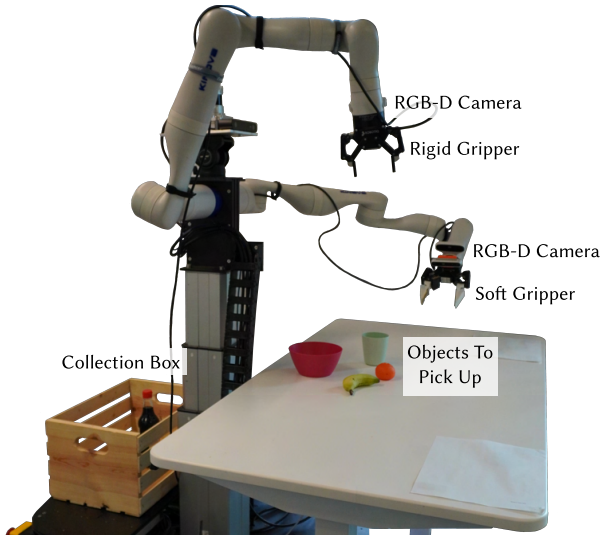


Figure 4.3: Overview of Robot Setup. The right robotic arm has a rigid gripper for non-delicate objects, while the left robotic arm has a soft gripper to manipulate delicate objects. In tandem, the two arms can move all the objects from the table to the collection box.

- 8 non-droppable objects (3 due to weight constraints)
- 4 robust objects with no special handling requirements (two being fake fruit matching delicate objects)

Each object in an image was annotated with a binary label for all of the handling constraints. On average, each object requires 1.06 handling constraints, with some objects having multiple constraints and others none. This distribution demonstrates the importance of property-aware manipulation, since missing a relevant constraint can lead to failure or damage, while selecting wrong constraints leads to unnecessarily slow and careful manipulation. The dataset was split into 80% training and 20% testing data. All reported numbers are averaged over 100 random shuffles of the data, with post-tuning training and calibration repeated for each shuffle.

Metrics We evaluate our system using three key metrics:

- **False Positive Rate (FP):** Rate of constraints incorrectly selected

- **False Negative Rate (FN):** Rate of missed necessary constraints (target: $\leq 10\%$)
- **Accuracy:** Ratio of correctly predicted constraints to total predictions

4.5.2 VLM Property Prediction Performance

We first assess the impact of the basic building blocks of our method: prompt format and conformal risk control, before investigating the impact of our alignment techniques.

Selection of Human-Designed Prompt Given that the prompt used can have a strong impact on the outputs of the VLM, we conducted a user study to collect various prompting approaches. This allows us to evaluate the performance of our methods over a large variety of prompts and acts as a baseline. Participants were instructed to create prompts that would help a VLM identify the manipulation-relevant properties of household objects for a robot. They were given a description of the task, sample images from objects in our dataset, and a list of constraints to consider. Thus, the main source of variance was the individual wording of the constraints and task. All seven participants were familiar with robotics and VLMs and wrote between one and three prompts, resulting in a total of 15 prompts. The performance of these prompts varied wildly, with the best one having a False Positive Rate of 0.60 ± 0.03 , while the median FP was 0.79 and the worst prompt had an FP of 1.0. A high FP means that too many constraints were selected, i.e. the robot has to be unnecessarily slow and careful.

Model Selection For our experiments, we used the open source VLM InternVL2-8B [16, 17], as we found it both achieved a good performance and its small size allowed it to be executed on the robot locally. To validate that the identified issues are common to all modern VLMs, we also evaluated the best prompt on InternVL2.5-8B [15] and Qwen2.5VL-8B [8], two models in the same size class, and GPT-4o [105], a current state of the art model. All of them perform worse than InternVL2-8B with false positive rates of 0.63 ± 0.03 , 0.98 ± 0.01 , and 0.96 ± 0.02 respectively. The difference in performance is surprisingly large and indicates that different models interpret the same terms and prompts significantly differently. It also shows that the problem of aligning with the real-world does not disappear with the general improvement in capabilities of larger or newer models.

Uncertainty Calibration To evaluate the benefit of our implementation of the uncertainty calibration we compare it to two baselines. The first approach is to directly ask the VLM to list all options which apply, without performing any

calibration. As seen in Table 4.1 under "Direct List", this does not allow us to control the false negative rate and therefore a lot of relevant manipulation constraints are missed. The bad performance of this baseline clearly shows that it is important to adapt the predictions of a generic VLM to the robot and task at hand using some domain-specific calibration data. Similarly to our approach, the second baseline uses such a calibration dataset to allow for calibration of the uncertainties. However, it uses a different approach to obtain the probabilities for each of the constraint options.

In the baseline ("Ask Individually"), the VLM is asked individually for each constraint whether it applies to avoid potential cross-coupling, whereas in our approach the VLM is asked to select the single best option. To implement the baseline, we extracted the individual constraint descriptions from the human prompts, and together with the original prompts pre- and postamble queried the VLM if this option applies. For each answer, we used the probability of the yes token as the score for our conformal prediction procedure. While this approach allows us to control the false negative rate using conformal calibration, doing so performs worse than our approach.

As shown in Table 4.1, our method achieves a better balance of minimizing FP while maintaining an acceptable FN rate. The Direct List approach fails to control the false negative rate, while the Ask Individually baseline leads to worse performance as the VLM has even less context to understand each individual option. Furthermore, we evaluated whether it would be enough to only use the initial scene image, instead of recording additional close-up images of each object. The results in Table 4.1 show that using only the scene image for constraint determination results in lower performance, with higher FP and lower accuracy. This shows that the additional images provide additional information to the VLM and are an important part of the pipeline.

VLM Alignment Approaches Post-tuning of VLM confidence scores provides the simplest approach to improve human-made prompts, reducing false positives (FP) by 16% through average constraint number optimization (Table 4.2). This demonstrates effective mitigation of systematic biases in the VLM’s predictions without significantly increasing computational costs. Building on this improvement, automated prompt optimization further reduces FP by an additional 17%, achieving a total reduction of 30%. The optimized prompt’s use of general language rather than detailed physical reasoning indicates that aligning with the VLM’s expectations is more effective. Finally, combining Self-Consistency Sampling with Post-Tuning achieves similar improvements to Prompt Optimization, but with a different computational trade-off. While Prompt Optimization re-

Method	FP ($\downarrow$)	FN ($=$)	Accuracy ($\uparrow$)
Direct List	0.83	0.781	0.57
Ask Individually (Conformal Risk)	0.83	0.095	0.43
Ours	0.79	0.095	0.45
Ours (Only Scene Image)	0.89	0.097	0.39

Table 4.1: Evaluation of uncertainty calibration. Our method achieves a better balance between maintaining a low false negative rate (FN) while minimizing the false positive rate (FP). The direct listing approach fails to control the FN, while individual evaluation leads to a higher FP. Using only the scene image reduces performance.

quires higher calibration time as it needs to evaluate 100 different prompts, Self-Consistency Sampling increases inference time by 4-16x due to its requirement for longer responses and up to ten iterations per question.

Method	FP ($\downarrow$)	FN ($=$)	Accuracy ($\uparrow$)
Human Baseline Prompts	0.79 ± 0.11	0.095 ± 0.009	0.45 ± 0.08
Post-Tuning (PT)	0.66 ± 0.13	0.090 ± 0.008	0.54 ± 0.09
Self-Consistency w. PT	0.59 ± 0.07	0.098 ± 0.024	0.59 ± 0.08
Prompt Optim. w. PT	0.55 ± 0.06	0.098 ± 0.021	0.61 ± 0.04

Table 4.2: Comparison of VLM alignment methods. All methods maintain a False Negative Rate (FN) below 10% while reducing False Positive Rate (FP). Post-tuning (PT) provides significant improvement, with prompt optimization and self-consistency sampling offering complementary benefits.

4.5.3 Real-world Manipulation Results

We evaluated our system in ten real-world manipulation scenarios, each involving transferring five objects with varying properties from a table to a box.

Baselines We compare our method against four baselines. The first baseline, MaxSpeed, uses the fastest available manipulation strategy for all objects. The second baseline, MaxCaution, employs the strategy, which has the lowest chance of causing damage. This means assuming that all objects are spillable containers and using the soft-gripper. The third baseline, Oracle, utilizes ground truth object properties to select the optimal strategy. The final baseline is to

directly query the PG-InstructBLIP model [33], a VLM finetuned to be physically grounded. While this approach does not use conformal prediction or our alignment approaches, the VLM benefits from being trained on a large dataset of properties including fragility, deformability, and the presence of content. For a fair comparison, we evaluated its performance on our calibration dataset using each of the prompts and selected the one that performed the best for the real world experiments.

Results Table 4.3 shows the results of the real-world experiments. Our method achieves a better balance between speed and safety compared to all the baselines. While both the MaxSpeed and PG-InstructBLIP baselines achieve comparable accuracies, they both have significantly higher false negative rates resulting in lower success rates.

Method	Time [s] (↓)	Succ. (↑)	FP (↓)	FN (↓)	Acc. (↑)
Ours	186	0.96	0.45	0.02	0.67
MaxSpeed	95	0.38	0.00	0.32	0.68
MaxCaution	185	0.60	1.00	0.00	0.32
Oracle	164	1.00	0.00	0.00	1.00
PG-InstructBLIP	201	0.46	0.17	0.75	0.65

Table 4.3: Real-world manipulation results across ten test scenarios with five objects each. Our method is able to maintain the required low false negative rate at the tradeoff of a higher false positive rate, allowing it to achieve a high success rate (Succ.). Time is reported as the average time for one successful object manipulation. Despite doing extra exploration steps, our method is not significantly slower than the ground-truth Oracle strategy.

4.6 Conclusion and Future work

In this work, we show the importance of considering uncertainties for real-world manipulation tasks to ensure safe operation. To this end, we introduce a calibration method based on conformal risk control to accurately estimate the certainty of the Vision-Language Model predictions. Furthermore, we highlight the importance of aligning the VLM’s understanding with the real-world task setup instead of directly relying on out-of-the-box VLM predictions. We propose strategies for prompt optimization and tuning of confidence scores that significantly improve performance. As a future direction, it would be interesting to investigate how

the morphology and basic capabilities of the robot could be added to the VLM prompt, such that it could potentially reason about skills and manipulation properties outside of the training set.

Learning Object-Object Affordances

5.1 Introduction

In the previous chapters, we focused on learning affordances for tasks involving interactions between a robot and a single object. As a next step, we investigate affordance learning for intricate tasks that require object-object interaction, for example hanging a coat hanger on a hook or stacking objects on top of each other, as visualized in Figure 5.1. Successfully performing these tasks demands a detailed understanding of the geometry of both objects, as well as the underlying physical principles governing their interaction. Affordances can serve as a powerful tool to represent the interaction information between two objects. This can be seen as a variation of the robot-object affordances discussed in previous chapters.

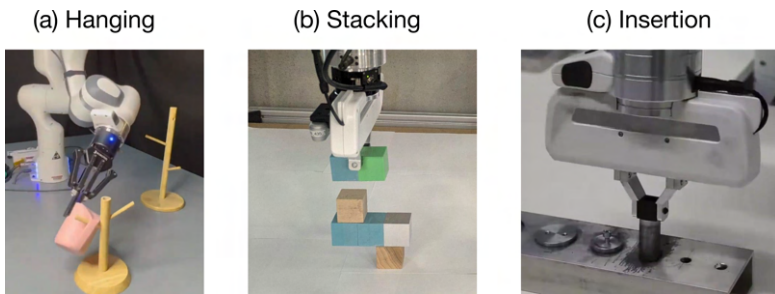


Figure 5.1: Examples of tasks requiring precise interaction between two objects. Image sources: (a) *RPDiff* [129] (b) Ours (c) *A Framework for Robot Manipulation* [58]

The question of how to effectively learn object-object affordances is still an active research topic. Current methods, however, often fall short of capturing a deeper understanding of the task and its underlying physics. Rather than generalizing across different tasks and objects, they typically focus on a single object category, such as hanging mugs of varying shapes on a rack [129]. For objects other than mugs that are not represented in the training data, this pipeline would fail.

Such specific, locally optimized approaches mean that for every new task or even object type a new pipeline has to be designed and trained, which is time consuming and tedious. Instead, we believe that the goal should be to find a method that has a higher-level understanding of physical interactions and object geometries. Extracting local geometry features and determining which of them are task-relevant is essential for generalizing to arbitrary objects.

Moreover, we demonstrate that for certain tasks, relying solely on an object’s appearance is insufficient, as physical properties such as friction and weight distribution also play an important role. Extracting such non-visual properties requires the robot to physically interact with the object, which underscores the importance of incorporating the robot into affordance learning.

In this chapter, we analyze the current state of the art in learning object-object affordances and propose potential directions toward developing a more generalist approach. In addition, we investigate the task of stacking objects with non-uniform density to highlight the importance of accounting for physical object properties.

5.2 Approaches for Object-Object Affordance Learning

Most state-of-the-art methods for tasks involving interactions between two objects can be divided into two main components. First, the shape of each object is captured through a geometric encoding in order to extract the task-relevant features. Second, the 6-Degree-of-Freedom (6-DoF) pose required to position the movable object correctly relative to the static one is predicted. In most cases, a conventional controller is then used to move the robot and place the object in the predicted pose. More recently, several end-to-end approaches have emerged that bypass these explicit intermediate steps and instead predict control outputs directly from perception input. This chapter reviews related work on object-object affordance learning, focusing on the two main aspects of geometry encoding and pose prediction.

5.2.1 Extracting Task-relevant Geometric Features

This section provides an overview of methods for encoding and extracting task-relevant geometric features from the two objects involved in the interaction. Figure 5.2 illustrates the different approaches. Some methods employ heuristic, task-specific rules to explicitly incorporate geometric and physical constraints relevant to the interaction. Others reduce the geometry to a set of semantic keypoints that are meaningful for the task. Finally, certain approaches encode the complete object shape using a point cloud encoder, relying on the downstream pose predictor to identify which geometric regions are relevant.

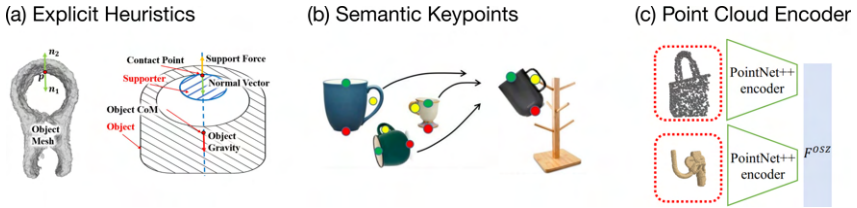


Figure 5.2: Overview of different approaches for encoding task-relevant object features. Image sources: (a) *I Get the Hang of It!* [81], (b) *kPAM* [87], (c) *Omnihang* [165].

Explicit Heuristics Some affordances depend on specific local geometries, for instance, flat surfaces for stacking, holes for insertion, or hooks for hanging. One approach is to identify such shape categories and segment the object into corresponding parts. However, this is challenging since defining suitable heuristics for segmenting and classifying geometric primitives is non-trivial, as acceptable deviations from ideal shapes vary by task and object. For example, a stacking surface may range from perfectly flat to slightly curved, depending on the objects involved. Moreover, considering only contact regions is insufficient, as the full object geometry must be taken into account to prevent unwanted intersections or collisions. Finally, manually defined heuristics are needed for each new task, limiting scalability to novel scenarios.

Recent work by Li et al. [81] proposed a learning-free method to infer stable relative poses for hanging objects. It uses heuristics based on surface normals and the direction of support forces to identify contact points and match them between objects, achieving high success rates for unseen objects in both simulation and real-world tests. However, since it relies on multiple task-specific heuristics, extending it to other interactions would require developing new rule sets.

Semantic Keypoints Semantic keypoints provide a powerful and efficient way to reduce complex object geometries to their task-relevant parts. The kPAM framework [87] introduced the use of semantic 3D keypoints for category-level manipulation, enabling simple task specification and robustness to shape variations in irrelevant regions. For hanging, for example, a keypoint on a mug would be in the center of the handle. This representation simplifies the task into the two steps of detecting the keypoints and then aligning the two keypoints on the interacting objects. However, focusing solely on keypoint alignment omits dense geometric information important for collision avoidance. kPAM-SC [35] addressed this by combining semantic keypoints with dense geometry, while kPAM 2.0 [36] extended the approach to closed-loop manipulation by including orientation and force/torque targets. A limitation of these methods is that they operate at the category level and require manually labeled keypoints.

In contrast, GIFT [136] uses a physics engine to simulate object interactions such as hammering and hooking, identifying resulting contact points as keypoints. This removes the need for manual labeling and incorporates the robot into the data collection loop. However, they focus only on predicting keypoints on individual tools and not on aligning them to a target object. SKT-Hang [72] addresses the complete hanging task by predicting both the target pose and the trajectory to reach it. Using simulated successful executions, it first learns to predict likely contact points on the support object and then generates a trajectory by deforming a template. While it generalizes to unseen support items, it still assumes known hanger objects.

Overall, semantic keypoints are a compact and effective representation for object-object interactions, but for more intricate tasks, they lack detailed shape information and are best complemented with dense geometry representations.

Learned Point Cloud Encoders While keypoints provide a compact representation of task-relevant object parts, they cannot capture the full object geometry, which may be important for interaction. The 3D geometry of an object is typically extracted using point cloud encoders, as point clouds can be directly obtained from sensors such as RGB-D cameras or LiDARs. Common encoders include PointNet++ [112] and DGCNN [144], while more recent transformer-based architectures [151, 157, 158] are better at capturing global dependencies and therefore improve benchmark performance.

However, transformers have typically more parameters and require greater computational resources, data, and training time. To mitigate this, pretrained encoders can be used as a starting point for finetuning. Geometric pretraining tasks such as part segmentation, classification, or autoencoding are often used

since they are easier to label than the downstream task. The key challenge is designing a pretraining task that enables the extraction of meaningful geometric features containing the information relevant for a variety of downstream tasks. In practice however, most related works for affordance learning train task-specific point cloud encoders from scratch, as this tends to achieve better performance on the individual task [129, 165, 170].

After encoding the objects individually, the next step is to combine their geometric features and capture the relationships between them. The simplest approach is to concatenate the features of the two point clouds and let the second part of the pipeline, the pose predictor, learn which features are relevant, as in Omnihang [165]. For tasks requiring detailed geometric interactions however, architectures that focus on learning the relationship between task-relevant object parts can be beneficial. Recent works have applied cross-attention mechanisms [129, 170], which learn attention weights reflecting the correlation between features of the two objects, thereby emphasizing the most relevant regions. Another approach, O2O-Afford [94], employs object-kernel point convolutions, sliding one object over the scene point cloud to produce a feature map encoding inter-object relations.

Overall, explicitly modeling the interrelation between object geometries allows the network to identify features critical for the task, which can improve performance for complex interactions.

5.2.2 Pose prediction

After encoding the object geometries, the next step is to predict the relative transform $T \in SE(3)$ between the two interacting objects. The predicted pose must ensure both proper alignment of the contact surfaces and physical stability under gravity. For example, in a hanging task, the object’s center of mass must lie vertically below the contact point to remain stable. Obtaining accurate pose predictions requires careful design of the prediction network. This chapter provides an overview of different approaches used for pose prediction, including MLP-based models, Reinforcement Learning, Imitation Learning, Vision-Language Action Models and Diffusion Models.

MLP Models A straight-forward approach to predicting the target pose is to use the combined geometry encoding and pass it through an Multi-Layer Perceptron (MLP). O2O-Afford [94] was among the first works to propose affordance learning for object-object interactions such as placing, fitting, and stacking. It uses a PointNet++ encoder [112] to encode both objects and applies convolution to obtain correlated features. Those features are passed through an MLP that predicts a point-wise affordance score indicating the likelihood of successful

interaction. The model is supervised using examples generated through physics simulation. The method performs well in their constrained setup, where object orientation is fixed. However, it mainly captures coarse spatial relations, focusing on object centers and sizes rather than detailed geometry. Consequently, it is not well suited for tasks requiring fine-grained 3D understanding, such as hanging.

OmniHang [165] specifically addresses hanging arbitrary objects on hooks, a task requiring precise pose prediction. The method first encodes both object point clouds using PointNet++ [112], concatenates the features, and uses an MLP to predict an initial pose. This is followed by predicting potential contact points on both objects and matching them to refine the pose. Finally, reinforcement learning is applied for local correction of residual errors. The network is supervised with ground-truth data from physics simulations, which include multiple valid hanging poses per object pair. The loss is computed as the distance between the predicted pose and the closest valid ground-truth pose, which may penalize feasible poses if the dataset lacks coverage. OmniHang achieves a 68% mean success rate, highlighting both the challenge of generalizing across object categories and the value of multi-stage pipelines for complex pose prediction tasks.

Both approaches demonstrate that, although using an MLP to predict the final pose is simple and effective, achieving a high success rate for complex tasks with a single-shot prediction remains challenging.

Reinforcement Learning Instead of relying on pre-collected datasets, a promising direction is to let robots learn affordances through direct interaction within simulated environments. Following Khetarpal et al. [62] and Yang et al. [162], affordances can be framed within a reinforcement learning (RL) paradigm as a Markov decision process (MDP). In this view, affordances represent the mapping between actions, their effects, and the states in which these effects are achievable. Khazatsky et al. [61] adopt this formulation to enable robots to finetune skills for new objects and environments.

While this approach focuses on single-object interactions, Xu et al. [159] use affordance to learn how to use a tool to manipulate other objects. They extend the concept of affordances to longer time horizons by predicting future possible actions, enabling the robot to plan sequences of actions toward long-horizon goals. Their model-based RL framework predicts future action affordances and learns an environment model via trial-and-error in simulation, enabling long-horizon reasoning and transfer to new tasks.

While the previous method uses affordances to select one of the parameterized motion primitives, Geng et al. [38] introduce an end-to-end reinforcement learning framework that directly learns affordance representations for robotic manipulation. Their approach trains an affordance module to predict contact maps of interest using contact information gathered during RL training. Al-

though the results are promising, the approach remains limited in performance and scalability, as RL on 3D point clouds is computationally demanding due to the high-frequency control requirements.

Imitation Learning When only a few demonstrations are available, imitation learning offers an effective way to learn a specific task. Simeonov et al. [128] demonstrate imitation learning for tasks involving spatial relations between two objects, for example hanging a cup on a hook, from only ten demonstrations. They represent objects using consistent local coordinate frames on task-relevant parts. To predict the location of this task-specific coordinate frame on an unseen object, they encode the geometry with pretrained, self-supervised Neural Descriptor Fields (NDFs) and need only a single annotated 3D keypoint as label. Next, an energy-based scheme predicts the target pose between keypoints. While effective within a category, NDFs must be retrained for new object types. To improve generalization, Chun et al. [19] propose Local Neural Descriptor Fields, which represent objects as voxel-based latent descriptors rather than modeling the entire object with a single network. This enables skill transfer to novel yet similar shapes, such as from a mug to a handled bottle.

ActionFlow [31] is another approach for learning object-object interaction tasks from demonstrations, producing both target poses and action sequences to move objects to the target. It leverages an SE(3)-invariant Transformer for spatial reasoning and employs Flow Matching, a generative model akin to diffusion models, for fast, high-quality action generation. The efficient inference enables learning robot policies from few demonstrations, successfully tackling tasks such as hanging or mounting light bulbs with closed-loop control.

In general, imitation learning has the advantage of being very data efficient, but this comes at the cost of limited generalization capabilities.

Vision-Language Action Models Recently, Vision-Language Action (VLA) models have emerged as powerful tools for learning generalist robotic manipulation policies from large-scale, diverse datasets [154]. OpenVLA [65], pretrained on 970k robot episodes spanning multiple embodiments and tasks [22], integrates visual and language backbones to enable multi-robot control and efficient adaptation to the local setup via parameter-efficient fine-tuning. Octo [104] employs a transformer-based architecture to process natural language instructions, goal images, and multi-modal actions through diffusion decoding. Discrete Policy [149] enhances multi-task learning by encoding action sequences into a discrete latent space for task-specific skill representation. While these models excel at tasks requiring semantic scene understanding, they remain limited in predicting 6-DoF poses for tasks demanding precise interactions between multiple objects, where accurate 3D shape reasoning is critical.

Diffusion-based Models Diffusion models have recently emerged as a promising approach for predicting precise 6-DoF poses by breaking down the task into incremental steps. Training involves gradually adding Gaussian noise to ground-truth poses, with the network learning to reverse this process by predicting incremental pose changes. Such diffusion-based methods have been applied to diverse manipulation tasks [18, 166], with DP3 [166] achieving high success rates from only ten demonstrations per task. While effective across variations in position, appearance, and viewpoint, these methods are limited to single objects and do not generalize to unseen categories or rely on fine geometric details.

RPDiff [129] extends diffusion-based methods to object-object affordances, for tasks such as hanging and stacking. Trained from demonstrations, it iteratively refines poses, with its de-noising process naturally producing multi-modal outputs as different initial poses converge to distinct final configurations. However, its generalization is limited to objects within the same category (e.g., mugs), indicating that the method overfits to object category rather than capturing the underlying task-relevant geometry.

AnyPlace [170] explicitly targets multi-modality, aiming to predict a wide range of feasible placement poses by combining diffusion-based predictions with high-level contextual reasoning from VLMs. The VLM first identifies a coarse placement location in a 2D image, and the corresponding 3D point cloud region is then cropped and processed by a low-level placement module. This two-stage approach reduces the complexity of the low-level task by focusing on relevant object geometry, enabling efficient multi-modal placement. However, the method requires large models and long training times, namely five days for multitask training.

5.2.3 Exploring Semantic Priors for Diffusion-based Pose Prediction

Motivation To improve the reliability and efficiency of object rearrangement, we seek ways to better guide diffusion-based pose generation toward successful outcomes. Inspired by Simeonov et al. [129] and Zhao et al. [170], we aim to investigate whether incorporating additional semantic information from large pretrained networks can improve success rates in rearrangement tasks. Specifically, we explore the idea that a semantic network could provide a coarse estimate of an object’s initial position, thereby initializing the diffusion model closer to the target pose. The intuition is that such networks can provide useful high-level cues about how an object should be roughly positioned for tasks such as hanging, even though their 2D image-based training limits their understanding of full 3D geometric interactions and precise 6-DoF poses. Diffusion models, by contrast, excel at refining poses and adjusting objects into their correct configurations. Motivated by this complementarity, we investigate whether a hybrid approach that combines semantic estimation with diffusion-based refinement can improve

task performance.

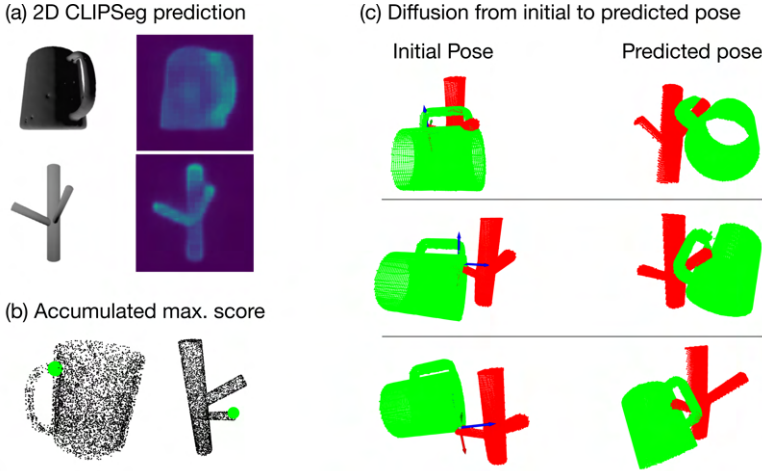


Figure 5.3: Experiment on the *hanging* task for integrating semantic priors into diffusion pose predictor. (a) Pixel-wise correspondence predictions from the pre-trained CLIPSeg model for the prompts “Tip of the hanging support” (rack) and “Hanging by handle” (mug). Lighter colors indicate higher correspondence scores. (b) Accumulated correspondence scores from multiple 2D views projected onto the 3D point cloud. The point with the highest aggregated score is highlighted with a green sphere. (c) Predicted *hanging* poses produced by the diffusion process, for various initial pose configurations.

Implementation To evaluate the potential benefit of this hybrid approach, we first need to establish whether diffusion-based pose prediction indeed performs better when initialized closer to the target pose, thereby simplifying the refinement process. To test this hypothesis, we designed an experiment to quantify how the quality of the initial pose affects the overall success rate. Our implementation builds on the diffusion-based model RPDiff [129], focusing on the task of hanging a mug on a rack. For their task ‘Mug/MedRack’, they report a success rate of 83%.

To incorporate the semantic knowledge to the pipeline, we implement the following steps: First, the two 3D objects involved in the task are rendered from different viewpoints and the rendered images are passed to a pretrained language-text model, namely CLIPSeg [86]. This model takes a user-defined text prompt

and an image as input and estimates the level of correspondence of each pixel in the image relative to that prompt. We are using the text prompt “hanging by handle” for the mug and “tip of hanging support” for the rack to obtain predictions for suitable contact points. An example prediction is visualized in Figure 5.3a.

For every image, we project the predicted correspondence values back onto the 3D point cloud, allowing us to aggregate information across multiple views. Then, we select the point with the highest score on the mug and the rack respectively, as seen in Figure 5.3b. These points are designated as contact points, i.e., locations where the two objects are assumed to touch. This constrains their relative position but leaves the orientation unconstrained; to ensure full coverage of the solution space, a random orientation is sampled. Finally, we use the pre-trained weights to apply the diffusion method from [129] without modification, except for initializing the object using our semantically guided pose. Figure 5.3c shows several example initializations and the resulting refined poses predicted by the diffusion model.

Experiments To verify the hypothesis, we evaluate and compare the success rate of three initialization strategies. As a baseline, we use the original implementation of Simeonov et al. [129], which initializes the mug pose by randomly sampling a point on the rack point cloud and aligning the mug’s center with that point using a random orientation. Secondly, following the proposed implementation described above, the initial object position is determined by the contact point predictions from the semantic network. Finally, as an upper bound, we include an oracle condition in which the mug is initialized directly in a valid hanging pose before the diffusion process begins.

Our results, shown in Table 5.1, indicate that the success rate remains largely unchanged across all methods. Even initializing the object in a valid target pose does not improve the success rate compared to the baseline. This suggests that the diffusion-based model already performs sufficient exploration to reliably converge to the target configuration. Moreover, since we used pretrained weights rather than retraining the model from scratch, the discrepancy between the training and testing initialization strategies may have limited the potential benefit of improved initialization. Retraining with consistent initialization would be a valuable direction for future work.

Discussion Overall, our results show that incorporating an initial pose estimate from a semantic network does not improve the performance of the diffusion model in the evaluated setup. Nevertheless, employing a semantic model to predict

Method	Success Rate
Rpdiff [129]	0.84 ± 0.065
Rpdiff w semantic init pose	0.84 ± 0.045
Rpdiff w oracle init pose	0.82 ± 0.043

Table 5.1: Average success rate and standard deviation for the mug-hanging task, averaged over five random seeds, using different initialization methods for the diffusion algorithm. Results indicate that the choice of initial pose has no significant effect on the overall success rate.

the initial pose could still be advantageous in scenarios where different types of placements are possible in the same location. For example, in the hanging task, a mug could be hung either by its handle or by its inner rim. In such cases, a user could specify the desired configuration via the text prompt to the semantic network, which would then generate an initial pose consistent with that intent. To validate this approach, the correlation between the initial pose and the predicted pose would need to be examined more rigorously. Unfortunately, testing such variations was not possible with the dataset from Simeonov et al. [129], as it only includes demonstrations of mugs hung by their handles. Consequently, even when the mug is initialized in a physically valid configuration where it is suspended by its inner rim, the model converges to a solution that hangs the mug by its handle, as seen in the second row of Figure 5.3c. This behavior suggests that the diffusion model does not develop a deep understanding of object geometry or physical interactions but rather learns to replicate the specific configurations present in its training data, thereby limiting its generalization to novel arrangements or object categories.

5.3 Incorporating Physical Properties into Affordance Learning

Most existing manipulation approaches are primarily based on visual appearance. However, complex real-world applications often require robots to think beyond geometry and account for physical properties, such as friction or mass distribution, that directly affect the affordance of an object. Humans naturally possess strong intuition about how objects interact with their environment, but current robotic systems remain far from achieving comparable physical understanding.

In this chapter, we explore the state-of-the-art methods for extracting physical object properties and using them to enhance affordance-based manipulation. Specifically, we focus on the task of stacking objects with non-uniform density, that is, objects composed of materials with different densities, such as a hammer. This task was chosen because knowledge of the center of mass is essential for

stable stacking and can only be inferred through interaction. To this end, we explore the role of physical object properties through a case study on stacking of composed objects.

5.3.1 Related Work

Learning Physical Object Representations Since physical properties cannot be estimated from static scenes, works that aim to learn physical object properties typically rely on interactions with objects using either scripted [139], random [141, 160] or explorative action sequences [29]. Different sensor modalities have been explored for learning, ranging from pure visual observations [141, 160] over pure tactile data [139] to a combination of both [29]. The authors of [29, 130, 139, 160] demonstrated how their learned latent representations can be decoded into explicit physical object properties such as mass, friction, and inertia. Pinto et al. [109] propose to use visual observations of physical interaction to learn an object-centric latent space which can then be used for different downstream task. Similarly, in our experiment, we propose to use a combined latent representation that contains the physical and visual properties of the object, as they are both crucial for the stacking task.

Appearance-based Approaches for Stacking Most existing works that address the task of stacking rely solely on visual input, thereby neglecting physical object features that can only be learned from interaction. Li et al. [80] propose an end-to-end approach that directly predicts the stability of stacked objects from appearance. Ahuja et al. [2] learn a physics intuition model that informs a robot of which objects can be safely extracted, similar to the game Jenga. These works use a latent representation to encode stability of an existing object stack, but do not focus on predicting new stable poses for additional objects. Groth et al. [47] train a visual classifier for binary stability prediction, which they then also use to actively construct stable object stacks. Furrer et al. [32] developed a method for stacking irregular natural stones on top of each other in a tabletop setting by scanning the stones and using a physics engine to find a stable pose. Their work was recently expanded by Johns et al. [59] to perform dry stone construction with an excavator, using a geometric planner to propose suitable placement positions. Lee et al. [76] use reinforcement learning for robot stacking with a vision-based policy using RGB images. Huang et al. [51] use the stacking and de-stacking of objects to perform mechanical search of cylindrical and rectangular objects on shelves. All these works assume simple objects with uniform mass distribution and consistent friction, where purely vision-based sensing provides sufficient information. In contrast, we explore stacking for objects with non-uniform mass distributions, requiring the center of mass to be inferred from interaction.

Stacking using Physical Properties To the best of our knowledge, no work directly addresses the task of stacking objects with non-uniform mass distribution. Liu et al. [85] estimate the center of mass based on RGB-D images and force/torque sensing, but only use this information to adapt the object grasp pose to improve grasp stability. Zhou et al. [173] address the task of stacking irregularly shaped stones and they propose to measure the center of mass with a force sensor after picking up the stone to compensate for grasping imprecision. However, their stacking strategy only considers the center of mass of the object to be placed, and not of the overall tower, which is important when considering objects composed of materials with different density.

5.3.2 Investigating Stacking of Multi-Material Objects

Motivation and Task setup In order to explore the importance of physical object properties, we designed an experiment that consists of stacking multi-material objects on top of each other to create a stable tower. The challenge is that since the objects are composed of multiple materials with different densities, the Center of Mass (CoM) of the objects does not correspond to the geometric center, as shown in Figure 5.4a. An example of such an object in the real world would be a hammer, where the CoM is strongly shifted towards the metal part. For the stacking task, both visual and physical properties are crucial, since a stacked object is stable as long as its CoM lies above the contact surface. While the size of the contact surface can be observed with visual sensors, observing the physical property requires interaction with the object. In the following paragraph, we describe how we first learn a latent representation of the objects and then in a second step use it to predict a stable stacking pose.

Implementation Our pipeline is split into two parts. First, we train an encoder that learns to predict an object embedding based on the observed shape and physical properties of that object. In a second step, we train a stacking policy that predicts where an object should be placed on an existing tower, using the embeddings from the tower objects and the new objects as input.¹

To train the object encoder, the robot collects force-torque observations from interactions with objects of different shape and mass distribution. We then employ a contrastive learning framework to encode the force-torque observations, using the ground-truth center of mass as supervision signal. The full object representation is obtained by combining the learned latent encodings with a parametric representation of the geometry. In this setup, the force-torque readings could

¹The stacking experiment was developed and executed in collaboration with Antonia Hüfner. She was responsible for the learning pipeline of the latent representations while I was working on the stacking pose predictor.

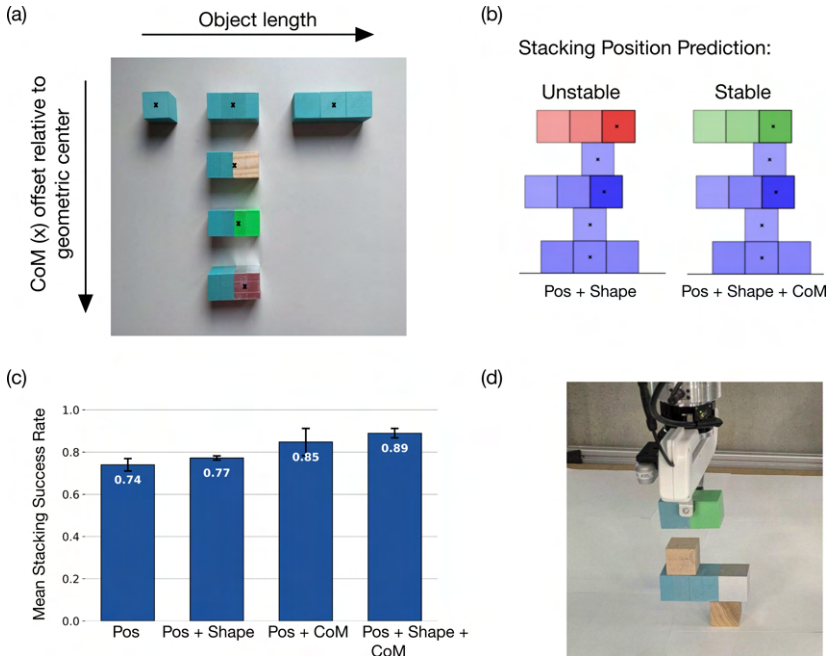


Figure 5.4: Experiment for stacking composed objects. (a) The objects used in the experiment are composed of materials with different density: Blue PLA with low infill 14 g, wood 41 g, green PLA with high infill 71 g and aluminum 170 g. The CoM is marked with an x . (b) The object (red/green) is placed in the predicted position on the given tower (blue). The shade of the cubes correlates with their densities. The right tower considers the CoM and therefore predicts a stable stacking position. (c) Mean success rate for stacking an object on an existing tower in simulation ($n=1000$). Each bar shows the mean and standard deviation over 5 random seeds. (d) Real-world qualitative experiment of stacking a tower using a robot arm.

also be used directly, sidestepping the need to learn a latent representation. However, we consider this work a proof of concept for learning an object-centric and task-independent embedding that integrates information from multiple modalities. This idea is analogous to training a point cloud encoder on an autoencoding task, where the resulting latent space captures the key characteristics of the object and can then be used for different downstream tasks.

The stacking pose predictor has the task of predicting the best pose to place an object on an existing tower based on the observations of the tower and the object. For this proof-of-concept experiment, we simplified the stacking task to a one-dimensional problem, in which the network predicts only the linear offset between the tower center and the cuboid-shaped object to be placed. To create a representation of the tower, which consists already of multiple stacked cuboid objects, for every individual object we concatenate its latent representation, its height index, and its position along the linear axis. This tower encoding and the encoding of the object to be placed are passed together into an MLP that predicts a linear position. To supervise the stacking network, the loss is calculated as the distance between the predicted position and the optimal position, which is determined in simulation to be the position furthest away from any position leading to collapse of the tower. The network is trained on the task of stacking one additional object on a randomly generated, borderline stable tower.

Experiments We evaluated the performance of our trained stacking policy to determine the benefit of our shared latent representation which contains the physical object encoding and the object shape. The objects in our setup consist of one to three connected cubes made from materials with varying masses, ranging from 14 g for 3D-printed plastic to 170 g for aluminum, as visualized in Figure 5.4a. For the baseline policy, we use a latent representation of the tower that includes only the centroid position of each object, without incorporating object shape (i.e., in our case, length) or CoM information. For uniformly dense objects, where the CoM coincides with the geometric center, the network could simply learn a heuristic of placing the new object directly above the tower’s center to ensure stability. However, as visualized in the example in Figure 5.4b, this heuristic fails for objects whose CoM does not align with their geometric center. In such cases, knowledge of the CoM becomes crucial for successful stacking.

Overall, our simulation experiments shown in Figure 5.4c demonstrate that using a combined latent representation, thereby taking CoM and shape into account, leads to a higher success rate in the stacking task. The baseline, which uses only centroid positions, achieves a success rate of 74%, as the simple heuristic of placing objects at the tower center often suffices. Adding shape information yields a modest improvement, while also including CoM information, increases the success rate to 89%.

Alongside the simulation experiments, we also implemented the experiment in a real-world environment, as visualized in Figure 5.4d. We used a 6-DoF robot arm with a wrist-mounted force/torque sensor to obtain the measurements that are required to train the object encoder. We queried the stacking pose predictor

and executed the task by using predefined motion primitives to grasp, move and place the object at the predicted position on the tower. With this setup, we demonstrate a promising direction for performing stacking of composed objects under real-world conditions.

Conclusion Overall, this experiment demonstrates that when stacking objects composed of different materials, stability can be improved by considering the center of mass rather than relying solely on visual appearance. Since mass distribution can only be inferred through physical interaction, this highlights the importance of incorporating the robot into the affordance learning pipeline. Future work could explore adding additional modalities such as friction and extending the stacking task to irregular objects, where surface interactions have a strong influence on stability.

5.4 Conclusion and Outlook on Object-Object Affordances

5.4.1 Conclusion

Solving tasks that require two objects to interact closely with each other is still an open research challenge for multiple reasons. For one, the predicted pose has to be very precise and the success of the task is binary, making it difficult to approximate a solution. Additionally, multiple solutions often exist, for example a clothes hanger could be hung on the hook part or on the inside triangle. Multimodality is difficult to represent in classical neural networks, since they tend to collapse to one of the solutions. Different approaches to solve the multimodality challenge range from adding random noise to the generative architecture to achieve a bigger spread of generated poses [165] to using randomly sampled initial conditions for diffusion models [129]. While these methods can prevent the network to collapse to a single solution, they also do not return an exhaustive list of possible poses.

Another major challenge is designing an efficient supervision scheme for the networks. One approach is to generate a dataset of valid solutions and compute the loss as the distance between the predicted pose and the closest ground-truth pose. Although computationally simple, this method could return a high loss for a valid prediction if the respective modality is not represented in the training dataset. Ideally, each predicted pose would be verified using a physics engine, but this is computationally expensive. Moreover, because most physics engines are non-differentiable, integrating their feedback as a supervision signal is non-trivial. Diffusion-based models offer an alternative: they begin from a ground-truth pose and iteratively add noise, training the network to predict small, incremental corrections toward the target pose. This can lead to more precise pose predictions

than using methods that estimate the final pose in a single step.

One of the key challenges is generalizing to unseen object geometries. Successfully predicting poses for object interaction requires recognizing local geometric features and developing an implicit understanding of physical interactions. Humans can easily handle objects they have never encountered before by abstracting from the overall shape to the relevant functional parts. Neural networks, however, still struggle with this. For complex tasks, they often perform well only within a single object category, and when they do generalize, they require large datasets and significant computational resources. Even then, they continue to struggle with tasks that require precise interaction between objects.

Lastly, most approaches addressing inter-object tasks rely exclusively on object shape or appearance as input. This neglects physical properties such as density or friction, which can strongly influence interaction outcomes. Our stacking experiment with objects of non-uniform density demonstrates that ignoring these physical properties leads to decreased performance. Observing and estimating physical properties requires the robot to interact with the object and measure resulting forces and torques. We therefore advocate for a more holistic approach to robotics that accounts not only for object geometry but also for its other physical characteristics.

5.4.2 Outlook

Overall, although recent works achieve impressive results on tasks involving objects similar to the training data, they still lack generalization capabilities. Ideally, neural networks should, like humans, efficiently extract task-relevant geometric features and combine them with an understanding of physics to enable generalization to unseen objects and novel tasks.

One potential approach to addressing the generalization challenge is to leverage foundation models. Recent advances in large language models and image generation have demonstrated that increasing both dataset size and model capacity can lead to remarkable results. In robotics, however, this is challenging because large datasets of robot-object interactions are not readily available and must be generated at significant cost, either through physics simulations or real-world experiments. A promising direction may therefore be to combine the semantic knowledge of large models trained on 2D images with the understanding of physics and 3D geometry learned from smaller robotic datasets. How best to achieve this remains an open question. Possible strategies could include using a semantic network to pre-filter the scene to identify relevant parts and only pass those on to the robot network, or employing a semantic network to evaluate and

guide predictions made by the robot network during training.

Alternatively, generalization to unseen shapes could be improved by explicitly encoding geometric features and physical laws into the network. For example, in a hanging task, we know that the object’s center of mass must lie vertically below the contact point. Such physics-based rules could be used to bootstrap learning or simplify the task by constraining certain degrees of freedom. However, relying on hand-crafted rules results in a policy tailored to a single task, and the approach does not scale efficiently to other tasks.

Another interesting challenge is multimodality, where the aim is not only to identify a single feasible solution but ideally the set of all possible solutions. One possible approach to predict a set of solutions could be to decompose the pipeline into multiple stages. The first stage would identify possible contact surfaces on each interacting object independently. In a second stage, a pose could then be predicted for every feasible combination of these contact surfaces. This approach may improve exploration and coverage of potential solutions.

Discussion and Outlook

6.1 Discussion

The following section highlights and discusses the key aspects investigated in this thesis. We examine the roles that affordances can play within a robotics pipeline and describe the different methods used to integrate the robot into the affordance learning process, ensuring that affordances are conditioned on the specific robot embodiment. Finally, we discuss how the concept of affordances can be extended to object–object interactions.

Role of Affordances in the Robotic Manipulation Pipeline This thesis demonstrates that affordance learning can effectively be used as a bridge between perception, reasoning, and control within robotic systems. By encoding the potential interactions between a robot and its environment, affordances offer a unified representation that supports both low-level control and high-level planning.

At the low level, we have shown that affordances can function as actionable cues that guide precise control of robotic motion. Point-wise affordance predictions specify where and how the robot should interact with objects, enabling fine-grained manipulation even for previously unseen objects. These local affordance cues are particularly advantageous for dynamic tasks, such as opening and closing articulated objects.

At the high level, affordances provide rich semantic information that characterizes which actions are feasible for specific objects within a scene. Such representations can inform symbolic reasoning and planning, allowing the robot to select and sequence actions to achieve long-horizon goals. Furthermore, we have

shown that by incorporating object affordances into decision-making, the system can ensure safe and context-aware interactions, particularly when handling delicate objects.

Incorporating the Robot into Affordance Learning A central insight of this work is that affordances should not be regarded merely as intrinsic object properties, but rather as relations that must be learned in the context of the robot’s embodiment and capabilities. This perspective ensures that predicted interactions are both physically realizable and contextually meaningful for a given robotic platform. Throughout this thesis, we have explored multiple approaches to learn affordances that guide how robots interact with objects.

One approach involves learning affordances through interaction in simulation. We demonstrated that incorporating the complete robot model into the learning process significantly enhances generalization and robustness. By engaging in simulated interactions, the robot autonomously generated training data that reflected its kinematic and dynamic constraints. This robot-centric formulation enabled the affordance model to predict feasible interaction poses, and even to re-evaluate predictions during execution in order to recover from unforeseen states. As a result, the learned affordances, combined with a sampling-based controller, proved effective for dynamic mobile manipulation tasks, and even allowed generalization to the real-world setup.

Extending this idea, we also explored affordance learning through simulation at the scene level rather than for isolated objects. We proposed an interactive exploration strategy that allows a robot to efficiently navigate in a simulation environment to collect interaction experiences. This data is then used to train a robot-specific object affordance model. Our method employs a reinforcement learning pipeline integrated with an object-level map, which enables the agent to re-identify objects across multiple viewpoints, leading to denser and more consistent annotations. Furthermore, by leveraging information from the object-level map during exploration, the agent improves both interaction success rates and data efficiency, ultimately producing higher-quality training data.

As an alternative to learning affordances through simulation, which is computationally expensive, we further investigated the potential of leveraging large pretrained models. We demonstrated that models trained on extensive web-scale datasets can be effectively adapted to local robotic setups using a small, robot-specific interaction dataset. By calibrating model uncertainty through a conformal risk control mechanism, we aligned the global semantic knowledge of pretrained models with the robot’s embodied physical constraints. This hybrid

approach preserves the semantic richness and generalization capacity of large models, while grounding them in robot-specific experience.

Extending Affordances to Interactions Between Objects While the previous approaches focused on robot–object interactions, we also explored affordance learning for multi-object interactions. Tasks such as hanging, stacking, or inserting objects require a precise understanding of relative object geometries and contact dynamics. Our analysis revealed that many state-of-the-art approaches are limited to highly specific tasks or objects and often fail to generalize across object categories. Future work could investigate how affordances can better capture the underlying principles governing interactions between objects.

Furthermore, we observed that most existing work relies solely on visual information, whereas incorporating physical properties, such as friction or mass distribution, can be critical for some multi-object tasks. Our case study on stacking objects of varying densities showed that including center-of-mass information in the object representation improves tower stability. These findings suggest that effective affordance reasoning for real-world manipulation benefits from direct interaction with objects and the integration of both perceptual and physical information.

Summary Overall, we have shown that affordance learning cannot be treated solely as a computer vision problem and instead it is essential to account for the robot’s capabilities and incorporate interaction-based experience. Integrating the robot into the affordance learning pipeline leads to improved performance on the evaluated tasks. However, learning affordances for generic objects across diverse tasks remains an open challenge that requires further research.

6.2 Outlook

One of the main challenges that remains to be addressed in the field of affordance learning is generalization to novel tasks and unseen objects. The methods presented in this thesis were designed to solve specific manipulation tasks and are tailored to particular robot morphologies. Designing a new pipeline and training it for each task and robot requires considerable effort. Therefore we propose different ideas for future work to improve generalization across robots and reduce training time.

Pretrained Models and Local Robot Adaptation A promising direction is the integration of large pretrained models to enhance task generalization. These models, trained on extensive and diverse datasets, can provide valuable semantic and contextual understanding. However, they currently struggle to achieve the precision required for manipulation tasks involving complex 3D geometries. Future work could explore hybrid approaches that combine the semantic reasoning power of large pretrained models with geometry-aware, robot-specific modules capable of predicting accurate interaction poses.

Additionally, future work should explore how pretrained, robot-independent modules can be effectively adapted to specific robots and their capabilities. This is particularly important for addressing the sim-to-real gap. Although simulations provide a scalable and controllable training environment, models trained solely in simulation often fail to generalize to real-world scenarios due to differences in visual appearance, physical properties, and sensor noise. Future research should therefore focus on strategies for fine-tuning simulation-pretrained networks using small real-world datasets, minimizing the need for extensive data collection. Developing such domain adaptation methods would significantly improve the practicality and robustness of affordance learning systems in real-world applications.

Decoupling Geometric Understanding from Task-Specific Learning Furthermore, to achieve robust generalization beyond imitation-style learning, it is important to focus on learning from geometric and physical principles rather than relying solely on data-driven imitation. Imitation learning approaches are inherently constrained by the diversity and size of the available datasets. In contrast, a model that truly learns to understand the underlying geometry and physics of interactions could extrapolate to unseen objects and tasks.

To this end, future work should investigate architectures that decouple the geometric encoder from the task-specific predictor. A pretrained encoder, trained

on general geometry- or physics-based objectives, could capture the essential spatial structure of objects, while a smaller task-specific module could be trained or fine-tuned for new manipulation objectives. This separation would encourage generalization, simplify the addition of new tasks, and promote a deeper understanding of object interactions.

Bibliography

- [1] B. Abbatematteo, S. Tellex, and G. Konidaris. Learning to Generalize Kinematic Models to Novel Objects. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2019.
- [2] S. Ahuja, H. Admoni, and A. Steinfeld. Learning vision-based physics intuition models for non-disruptive object extraction. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020.
- [3] J.-B. Alayrac, I. Laptev, J. Sivic, and S. Lacoste-Julien. Joint discovery of object states and manipulation actions. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [4] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster. Conformal Risk Control. *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2024.
- [5] P. Ardón, È. Pairet, K. S. Lohan, S. Ramamoorthy, and R. P. A. Petrick. Affordances in Robotic Tasks – A Survey. *arXiv preprint arXiv:2004.07400*, 2020.
- [6] M. Arduengo, C. Torras, and L. Sentis. A versatile framework for robust and adaptive door operation with a mobile manipulator robot. *arXiv*, 02 2019. doi: 10.48550/arXiv.1902.09051.
- [7] D. Baek, B. Peng, S. Gupta, and J. Ramos. Online Learning-Based Inertial Parameter Identification of Unknown Object for Model-Based Control of Wheeled Humanoids. *IEEE Robotics and Automation Letters*, 9(12), 2024.
- [8] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, H. Zhong, Y. Zhu, M. Yang, Z. Li, J. Wan, P. Wang, W. Ding, Z. Fu, Y. Xu, J. Ye, X. Zhang, T. Xie, Z. Cheng, H. Zhang, Z. Yang, H. Xu, and J. Lin. Qwen2.5-vl technical report. *arXiv*, abs/2502.13923, 2025.
- [9] J. Borja-Diaz, O. Mees, G. Kalweit, L. Hermann, J. Boedecker, and W. Burgard. Affordance learning from play for sample-efficient policy learning. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2022.

- [10] A. M. Castro, A. Qu, N. Kuppuswamy, A. Alspach, and M. Sherman. A Transition-Aware Method for the Simulation of Compliant Contact with Regularized Friction. *IEEE Robotics and Automation Letters*, 2020.
- [11] D. S. Chaplot, D. Gandhi, S. Gupta, A. Gupta, and R. Salakhutdinov. Learning to explore using active neural slam. In *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2020.
- [12] D. S. Chaplot, M. Dalal, S. Gupta, J. Malik, and R. R. Salakhutdinov. Seal: Self-supervised embodied active learning using exploration and 3d consistency. *Advances in neural information processing systems*, 34:13086–13098, 2021.
- [13] A. Chaudhry, S. Thiagarajan, and D. Gorur. Finetuning Language Models to Emit Linguistic Expressions of Uncertainty. *arXiv*, abs/2401.00004, 2024.
- [14] T. Chen, S. Gupta, and A. Gupta. Learning exploration policies for navigation. In *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2019.
- [15] Z. Chen, W. Wang, Y. Cao, Y. Liu, Z. Gao, E. Cui, J. Zhu, S. Ye, H. Tian, Z. Liu, et al. Expanding Performance Boundaries of Open-Source Multimodal Models with Model, Data, and Test-Time Scaling. *arXiv*, abs/2412.05271, 2024.
- [16] Z. Chen, W. Wang, H. Tian, S. Ye, Z. Gao, E. Cui, W. Tong, K. Hu, J. Luo, Z. Ma, et al. How Far Are We to GPT-4V? Closing the Gap to Commercial Multimodal Models with Open-Source Suites. *Science China Information Sciences*, 2024.
- [17] Z. Chen, J. Wu, W. Wang, W. Su, G. Chen, S. Xing, M. Zhong, Q. Zhang, X. Zhu, L. Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [18] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song. Diffusion policy: Visuomotor policy learning via action diffusion. *Proc. of Robotics: Science and Systems (RSS)*, 2023.
- [19] E. Chun, Y. Du, A. Simeonov, T. Lozano-Perez, and L. Kaelbling. Local neural descriptor fields: Locally conditioned object representations for manipulation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2023.

- [20] J. J. Chung, J. Förster, P. Wulkop, N. Lawrance, L. Ott, and R. Siegwart. It's just semantics: How to get robots to understand the world the way we do. In *Int. Symposium on Robotics Research*, 2022.
- [21] J. J. Chung, J. Förster, P. Wulkop, L. Ott, N. Lawrance, and R. Siegwart. It's Just Semantics: How to Get Robots to Understand the World the Way We Do. *Robotics Research*, 27:3, 2023.
- [22] E. Collaboration, A. O'Neill, A. Rehman, A. Gupta, A. Maddukuri, A. Gupta, A. Padalkar, A. Lee, A. Pooley, A. Gupta, A. Mandlekar, A. Jain, A. Tung, A. Bewley, A. Herzog, A. Irpan, A. Khazatsky, A. Rai, A. Gupta, A. Wang, A. Kolobov, A. Singh, A. Garg, A. Kembhavi, A. Xie, A. Brohan, A. Raffin, A. Sharma, A. Yavary, A. Jain, A. Balakrishna, A. Wahid, B. Burgess-Limerick, B. Kim, B. Schölkopf, B. Wulfe, B. Ichter, C. Lu, C. Xu, C. Le, C. Finn, C. Wang, C. Xu, C. Chi, C. Huang, C. Chan, C. Agia, C. Pan, C. Fu, C. Devin, D. Xu, D. Morton, D. Driess, D. Chen, D. Pathak, D. Shah, D. Büchler, D. Jayaraman, D. Kalashnikov, D. Sadigh, E. Johns, E. Foster, F. Liu, F. Ceola, F. Xia, F. Zhao, F. V. Frujeri, F. Stulp, G. Zhou, G. S. Sukhatme, G. Salhotra, G. Yan, G. Feng, G. Schiavi, G. Berseth, G. Kahn, G. Yang, G. Wang, H. Su, H.-S. Fang, H. Shi, H. Bao, H. B. Amor, H. I. Christensen, H. Furuta, H. Bharadhwaj, H. Walke, H. Fang, H. Ha, I. Mordatch, I. Radosavovic, I. Leal, J. Liang, J. Abou-Chakra, J. Kim, J. Drake, J. Peters, J. Schneider, J. Hsu, J. Vakil, J. Bohg, J. Bingham, J. Wu, J. Gao, J. Hu, J. Wu, J. Wu, J. Sun, J. Luo, J. Gu, J. Tan, J. Oh, J. Wu, J. Lu, J. Yang, J. Malik, J. Silvério, J. Hejna, J. Booher, J. Thompson, J. Yang, J. Salvador, J. J. Lim, J. Han, K. Wang, K. Rao, K. Pertsch, K. Hausman, K. Go, K. Gopalakrishnan, K. Goldberg, K. Byrne, K. Oslund, K. Kawaharazuka, K. Black, K. Lin, K. Zhang, K. Ehsani, K. Lekkala, K. Ellis, K. Rana, K. Srinivasan, K. Fang, K. P. Singh, K.-H. Zeng, K. Hatch, K. Hsu, L. Itti, L. Y. Chen, L. Pinto, L. Fei-Fei, L. Tan, L. J. Fan, L. Ott, L. Lee, L. Weihs, M. Chen, M. Lepert, M. Memmel, M. Tomizuka, M. Itkina, M. G. Castro, M. Spero, M. Du, M. Ahn, M. C. Yip, M. Zhang, M. Ding, M. Heo, M. K. Srirama, M. Sharma, M. J. Kim, M. Z. Irshad, N. Kanazawa, N. Hansen, N. Heess, N. J. Joshi, N. Suenderhauf, N. Liu, N. D. Palo, N. M. M. Shafiullah, O. Mees, O. Kroemer, O. Bastani, P. R. Sanketi, P. T. Miller, P. Yin, P. Wohlhart, P. Xu, P. D. Fagan, P. Mitranio, P. Sermanet, P. Abbeel, P. Sundaresan, Q. Chen, Q. Vuong, R. Rafailov, R. Tian, R. Doshi, R. Martín-Martín, R. Baijal, R. Scalise, R. Hendrix, R. Lin, R. Qian, R. Zhang, R. Mendonca, R. Shah, R. Hoque, R. Julian, S. Bustamante, S. Kirmani, S. Levine, S. Lin, S. Moore, S. Bahl, S. Dass, S. Sonawani, S. Tulsiani, S. Song, S. Xu, S. Haldar, S. Karam-

- cheti, S. Adebola, S. Guist, S. Nasiriany, S. Schaal, S. Welker, S. Tian, S. Ramamoorthy, S. Dasari, S. Belkhale, S. Park, S. Nair, S. Mirchandani, T. Osa, T. Gupta, T. Harada, T. Matsushima, T. Xiao, T. Kollar, T. Yu, T. Ding, T. Davchev, T. Z. Zhao, T. Armstrong, T. Darrell, T. Chung, V. Jain, V. Kumar, V. Vanhoucke, V. Guizilini, W. Zhan, W. Zhou, W. Burgard, X. Chen, X. Chen, X. Wang, X. Zhu, X. Geng, X. Liu, X. Liangwei, X. Li, Y. Pang, Y. Lu, Y. J. Ma, Y. Kim, Y. Chebotar, Y. Zhou, Y. Zhu, Y. Wu, Y. Xu, Y. Wang, Y. Bisk, Y. Dou, Y. Cho, Y. Lee, Y. Cui, Y. Cao, Y.-H. Wu, Y. Tang, Y. Zhu, Y. Zhang, Y. Jiang, Y. Li, Y. Li, Y. Iwasawa, Y. Matsuo, Z. Ma, Z. Xu, Z. J. Cui, Z. Zhang, Z. Fu, and Z. Lin. Open X-Embodiment: Robotic Learning Datasets and RT-X Models. *arXiv preprint. arXiv:2310.08864*, 2025.
- [23] A. Das, S. Datta, G. Gkioxari, S. Lee, D. Parikh, and D. Batra. Embodied question answering. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [24] A. Delitzas, A. Takmaz, F. Tombari, R. Sumner, M. Pollefeys, and F. Engelmann. SceneFun3D: Fine-Grained Functionality and Affordance Understanding in 3D Scenes. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [25] A. Delitzas, A. Takmaz, F. Tombari, R. Sumner, M. Pollefeys, and F. Engelmann. SceneFun3D: Fine-Grained Functionality and Affordance Understanding in 3D Scenes. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [26] S. Deng, X. Xu, C. Wu, K. Chen, and K. Jia. 3D AffordanceNet: A Benchmark for Visual Object Affordance Understanding. *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- [27] K. Desingh, S. Lu, A. Opipari, and O. C. Jenkins. Factored Pose Estimation of Articulated Objects using Efficient Nonparametric Belief Propagation. *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2019.
- [28] T.-T. Do, A. Nguyen, and I. Reid. Affordancenet: An end-to-end deep learning approach for object affordance detection. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2018.
- [29] A. Dutta, E. Burdet, and M. Kaboli. Push to know! – Visuo-Tactile based Active Object Parameter Inference with Dual Differentiable Filtering. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2023.

-
- [30] K. Fang, T.-L. Wu, D. Yang, S. Savarese, and J. J. Lim. Demo2vec: Reasoning object affordances from online videos. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
 - [31] N. Funk, J. Urain, J. Carvalho, V. Prasad, G. Chalvatzaki, and J. Peters. Actionflow: Equivariant, accurate, and efficient policies with spatially symmetric flow matching. *arXiv preprint arXiv:2409.04576*, 2024.
 - [32] F. Furrer, M. Wermelinger, H. Yoshida, F. Gramazio, M. Kohler, R. Siegwart, and M. Hutter. Autonomous robotic stone stacking with online next best object target pose planning. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*, page 2350–2356. IEEE Press, 2017.
 - [33] J. Gao, B. Sarkar, F. Xia, T. Xiao, J. Wu, B. Ichter, A. Majumdar, and D. Sadigh. Physically Grounded Vision-Language Models for Robotic Manipulation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2024.
 - [34] J. Gao, B. Sarkar, F. Xia, T. Xiao, J. Wu, B. Ichter, A. Majumdar, and D. Sadigh. Physically grounded vision-language models for robotic manipulation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2024.
 - [35] W. Gao and R. Tedrake. kpm-sc: Generalizable manipulation planning using keypoint affordance and shape completion. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2021.
 - [36] W. Gao and R. Tedrake. kpm 2.0: Feedback control for category-level robotic manipulation. In *IEEE Robotics and Automation Letters (RA-L)*, 2021.
 - [37] X. Gao, R. Gong, T. Shu, X. Xie, S. Wang, and S.-C. Zhu. VRKitchen: an Interactive 3D Virtual Environment for Task-oriented Learning. *arXiv preprint arXiv:1903.05757*, 2019.
 - [38] Y. Geng, B. An, H. Geng, Y. Chen, Y. Yang, and H. Dong. Rlafford: End-to-end affordance learning for robotic manipulation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2023.
 - [39] J. Gerigk, P. Wulkop, P. Bajaria, R. Siegwart, F. Shkurti, M. Itkina, and I. Gilitschenski. Handle With Care: Calibrating Vision-Language Models for Safe and Efficient Object Manipulation. *In submission*, 2025.
 - [40] T. Gervet, S. Chintala, D. Batra, J. Malik, and D. S. Chaplot. Navigating to objects in the real world. *Science Robotics*, 8(79), 2023.

- [41] J. J. Gibson. The theory of affordances. *Perceiving, acting, and knowing*, 1977. doi: 10.4324/9781315740218-18.
- [42] J. J. Gibson. *The Ecological Approach to Visual Perception*. Houghton Mifflin, Boston MA, U.S.A, 1979.
- [43] D. Gordon, A. Kembhavi, M. Rastegari, J. Redmon, D. Fox, and A. Farhadi. Iqa: Visual question answering in interactive environments. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [44] A. Grattafori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Vaughan, A. Yang, A. Fan, A. Goyal, A. Hartshorn, A. Yang, A. Mitra, A. Sravankumar, A. Korenev, A. Hinsvark, A. Rao, A. Zhang, A. Rodriguez, A. Gregerson, A. Spataru, B. Roziere, B. Biron, B. Tang, B. Chern, C. Caucheteux, C. Nayak, C. Bi, C. Marra, C. McConnell, C. Keller, C. Touret, C. Wu, C. Wong, C. Canton Ferrer, C. Nikolaidis, D. Allonsius, D. Song, D. Pintz, D. Livshits, D. Wyatt, D. Esiobu, D. Choudhary, D. Mahajan, D. Garcia-Olano, D. Perino, D. Hupkes, E. Lakomkin, E. AlBadawy, E. Lobanova, E. Dinan, E. M. Smith, F. Radenovic, F. Guzmán, F. Zhang, G. Synnaeve, G. Lee, G. Lewis Anderson, G. Thattai, G. Nail, G. Mialon, G. Pang, G. Cucurell, H. Nguyen, H. Korevaar, H. Xu, H. Touvron, I. Zarov, I. Arrieta Ibarra, I. Kloumann, I. Misra, I. Evtimov, J. Zhang, J. Copet, J. Lee, J. Geffert, J. Vranes, J. Park, J. Mahadeokar, J. Shah, J. van der Linde, J. Billock, J. Hong, J. Lee, J. Fu, J. Chi, J. Huang, J. Liu, J. Wang, J. Yu, J. Bitton, J. Spisak, J. Park, J. Rocca, J. Johnstun, J. Saxe, J. Jia, K. V. Alwala, K. Prasad, K. Upasani, K. Plawiak, K. Li, K. Heafield, K. Stone, K. El-Arini, K. Iyer, K. Malik, K. Chiu, K. Bhalla, K. Lakhotia, L. Rantala-Young, L. van der Maaten, L. Chen, L. Tan, L. Jenkins, L. Martin, L. Madaan, L. Malo, L. Blecher, L. Landzaat, L. de Oliveira, M. Muzzi, M. Pasupuleti, M. Singh, M. Paluri, M. Kardas, M. Tsimpoukelli, M. Oldham, M. Rita, M. Pavlova, M. Kambadur, M. Lewis, M. Si, M. Kumar Singh, M. Hassan, N. Goyal, N. Torabi, N. Bashlykov, N. Bogoychev, N. Chatterji, N. Zhang, O. Duchenne, O. Çelebi, P. Alrassy, P. Zhang, P. Li, P. Vasic, P. Weng, P. Bhargava, P. Dubal, P. Krishnan, P. S. Koura, P. Xu, Q. He, Q. Dong, R. Srinivasan, R. Ganapathy, R. Calderer, R. S. Cabral, R. Stojnic, R. Raileanu, R. Maheswari, R. Girdhar, R. Patel, R. Sauvestre, R. Polidoro, R. Sumbaly, R. Taylor, R. Silva, R. Hou, R. Wang, S. Hosseini, S. Chennabasappa, S. Singh, S. Bell, S. S. Kim, S. Edunov, S. Nie, S. Narang, S. Raparthy, S. Shen, S. Wan, S. C. Zha, S. Patil, S. Shankar, S. Zhang, S. Wang, S. Agarwal, S. Sajuyigbe, S. Chintala, S. Max, S. Chen, S. Kehoe, S. Satterfield, S. Govindaprasad, S. Gupta, S. Deng, S. Cho,

- S. Virk, S. Subramanian, S. Choudhury, S. Goldman, T. Remez, T. Glaser, T. Best, T. Koehler, T. Robinson, T. Li, T. Zhang, T. Matthews, T. Chou, T. Shaked, V. Vontimitta, V. Ajayi, V. Montanez, V. Mohan, V. S. Kumar, V. Mangla, V. Ionescu, V. Poenaru, V. T. Mihailescu, V. Ivanov, W. Li, W. Wang, W. Jiang, W. Bouaziz, W. Constable, X. Tang, X. Wu, X. Wang, X. Wu, X. Gao, Y. Kleinman, Y. Chen, Y. Hu, Y. Jia, Y. Li, Y. Zhang, Y. Zhang, Y. Adi, Y. Nam, Y. Wang, Y. Zhao, Y. Hao, Y. Qian, Y. Li, Y. He, Z. Rait, Z. DeVito, Z. Rosnbrick, Z. Wen, Z. Yang, Z. Zhao, and Z. Ma. The Llama 3 Herd of Models. Meta AI, 2024.
- [45] M. Grinvald, F. Tombari, R. Siegwart, and J. Nieto. TSDF++: A multi-object formulation for dynamic object tracking and reconstruction. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2021.
- [46] T. Groot and M. Valdenegro-Toro. Overconfidence is Key: Verbalized Uncertainty Evaluation in Large Language and Vision-Language Models. In *Proceedings of the 4th Workshop on Trustworthy Natural Language Processing (TrustNLP)*, 2024.
- [47] O. Groth, F. B. Fuchs, I. Posner, and A. Vedaldi. Shapestacks: Learning vision-based physical intuition for generalised object stacking. In *Proc. of the European Conference on Computer Vision (ECCV)*, 2018.
- [48] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *Proc. of the International Conference on Machine Learning (ICML)*, 2017.
- [49] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger. On calibration of modern neural networks. In *International conference on machine learning*, pages 1321–1330. Proc. of of Machine Learning Research (PMLR), 2017.
- [50] M. Hassanin, S. Khan, and M. Tahtali. Visual affordance and function understanding: A survey. *Association for Computing Machinery (ACM) Computing Surveys*, 54(3), 2021.
- [51] H. Huang, L. Fu, M. Danielczuk, C. M. Kim, Z. Tam, J. Ichnowski, A. Angelova, B. Ichter, and K. Goldberg. Mechanical search on shelves with efficient stacking and destacking of objects. In *Proc. of the Int. Symposium of Robotics Research (ISRR)*, 2022.
- [52] IKEA. Furniture Models. URL <https://www.ikea.com/ch/en/planners>. Accessed: 2025-04-14.

- [53] A. Irpan, A. Herzog, A. T. Toshev, A. Zeng, A. Brohan, B. A. Ichter, B. David, C. Parada, C. Finn, C. Tan, D. Reyes, D. Kalashnikov, E. V. Jang, F. Xia, J. L. Rettinghouse, J. C. Hsu, J. L. Quiambao, J. Ibarz, K. Rao, K. Hausman, K. Gopalakrishnan, K.-H. Lee, K. A. Jeffrey, L. Luu, M. Yan, M. S. Ahn, N. Sievers, N. J. Joshi, N. Brown, O. E. E. Cortes, P. Xu, P. P. Sampedro, P. Sermanet, R. J. Ruano, R. C. Julian, S. A. Jesmonth, S. Levine, S. Xu, T. Xiao, V. O. Vanhoucke, Y. Lu, Y. Chebotar, and Y. Kuang. Do as i can, not as i say: Grounding language in robotic affordances. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2022.
- [54] A. Jain and S. Niekum. Learning Hybrid Object Kinematics for Efficient Hierarchical Planning Under Uncertainty. *Proc. of the IEEE/RSSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020.
- [55] A. Jain, R. Lioutikov, and S. Niekum. ScrewNet: Category-Independent Articulation Model Estimation From Depth Images Using Screw Theory. *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2021.
- [56] L. Jamone, E. Ugur, A. Cangelosi, L. Fadiga, A. Bernardino, J. Piater, and J. Santos-Victor. Affordances in psychology, neuroscience, and robotics: A survey. *IEEE Transactions on Cognitive and Developmental Systems*, 2016.
- [57] Z. Jiang, C.-C. Hsu, and Y. Zhu. Ditto: Building Digital Twins of Articulated Objects from Interaction. *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [58] L. Johannsmeier, M. Gerchow, and S. Haddadin. A framework for robot manipulation: Skill formalism, meta learning and adaptive control. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2019.
- [59] R. L. Johns, M. Wermelinger, R. Mascaro, D. Jud, I. Hurkxkens, L. Vasey, M. Chli, F. Gramazio, M. Kohler, and M. Hutter. A framework for robotic excavation and dry stone construction using on-site materials. *Science Robotics*, 8(84):eabp9758, 2023.
- [60] Z. Khalifa and S. A. A. Shah. A large scale multi-view rgb-d visual affordance learning dataset. *arXiv preprint arXiv:2203.14092*, 2023.
- [61] A. Khazatsky, A. Nair, D. Jing, and S. Levine. What can i do here? learning new skills by imagining visual affordances. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2021.
- [62] K. Khetarpal, Z. Ahmed, G. Comanici, D. Abel, and D. Precup. What can i do here? A theory of affordances in reinforcement learning. In *Proc. of the International Conference on Machine Learning (ICML)*, 2020.

-
- [63] K. Khetarpal, Z. Ahmed, G. Comanici, D. Abel, and D. Precup. What can i do here? A Theory of Affordances in Reinforcement Learning. In *Proc. of the International Conference on Machine Learning (ICML)*, 2020.
 - [64] D. Kim and G. Sukhatme. Interactive affordance map building for a robotic task. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2015.
 - [65] M. J. Kim, K. Pertsch, S. Karamcheti, T. Xiao, A. Balakrishna, S. Nair, R. Rafailov, E. Foster, G. Lam, P. Sanketi, et al. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.
 - [66] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A. C. Berg, W.-Y. Lo, P. Dollár, and R. Girshick. Segment anything. In *Proc. of the IEEE Int. Conf. on Computer Vision (ICCV)*, 2023.
 - [67] M. Kokic, J. A. Stork, J. A. Haustein, and D. Kragic. Affordance detection for task-specific grasping using deep learning. In *IEEE-RAS Int. Conf. on Humanoid Robotics (Humanoids)*, 2017.
 - [68] E. Kolve, R. Mottaghi, W. Han, E. VanderBilt, L. Weihs, A. Herrasti, D. Gordon, Y. Zhu, A. Gupta, and A. Farhadi. AI2-THOR: An Interactive 3D Environment for Visual AI. *arXiv*, 2017.
 - [69] H. S. Koppula, R. Gupta, and A. Saxena. Learning Human Activities and Object Affordances from RGB-D Videos. *The International Journal of Robotics Research*, 32(8):951–970, 2013. doi: 10.1177/0278364913478446.
 - [70] S. Krishna, C. Agarwal, and H. Lakkaraju. Understanding the Effects of Iterative Prompting on Truthfulness. In *Proc. of the International Conference on Machine Learning (ICML)*, 2024.
 - [71] J. Kuffner and S. LaValle. RRT-connect: An efficient approach to single-query path planning. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2000.
 - [72] C.-L. Kuo, Y.-W. Chao, and Y.-T. Chen. Skt-hang: Hanging everyday objects via object-agnostic semantic keypoint trajectory generation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2024.
 - [73] P. Laban, L. Murakhovs’ka, C. Xiong, and C.-S. Wu. Are You Sure? Challenging LLMs Leads to Performance Drops in The FlipFlop Experiment. *arXiv*, abs/2311.08596, 2024.

- [74] A. Lambert, A. Fishman, D. Fox, B. Boots, and F. Ramos. Stein Variational Model Predictive Control. *Proc. of the Conf. on Robot Learning (CoRL)*, 2020.
- [75] L. K. Le Goff, O. Yaakoubi, A. Coninx, and S. Doncieux. Building an affordances map with interactive perception. *Frontiers in Neurorobotics*, 16, 2022.
- [76] A. X. Lee, C. Devin, Y. Zhou, T. Lampe, K. Bousmalis, J. T. Springenberg, A. Byravan, A. Abdolmaleki, N. Gileadi, D. Khosid, C. Fantacci, J. E. Chen, A. Raju, R. Jeong, M. Neunert, A. Laurens, S. Saliceti, F. Casarini, M. Riedmiller, R. Hadsell, and F. Nori. Beyond pick-and-place: Tackling robotic stacking of diverse shapes. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2021.
- [77] G. Li, D. Sun, L. Sevilla-Lara, and V. Jampani. One-shot open affordance learning with foundation models. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [78] J. Li, Y. Zhu, Z. Xu, J. Gu, M. Zhu, X. Liu, N. Liu, Y. Peng, F. Feng, and J. Tang. MMRo: Are Multimodal LLMs Eligible as the Brain for In-Home Robotics? *arXiv*, abs/2401.00001, 2024.
- [79] Q. Li, K. Mo, Y. Yang, H. Zhao, and L. Guibas. IFR-Explore: Learning Inter-object Functional Relationships in 3D Indoor Scenes. In *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2022.
- [80] W. Li, A. Leonardis, and M. Fritz. Visual stability prediction for robotic manipulation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2017.
- [81] W. Li, L. Pan, B. Jiang, Y. Wu, W. Liu, and G. S. Chirikjian. I get the hang of it! a learning-free method to predict hanging poses for previously unseen objects. *IEEE Robotics and Automation Letters*, 9(4), 2024.
- [82] X. Li, H. Wang, L. Yi, L. J. Guibas, A. L. Abbott, and S. Song. Category-level articulated object pose estimation. *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [83] J. Liang, W. Huang, F. Xia, P. Xu, K. Hausman, B. Ichter, P. Florence, and A. Zeng. Code as Policies: Language Model Programs for Embodied Control. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2023.

-
- [84] X. Liang, A. Han, W. Yan, A. Raghunathan, and P. Abbeel. ALP: Action-Aware Embodied Learning for Perception. *arXiv preprint arXiv:2306.10190*, 2023.
- [85] S. Liu, X. Wei, L. Wang, J. Zhang, B. Li, and H. Yue. Center-of-mass-based robust grasp pose adaptation using rgbd camera and force/torque sensing. In *Advances in Guidance, Navigation and Control*, pages 2699–2708, Singapore, 2023. Springer Nature Singapore.
- [86] T. Lüddecke and A. S. Ecker. Image segmentation using text and image prompts. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [87] L. Manuelli, W. Gao, P. Florence, and R. Tedrake. kpam: Keypoint affordances for category-level robotic manipulation. In *Proc. of the Int. Symposium of Robotics Research (ISRR)*, 2019.
- [88] R. Martín-Martín and O. Brock. Coupled recursive estimation for online interactive perception of articulated objects. *The International Journal of Robotics Research*, 2019.
- [89] H. Min, C. Yi, R. Luo, J. Zhu, and S. Bi. Affordance research in developmental robotics: A survey. *IEEE Transactions on Cognitive and Developmental Systems*, 2016.
- [90] R. Mirjalili, M. Krawez, S. Silenzi, Y. Blei, and W. Burgard. Lan-grasp: Using Large Language Models for Semantic Object Grasping. In *Proc. of the Int. Symposium of Robotics Research (ISRR)*, 2024.
- [91] M. Mittal, D. Hoeller, F. Farshidian, M. Hutter, and A. Garg. Articulated Object Interaction in Unknown Scenes with Whole-Body Mobile Manipulation. *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2022.
- [92] K. Mo, S. Zhu, A. X. Chang, L. Yi, S. Tripathi, L. J. Guibas, and H. Su. PartNet: A Large-Scale Benchmark for Fine-Grained and Hierarchical Part-Level 3D Object Understanding. *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [93] K. Mo, L. J. Guibas, M. Mukadam, A. Gupta, and S. Tulsiani. Where2Act: From Pixels to Actions for Articulated 3D Objects. In *Proc. of the IEEE Int. Conf. on Computer Vision (ICCV)*, 2021.
- [94] K. Mo, Y. Qin, F. Xiang, H. Su, and L. J. Guibas. O2O-Afford: Annotation-Free Large-Scale Object-Object Affordance Learning. *Proc. of the Conf. on Robot Learning (CoRL)*, 2021.

- [95] J. Mu, W. Qiu, A. Kortylewski, A. Yuille, N. Vasconcelos, and X. Wang. A-SDF: Learning Disentangled Signed Distance Functions for Articulated Shape Representation. *Proc. of the IEEE Int. Conf. on Computer Vision (ICCV)*, 2021.
- [96] J. F. Mullen Jr. and D. Manocha. LAP, Using Action Feasibility for Improved Uncertainty Alignment of Large Language Model Planners. *arXiv*, 2024.
- [97] A. Myers, C. L. Teo, C. Fermüller, and Y. Aloimonos. Affordance detection of tool parts from geometric features. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2015.
- [98] T. Nagarajan and K. Grauman. Learning affordance landscapes for interaction exploration in 3d environments. *Proc. of Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [99] T. Nagarajan, C. Feichtenhofer, and K. Grauman. Grounded human-object interaction hotspots from video. In *Proc. of the IEEE Int. Conf. on Computer Vision (ICCV)*, 2019.
- [100] T. Nagarajan, Y. Li, C. Feichtenhofer, and K. Grauman. Ego-topo: Environment affordances from egocentric video. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [101] A. Nguyen, D. Kanoulas, D. G. Caldwell, and N. G. Tsagarakis. Detecting object affordances with convolutional neural networks. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2016.
- [102] A. Nguyen, D. Kanoulas, D. G. Caldwell, and N. G. Tsagarakis. Object-based affordances detection with convolutional neural networks and dense conditional random fields. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2017.
- [103] T.-L. Nguyen, D.-V. Nguyen, and T.-H. Le. Reinforcement learning based navigation with semantic knowledge of indoor environments. In *Int. Conf. on Knowledge and Systems Engineering (KSE)*, 2019.
- [104] Octo Model Team, D. Ghosh, H. Walke, K. Pertsch, K. Black, O. Mees, S. Dasari, J. Hejna, C. Xu, J. Luo, T. Kreiman, Y. Tan, L. Y. Chen, P. Sanketi, Q. Vuong, T. Xiao, D. Sadigh, C. Finn, and S. Levine. Octo: An open-source generalist robot policy. In *Proceedings of Robotics: Science and Systems*, 2024.

-
- [105] OpenAI. Gpt-4o system card. URL <https://openai.com/index/gpt-4o-system-card/>. Accessed: 2025-04-14.
 - [106] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman. Inductive confidence machines for regression. In *Proc. of the European Conf. on Machine Learning (ECML)*. Springer, 2002.
 - [107] J. Pavlasek, S. Lewis, K. Desingh, and O. C. Jenkins. Parts-Based Articulated Object Localization in Clutter Using Belief Propagation. *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020.
 - [108] B. Peng, D. Baek, Q. Wang, and J. Ramos. FSL-LVLM: Friction-Aware Safety Locomotion using Large Vision Language Model in Wheeled Robots. *arXiv*, abs/2401.00002, 2024.
 - [109] L. Pinto, D. Gandhi, Y. Han, Y.-L. Park, and A. Gupta. The curious robot: Learning visual representations via physical interactions. In *Proc. of the European Conference on Computer Vision (ECCV)*, 2016.
 - [110] C. Pohl, K. Hitzler, R. Grimm, A. Zea, U. D. Hanebeck, and T. Asfour. Affordance-based grasping and manipulation in real world applications. *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020.
 - [111] X. Puig, K. Ra, M. Boben, J. Li, T. Wang, S. Fidler, and A. Torralba. Virtualhome: Simulating household activities via programs. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
 - [112] C. R. Qi, L. Yi, H. Su, and L. J. Guibas. PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space. *Proc. of Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
 - [113] W. Qi, R. T. Mullapudi, S. Gupta, and D. Ramanan. Learning to move with affordance maps. In *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2020.
 - [114] S. K. Ramakrishnan, D. Jayaraman, and K. Grauman. An exploration of embodied visual exploration. *Int. Journal of Computer Vision*, 2021.
 - [115] N. Ravi, V. Gabeur, Y.-T. Hu, R. Hu, C. Ryali, T. Ma, H. Khedr, R. Rädle, C. Rolland, L. Gustafson, E. Mintun, J. Pan, K. V. Alwala, N. Carion, C.-Y. Wu, R. Girshick, P. Dollár, and C. Feichtenhofer. SAM 2: Segment Anything in Images and Videos. *arXiv*, abs/2408.00714, 2024.

- [116] A. Z. Ren, A. Dixit, A. Bodrova, S. Singh, S. Tu, N. Brown, P. Xu, L. Takayama, F. Xia, J. Varley, Z. Xu, D. Sadigh, A. Zeng, and A. Majumdar. Robots That Ask For Help: Uncertainty Alignment for Large Language Model Planners. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2023.
- [117] A. Z. Ren, J. Clark, A. Dixit, M. Itkina, A. Majumdar, and D. Sadigh. Explore until Confident: Efficient Exploration for Embodied Question Answering. *Proc. of Robotics: Science and Systems (RSS)*, 2024.
- [118] G. M. Rizzi, J. J. Chung, A. R. Gaweł, L. Ott, M. Tognon, and R. Siegwart. Robust sampling-based control of mobile manipulators for interaction with articulated objects. *IEEE Transactions on Robotics*, 39(3), 2023.
- [119] A. Rofer, G. Bartels, W. Burgard, A. Valada, and M. Beetz. Kineverse: A symbolic articulation model framework for model-agnostic mobile manipulation. *IEEE Robotics and Automation Letters*, 7(2):3372–3379, 2022.
- [120] O. Ronneberger, P. Fischer, and T. Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention*, 2015.
- [121] E. Ruiz and W. W. Mayol-Cuevas. Geometric Affordance Perception: Leveraging Deep 3D Saliency With the Interaction Tensor. *Frontiers in Neurobotics*, 2020.
- [122] G. Schiavi, P. Wulkoop, G. Rizzi, L. Ott, R. Siegwart, and J. J. Chung. Learning Agent-Aware Affordances for Closed-Loop Interaction with Articulated Objects. In *International Conference on Robotics and Automation (ICRA)*, 2023.
- [123] L. Schmid, J. Delmerico, J. Schönberger, J. Nieto, M. Pollefeys, R. Siegwart, and C. Cadena. Panoptic Multi-TSDFs: a Flexible Representation for Online Multi-resolution Volumetric Mapping and Long-term Dynamic Scene Consistency. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2022.
- [124] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov. Proximal policy optimization algorithms. *Computing Research Repository (CoRR)*, abs/1707.06347, 2017.
- [125] R. Shao, W. Li, L. Zhang, R. Zhang, Z. Liu, R. Chen, and L. Nie. Large VLM-based Vision-Language-Action Models for Robotic Manipulation: A Survey. *arXiv preprint arXiv:2508.13073*, 2025.

-
- [126] O. Shorinwa, Z. Mei, J. Lidard, A. Z. Ren, and A. Majumdar. A Survey on Uncertainty Quantification of Large Language Models: Taxonomy, Open Research Challenges, and Future Directions. *ACM Computing Surveys*, 2024.
- [127] M. Shridhar, J. Thomason, D. Gordon, Y. Bisk, W. Han, R. Mottaghi, L. Zettlemoyer, and D. Fox. Alfred: A benchmark for interpreting grounded instructions for everyday tasks. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [128] A. Simeonov, Y. Du, L. Yen-Chen, A. Rodriguez, L. P. Kaelbling, T. Lozano-Perez, and P. Agrawal. SE(3)-Equivariant Relational Rearrangement with Neural Descriptor Fields. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2022.
- [129] A. Simeonov, A. Goyal, L. Manuelli, L. Yen-Chen, A. Sarmiento, A. Rodriguez, P. Agrawal, and D. Fox. Shelving, stacking, hanging: Relational pose diffusion for multi-modal rearrangement. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2023.
- [130] C. Song and A. Boularias. A probabilistic model for planar sliding of objects with unknown material properties: Identification and robust planning. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020.
- [131] J. Sturm, A. Jain, C. Stachniss, C. C. Kemp, and W. Burgard. Operating articulated objects based on experience. *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2010.
- [132] D. Stutz, A. T. Cemgil, A. Doucet, et al. Learning optimal conformal classifiers. In *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2022.
- [133] B. Talbot, F. Dayoub, P. Corke, and G. Wyeth. Robot navigation in unseen spaces using an abstract map. *IEEE Transactions on Cognitive and Developmental Systems*, 2020.
- [134] H. Tan, L. Yu, and M. Bansal. Learning to navigate unseen environments: Back translation with environmental dropout. In *Conference of the North American Chapter of the Association for Computational Linguistics*, 2019.
- [135] W. Tu, W. Deng, D. Campbell, S. Gould, and T. Gedeon. An empirical study into what matters for calibrating vision-language models. In *Proc. of the International Conference on Machine Learning (ICML)*, 2024.

- [136] D. Turpin, L. Wang, S. Tsogkas, S. Dickinson, and A. Garg. GIFT: Generalizable Interaction-aware Functional Tool Affordances without Labels. In *Proceedings of Robotics: Science and Systems*, 2021.
- [137] V. Vovk, A. Gammernan, and G. Shafer. *Algorithmic learning in a random world*, volume 29. Springer, 01 2005. doi: 10.1007/b106715.
- [138] A. D. Vuong, M. N. Vu, B. Huang, N. Nguyen, H. Le, T. Vo, and A. Nguyen. Language-driven Grasp Detection. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [139] C. Wang, S. Wang, B. Romero, F. Veiga, and E. Adelson. Swingbot: Learning physical features from in-hand tactile exploration for dynamic swing-up manipulation. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2020.
- [140] H. Wang, S. Sridhar, J. Huang, J. Valentin, S. Song, and L. J. Guibas. Normalized Object Coordinate Space for Category-Level 6D Object Pose and Size Estimation. *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [141] J. Wang, C. Hu, Y. Wang, and Y. Zhu. Dynamics learning with object-centric interaction networks for robot manipulation. *IEEE Access*, 9:68277–68288, 2021.
- [142] X. Wang, B. Zhou, Y. Shi, X. Chen, Q. Zhao, and K. Xu. Shape2Motion: Joint Analysis of Motion Parts and Attributes From 3D Shapes. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [143] X. Wang, Y. Liu, X. Song, Y. Liu, S. Zhang, and S. Jiang. An interactive navigation method with effect-oriented affordance. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [144] Y. Wang, Y. Sun, Z. Liu, S. E. Sarma, M. M. Bronstein, and J. M. Solomon. Dynamic graph cnn for learning on point clouds. *ACM Trans. Graph.*, 38 (5), Oct. 2019. ISSN 0730-0301. doi: 10.1145/3326362.
- [145] Y. Wang, R. Wu, K. Mo, J. Ke, Q. Fan, L. Guibas, and H. Dong. AdaAfford: Learning to adapt manipulation affordance for 3d articulated objects via few-shot interactions. In *Proc. of the European Conference on Computer Vision (ECCV)*, 2022.

-
- [146] Y. Weng, H. Wang, Q. Zhou, Y. Qin, Y. Duan, Q. Fan, B. Chen, H. Su, and L. J. Guibas. CAPTRA: CAtegory-level Pose Tracking for Rigid and Articulated Objects from Point Clouds. *Proc. of the IEEE Int. Conf. on Computer Vision (ICCV)*, 2021.
- [147] G. Williams, A. Aldrich, and E. A. Theodorou. Model Predictive Path Integral Control: From Theory to Parallel Computation. *Journal of Guidance, Control, and Dynamics*, 40(2):344–357, 2017.
- [148] G. Williams, N. Wagener, B. Goldfain, P. Drews, J. M. Rehg, B. Boots, and E. A. Theodorou. Information theoretic MPC for model-based reinforcement learning. *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2017.
- [149] K. Wu, Y. Zhu, J. Li, J. Wen, N. Liu, Z. Xu, and J. Tang. Discrete policy: Learning disentangled action space for multi-task robotic manipulation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2025.
- [150] R. Wu, Z. Guo, Q. Fan, Y. Zhao, X. Chen, L. Guibas, H. Dong, Y. Wang, T. Wu, and K. Mo. VAT-Mart: Learning Visual Action Trajectory Proposals for Manipulating 3D ARTiculated Objects. *Proc. of the Int. Conf. on Learning Representations (ICLR)*, 2021.
- [151] X. Wu, L. Jiang, P.-S. Wang, Z. Liu, X. Liu, Y. Qiao, W. Ouyang, T. He, and H. Zhao. Point transformer v3: Simpler, faster, stronger. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- [152] P. Wulkoop, H. U. Özdemir, A. Hübner, J. J. Chung, R. Siegwart, and L. Ott. Learning Affordances from Interactive Exploration using an Object-level Map. In *International Symposium of Robotics Research (ISRR)*, 2024.
- [153] F. Xiang, Y. Qin, K. Mo, Y. Xia, H. Zhu, F. Liu, M. Liu, H. Jiang, Y. Yuan, H. Wang, L. Yi, A. X. Chang, L. J. Guibas, and H. Su. SAPIEN: A Simulated Part-based Interactive ENvironment. *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [154] X. Xiao, J. Liu, Z. Wang, Y. Zhou, Y. Qi, S. Jiang, B. He, and Q. Cheng. Robot learning in the era of foundation models: a survey. *Neurocomputing*, 638:129963, 2025.
- [155] J. Xie, A. S. Chen, Y. Lee, E. Mitchell, and C. Finn. Calibrating language models with adaptive temperature scaling. *Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2024.

- [156] W. Xie, M. Valentini, J. Lavering, and N. Correll. DeliGrasp: Inferring Object Properties with LLMs for Adaptive Grasp Policies. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2024.
- [157] Z. Xie, Z. Wei, S. Liu, and J. Jia. Pointbert: Pre-training 3d point cloud transformers with masked point modeling. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [158] Z. Xie, Z. Wei, S. Liu, and J. Jia. Point-mae: Masked autoencoders are scalable vision learners for 3d point clouds. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- [159] D. Xu, A. Mandlekar, R. Martín-Martín, Y. Zhu, S. Savarese, and L. Fei-Fei. Deep affordance foresight: Planning through what can be done in the future. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2021.
- [160] Z. Xu, J. Wu, A. Zeng, J. B. Tenenbaum, and S. Song. Densephysnet: Learning dense physical object representations via multi-step dynamic interactions. In *Proc. of Robotics: Science and Systems (RSS)*, 2019.
- [161] Z. Xu, Z. He, and S. Song. Universal manipulation policy network for articulated objects. *IEEE Robotics and Automation Letters*, 7(2):2447–2454, Apr. 2022. ISSN 2377-3774. doi: 10.1109/lra.2022.3142397.
- [162] X. Yang, Z. Ji, J. Wu, and Y.-K. Lai. Recent advances of deep robotic affordance learning: A reinforcement learning perspective. *IEEE Transactions on Cognitive and Developmental Systems*, 15(3):1139–1149, 2023.
- [163] F. Ye, M. Yang, J. Pang, L. Wang, D. F. Wong, E. Yilmaz, S. Shi, and Z. Tu. Benchmarking LLMs via Uncertainty Quantification. In *Proc. of Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [164] L. Yen-Chen, A. Zeng, S. Song, P. Isola, and T.-Y. Lin. Learning to See before Learning to Act: Visual Pre-training for Manipulation. *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2020.
- [165] Y. You, L. Shao, T. Migimatsu, and J. Bohg. Omnihang: Learning to hang arbitrary objects using contact point correspondences and neural collision estimation. In *Proc. of the IEEE Int. Conf. on Robotics & Automation (ICRA)*, 2021.
- [166] Y. Ze, G. Zhang, K. Zhang, C. Hu, M. Wang, and H. Xu. 3D Diffusion Policy: Generalizable Visuomotor Policy Learning via Simple 3D Representations. *Proc. of Robotics: Science and Systems (RSS)*, 2024.

-
- [167] A. Zeng, S. Song, S. Welker, J. Lee, A. Rodriguez, and T. Funkhouser. Learning Synergies Between Pushing and Grasping with Self-Supervised Deep Reinforcement Learning. *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2018.
 - [168] V. Zeng, T. E. Lee, J. Liang, and O. Kroemer. Visual Identification of Articulated Object Parts. In *Proc. of the IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2021.
 - [169] J. Zhang, L. Dai, F. Meng, Q. Fan, X. Chen, K. Xu, and H. Wang. 3D-Aware Object Goal Navigation via Simultaneous Exploration and Identification. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.
 - [170] Y. Zhao, M. Bogdanovic, C. Luo, S. Tohme, K. Darvish, A. Aspuru-Guzik, F. Shkurti, and A. Garg. Anyplace: Learning generalizable object placement for robot manipulation. In *Proc. of the Conf. on Robot Learning (CoRL)*, 2025.
 - [171] Y. Zheng, L. Yao, Y. Su, Y. Zhang, Y. Wang, S. Zhao, Y. Zhang, and L.-P. Chau. A survey of embodied learning for object-centric robotic manipulation. *Machine Intelligence Research*, 22(4):588–626, June 2025. ISSN 2731-5398.
 - [172] E. Zhou, Q. Su, C. Chi, Z. Zhang, Z. Wang, T. Huang, L. Sheng, and H. Wang. Code-as-Monitor: Constraint-aware Visual Programming for Reactive and Proactive Robotic Failure Detection. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.
 - [173] X. Zhou, G. Zhao, Y. Xing, J. Wu, and Z. Xiong. Robotic stacking of irregular objects with load position identification and compensation. In *IEEE Int. Conf. on Automation Science and Engineering (CASE)*, 2023.
 - [174] Y. Zhu, C. Jiang, Y. Zhao, D. Terzopoulos, and S.-C. Zhu. Inferring forces and learning human utilities from videos. In *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

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